The normalized Lmin core selection on exact games

Draft fragment: boundary theorems and exact monotonicity counterexamples

Maxwell Flanagan

August 2026 — working fragment

Status: mathematical core of a paper in progress. Singleton bumps are proved monotone even on balanced games. Grand-coalition bumps remain on an exact path but labeled resource monotonicity is disproved by an exact five-player example. Proper nonsingleton coalitional monotonicity is also disproved, at the minimum possible dimension, by an exact five-player lower-envelope pair with a complete hand-checkable leximin path. An exact symbolic classification proves that every singleton-normalized exact game through four players is an exact partition game at its Lmin allocation, and hence that Lmin equals Lmax on this domain. A dimension-free protector-packing inequality gives a human sacrifice-budget explanation of the sharp unique-richest threshold: it is impossible through four players and attained by one indecomposable four-generator integral family in every dimension from five onward. Its closed-form normalized Lmin–Lmax distance tends to one, which cannot be obtained by null-player padding and makes the four-player equality threshold sharp dimension-free. A secondary two-parameter amplification gives the same phenomenon with a tied richest stratum of arbitrary size, with asymptotics stated only at fixed block size. A dimension-free balanced-trace transport theorem shows that every slack richer Lmin suffix must be carried by a minimal balanced rich-side circuit and opposing poor-side displacements. For two poorer players this forces tightness through three richer players. More generally, every two-poorer failure in every dimension forces a nonintegral active trace hypergraph and an odd balanced-matrix obstruction. More sharply, no rank-two (graph-like) active trace hypergraph can support a two-poorer failure: every such slack boundary has a tight coalition meeting the richer suffix in at least three players. A real-valued exact-face overload budget unifies the sharp unique-richest threshold with all subsequent rounding contradictions; its integer singleton specialization says that exact faces forbid every tight cover with rich load $\mathbf{1}_H + ke_d$ and safe zero–one poor loads. An idempotent tight-skeleton retraction gives a support-independent converse reduction: every singleton-normalized exact game can be replaced by an exact game defined only by its Lmin-tight coalition inequalities, while preserving Lmin and the tight/slack status of every coalition. Thus the exact-partition question loses no information when reduced to the labeled tight hypergraph. A boundary-local laminar-cover theorem then gives a dimension-free positive domain: if a strict richer-side Lmin cover can be chosen with laminar support at every payoff boundary, every richer suffix is tight and the game is exact partition. Equivalently, every slack suffix forces crossing coalitions in the support of every strict cover. A cover-free convex-geometric theorem is stronger: for every compact convex efficient allocation set, a slack richer face at its leximin point has a one-coordinate poorer-side extension with strictly larger face deficit. For exact-game cores this is precisely coalition-slack ascent, equivalently an adjacent marginal deficit. Moreover, maximum slack over the Boolean interval above the face equals the maximum one-sided total variation of the poorer-side core projection. Terminal coalitions are exactly the sign cells of the maximiz-

ing core points, and a maximizing exactifier may be chosen at a core vertex. Thus every failed suffix produces a payoff-inverted punctured suffix; the strict-cover machinery is needed only for the sharper deficit-weighted amplification and singleton-sacrifice budgets. The theorem covers both pure-star and trace-coupled branches and is saturated by three independently recomputed obstruction families. Its correctness and novelty remain subject to specialist review; no claim of external validation or unrestricted resolution is made. An exact-face uncrossing theorem applies to any core point of any exact game: a uniformly loaded, outside-underloading tight cover forces its target face whenever the binding coalitions are closed under uncrossing. At Lmin this broadens the positive domain to games supermodular only on binding pairs; conversely, every failed boundary has a canonical broken tight diamond with a strictly positive local supermodularity defect. A quantitative uncrossing conservation law strengthens this obstruction: the signed cumulative local defects along any integral laminarization equal the final weighted core slack and bound the slack of the target exact face. An independent deficit-completion theorem replaces uncovered singleton payoff by the minimum actual slack cost of a poorer-side coalition cover. Its zero-cost cone forces exact partition, and both the support-two and support-three obstruction families are sharp equality cases. Every slack boundary furthermore reduces conformally to one pure minimal semi-balanced circuit with the failed suffix exceptional, at most n proper coalitions, and a genuinely costly poorer singleton. This turns the general obstruction problem into the classification of level-oriented signed circuits. The costly singleton also forces every such atom to be decomposable in the established exact-cone sense, so Lmin failures systematically avoid facet-defining atoms and require a relative, label-preserving decomposition theory. More generally, exactifying any coalition H proves that its marginal game $Q \mapsto v(H \cup Q) - v(H)$ on the complement is balanced, and H is tight at a core point exactly when the complementary restriction is a core allocation of that marginal game. Exact partition is therefore a marginal-core compatibility property, while a failed Lmin boundary is an under-efficient allocation of a balanced marginal game with deficit equal to the face slack. Absorbing that forced completion triangle yields a stronger rooted invariant: the poorer-side completion is charged by its marginal worth over the failed suffix, subtracting the full superadditivity surplus. Every failure reduces again to a bounded rooted circuit. Rooting produces an indecomposable facet atom for the five-player diamond but leaves one residual decomposition in the seven-player family, giving a precise bridge—not an identification—between Lmin obstructions and exact-cone facets. For the entire three-trace family that residual is classified: the rooted atom is exactly the sum of a complementary-diamond facet and an internal three-block balanced facet, and both inequalities are saturated. Thus the first two obstruction tiers admit saturated facet decompositions of lengths one and two, respectively. The full two-column obstruction tier has a normal form: its two tight coalitions meet exactly in the failed richer suffix, and the exterior coalition's core slack must finance the target slack. A sharp converse lower-envelope family realizes precisely the permitted capacity range. This gives a direct Boolean-lattice and slack-conservation explanation of the four/five threshold. An explicit scale-separated set-cover objective then shows that any minor lift preserving one strict weighted gap is automatically exact on all rich rows, has disjoint label-free contractions, and retains the gaining-player deficit; the unresolved step is precisely preservation of that objective by a relative minimally-nonideal reduction. At positive Lehman trace cores, the classical dual identity generates these rooted covers canonically, and a two-label density theorem subsumes the odd-cycle and coarse projective-plane roundings. A two-sided label residual and its subset face budget unify singleton row exclusions with the canonical internal-face amplification and provide a candidate recursive invariant; its transpose transport law forces exact cancellation on every

label-free trace. A capacity–parity lemma excludes all packets with at most three gaining-label columns and reduces the first thin four-label case to six local trace tiles. The classical triple-intersection bound then packs all negative-extreme rows into disjoint primal supports and forces a uniform companion star around each four-label-simple row. A global incidence count identifies every heavy losing-label row with one canonical dual column and excludes all dual-cubic cores. At the first feasible four-player richer suffix, a human four-equation classification shows that the only surviving fractional packet is the four-co-singleton cover with weights one third, exactly matching the sharp family. A further star-rounding theorem excludes projective-plane incidence packets whenever a simple-point density inequality holds; a two-sided convex Lehman-row interpolation eliminates every finite projective-plane packet, in every order. The unique three-player nonpartition packet is eliminated by exact singleton faces, while the four-richer family realizes the first surviving odd trace cycle. A dimension-free integral continuation realizes a slack boundary throughout the half-plane with at least two poorer and at least four richer players, even with Lmin equal to Lmax. A second universal integral family classifies the entire three-column obstruction tier as a blown-up triangle, proves its sharp atom-size threshold, and realizes every admissible shape from five rich players onward. This disproves reduction of all larger failures to the first four-co-singleton packet. Another universal family realizes every boundary shape with at least three poorer and at least two richer players. Together with the unique-richest family and the human impossibility theorems, these constructions give a complete cardinality classification: the minimum poorer-side size for a slack richer suffix is four for a one-player suffix, three for suffix size two or three, and two for suffix size at least four. The new three-poorer family also has a closed-form distinct Lmax in every dimension. The separate exact-bump capacity implication is proved for every two-level failure dual, including all two-player coalitions, and a sharp quantitative carrier–star inequality is proved for two-player coalitions. The quantitative carrier inequality for arbitrary target coalitions is proved exactly through six players and disproved by exact three-generator games in every dimension from seven onward. Balanced nonexact counterexamples provide additional sharpness results.

1 The normalized Lmin core selection

Definition 1.1 (games, cores, exactness, and balanced collections)**.** *A transferable-utility game is a set function $v : 2^N \to \mathbb{R}$ with $v(\emptyset) = 0$. Its core is*

$$\mathrm{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(T) \geq v(T) \text{ for every } T \subsetneq N\}.$$

The game is balanced *when its core is nonempty and* exact *when, for every coalition T, some $x \in \mathrm{Core}(v)$ satisfies $x(T) = v(T)$. A* balanced collection *is a family of coalitions with positive weights α_T satisfying $\sum_T \alpha_T \mathbf{1}_T = \mathbf{1}_N$; it is* minimal *when no proper subfamily is balanced.*

Let (N, v) be a balanced TU game, $M = v(N)$, and floors $m_i := v(\{i\})$. For $x \in \mathbb{R}^N$ write the *gains* $g_i(x) = x_i - m_i$. The *leximin-above-floors* selection is

$$\phi(v) := \arg \operatorname*{leximin}_{x \in \mathrm{Core}(v)} \big(g_1(x), \ldots, g_n(x)\big),$$

the unique core allocation whose ascending-sorted gain vector is lexicographically largest. It is single-valued (leximin over a polytope) and is computed by the Maschler-type scheme over player coordinates: repeatedly raise a common lower level s of the free players' gains as far as the

core allows, fix ("pin") every free player whose payoff cannot exceed $m_i + s$ on the level-optimal face, and recurse on the rest.

This is a normalized version of the standard Lmin solution of Arin and Iñarra (2001), not a new unnamed egalitarian selector. Put

$$v^0(T) = v(T) - \sum_{i \in T} m_i, \qquad m = (m_i)_{i \in N}. \tag{LN}$$

Then $\mathrm{Core}(v^0) = \mathrm{Core}(v) - m$ and therefore

$$\phi(v) = m + \mathrm{Lmin}(v^0). \tag{L0}$$

Thus we call ϕ the *normalized Lmin* selection. Whenever the singleton values are zero—as in the exact proper-coalition counterexample below—ϕ is precisely the ordinary Lmin solution. The normalization makes the individually rational gains invariant under additive translations of the game.

On *exact* games $m_i = \min_{\mathrm{Core}} x_i$, so $g_i \geq 0$ and ϕ egalitarianly distributes the surplus $x_i - v(\{i\})$ inside the core; on convex games it is the constrained egalitarian solution of Dutta and Ray (1989). Implementation (exact rational + float) in `src/leximin.py`, validated against those two coincidences.

We ask whether ϕ is *coalitionally monotonic*: if $v' = v + d\,\mathbf{1}_S$ for a single coalition S and $d > 0$, is $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$? A positive answer on exact games would exhibit an explicit coalitionally monotonic core selection on that class. The reduction in Section 5 addresses only proper nonsingleton coalitions. Singleton bumps change the floors defining ϕ, and a grand-coalition bump changes the efficiency hyperplane. The singleton case is resolved next. The grand-coalition case has a negative answer even on a simple exact five-player game (Theorem 3.2).

2 Singleton bumps are monotone

We first isolate an elementary property of the leximin order. Write $a \succ_{\mathrm{lx}} b$ when the ascending rearrangement of a is lexicographically larger than that of b.

Lemma 2.1 (one-coordinate translation)**.** *If $a \succ_{\mathrm{lx}} b$, $a_i < b_i$, and $d > 0$, then $a + de_i \succ_{\mathrm{lx}} b + de_i$.*

Proof. For a finite vector z, put $L_\lambda(z) = \sum_j e^{-\lambda z_j}$. The first unequal coordinate of the ascending rearrangements dominates this sum as $\lambda \to \infty$; hence $a \succ_{\mathrm{lx}} b$ if and only if $L_\lambda(a) < L_\lambda(b)$ for all sufficiently large λ. Moreover,

$$\begin{aligned} L_\lambda(b + de_i) - L_\lambda(a + de_i) = L_\lambda(b) - L_\lambda(a) \\ + (1 - e^{-\lambda d})(e^{-\lambda a_i} - e^{-\lambda b_i}). \end{aligned}$$

Both terms on the right are positive for all sufficiently large λ, because $a_i < b_i$. The same characterization gives the claim. □

Theorem 2.2 (singleton coalitional monotonicity)**.** *Let v and $v' = v + d\,\mathbf{1}_{\{i\}}$ be balanced games, with $d > 0$. Then*

$$\phi_i(v') \geq \phi_i(v).$$

Thus singleton bumps are monotone even without exactness.

Proof. Let $m_j = v(\{j\})$ and express the old core in gain coordinates

$$G = \{(x_j - m_j)_{j\in N} : x \in \mathrm{Core}(v)\}.$$

If the singleton floor of player i is raised by d, the new feasible gain set is

$$H = \{g - de_i : g \in G,\ g_i \geq d\}.$$

Let z be the leximin point of G and h the leximin point of H. The desired inequality is $h_i + d \geq z_i$. If $z_i < d$, this is immediate from $h_i \geq 0$. Suppose $z_i \geq d$. Then $b = z - de_i$ belongs to H, so h is either b or $h \succ_{\mathrm{lx}} b$. If, contrary to the claim, $h_i + d < z_i$, then $h_i < b_i$ and the equality case is impossible. Lemma 2.1 gives

$$h + de_i \succ_{\mathrm{lx}} b + de_i = z.$$

But $h + de_i \in G$, contradicting the definition of z as the leximin point of G. □

3 Grand-coalition bumps: an exact path but no resource monotonicity

The grand-coalition case also has an exact path, but its geometry differs from the nested-core path of Section 5.

Lemma 3.1 (grand-bump exact path)**.** *Let v be exact and, for $r \geq 0$, let v^r agree with v on every proper coalition and satisfy $v^r(N) = v(N) + r$. Then v^r is exact for every $r \geq 0$. Moreover,*

$$\mathrm{Core}(v) + r\Delta_N \subseteq \mathrm{Core}(v^r), \qquad \Delta_N = \{a \in \mathbb{R}^N_+ : a(N) = 1\}.$$

Proof. If $x \in \mathrm{Core}(v)$ and $a \in r\Delta_N$, then $x + a$ has total $v(N) + r$ and weakly raises every coalition sum, proving the inclusion and balancedness. Fix a proper coalition T. Exactness of v supplies $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$. Choose $j \notin T$. Then $x^T + re_j \in \mathrm{Core}(v^r)$ by the inclusion and remains tight at T. Thus

$$\min_{x\in\mathrm{Core}(v^r)} x(T) = v(T) = v^r(T).$$

The grand coalition is tight throughout its core by efficiency, so v^r is exact at every coalition. □

The exact path does not rescue labeled resource monotonicity.

Theorem 3.2 (grand resource monotonicity fails on exact games)**.** *There is an exact five-player game v such that, when only $v(N)$ is raised, one player's leximin-above-floors payoff strictly falls. Indeed, let*

$$p = (0, 0, 7, 0, 7), \qquad q = (2, 10, 0, 2, 0),$$

put $v(N) = 14$, and for every proper T put $v(T) = \min\{p(T), q(T)\}$. If $v^r(N) = 14 + r$ and $v^r(T) = v(T)$ for $T \subsetneq N$, then for $0 \leq \varepsilon \leq 3/2$,

$$\phi(v^{2+\varepsilon}) = \left(3 + \frac{\varepsilon}{3}, 4 - \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}\right).$$

Thus player 2 falls at rate $1/3$ while the grand worth rises. For example, raising the grand worth from 16 to $129/8$ changes that player's payoff from 4 to $95/24$.

Proof. Both p and q have total 14 and belong to $\mathrm{Core}(v)$. At every proper coalition one of them attains $v(T)$, so v is exact; every v^r is exact by Lemma 3.1. All singleton worths are zero.

Write $s = 3+\varepsilon/3$ and $z = (s, 7-s, s, s, s)$. Direct substitution in the lower-envelope definition of v shows that $z \in \mathrm{Core}(v^{2+\varepsilon})$ throughout the stated interval. (At $\varepsilon = 0$ every core slack is nonnegative; the only coalition sum with negative slope is the singleton of player 2, whose slack remains positive. Every other slope is nonnegative.)

It remains to identify the leximin point. The two pair constraints

$$x_2 + x_3 \geq 7, \qquad x_2 + x_5 \geq 7$$

belong to the core. If every coordinate of a core point is at least t, then efficiency and the first constraint give $16 + \varepsilon = x(N) \geq 7 + 3t$, hence $t \leq s$. So z attains the largest possible minimum coordinate. On that level-optimal face put $d_i = x_i - s \geq 0$ and $D = 1 - 2\varepsilon/3 \geq 0$. Efficiency gives $\sum_i d_i = D$, while the two displayed constraints give $d_2 + d_3 \geq D$ and $d_2 + d_5 \geq D$. Nonnegativity forces $d_2 = D$ and every other $d_i = 0$. Hence the level-optimal face is the singleton $\{z\}$, proving the formula and the strict drop. □

The counterexample pinpoints the failure mechanism: four players form the lowest leximin stratum, and additional total resource raises them jointly only by taking payoff from a higher-stratum player pinned to two cross-stratum pair constraints. The feasible comparison points supplied by $\mathrm{Core}(v) + r\Delta_N$ therefore do not control labeled coordinates.

4 Exactness is necessary

Coalitional monotonicity of ϕ fails on balanced games in general—already at $n = 4$, and even at $n = 5$ where every floor is core-attained. So exactness cannot be weakened to balancedness.

Theorem 4.1. *There is a balanced (non-exact) five-player game v, a coalition S with $|S| = 3$, and $d > 0$ with $v + d\,\mathbf{1}_S$ balanced, such that some $i \in S$ has $\phi_i(v + d\,\mathbf{1}_S) < \phi_i(v)$. The same holds for a four-player game.*

Example 4.2 ($n = 5$; floors attained)**.** *Let $N = \{1, \ldots, 5\}$ with $v(N) = 7$ and (all other coalitions as listed in the supplement) in particular*

$$v(\{1,2,3\}) = 3, \quad v(\{2,3\}) = 3, \quad v(\{1,3,5\}) = v(\{3,5\}) = 4, \quad v(\{1,5\}) = -1.$$

Floors are $m = (0, 1, 1, 1, 2)$, and $m_i = \min_{\mathrm{Core}} x_i$ for every *i (all floors attained); the game is balanced but not exact (e.g. $\min_{\mathrm{Core}} x(\{1,5\}) = 1 > -1$). Then*

$$\phi(v) = \left(\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}, 1, \tfrac{5}{2}\right), \qquad \phi(v + \mathbf{1}_{\{1,2,3\}}) = (1, 1, 2, 1, 2),$$

both in their respective cores, and player $2 \in \{1,2,3\}$ drops *from $\frac{3}{2}$ to 1 when $v(\{1,2,3\})$ is raised from 3 to 4. Thus exactness at the singletons (floor attainment) does not suffice: it is exactness at higher coalitions that the positive theorem needs.*

Example 4.3 ($n = 4$; floor unattained)**.** *Let $N = \{1,2,3,4\}$, $v(N) = 20$, with $v(\{1\}) = -3$, $v(\{1,2\}) = 11$, $v(\{1,3\}) = 7$, $v(\{2,3\}) = 4$, $v(\{1,2,3\}) = 13$, $v(\{2,4\}) = 10$, $v(\{1,3,4\}) = 9$, $v(\{2,3,4\}) = 13$, and the remaining values in the supplement. Here $v(\{1\}) = -3 < 1 = \min_{\mathrm{Core}} x_1$. Then $\phi(v) = (\frac{5}{2}, \frac{17}{2}, \frac{9}{2}, \frac{9}{2})$ and $\phi(v + 3\,\mathbf{1}_{\{1,2,3\}}) = (2, 9, 5, 4)$: player $1 \in \{1,2,3\}$ drops from $\frac{5}{2}$ to 2.*

Both examples are certified in exact rational arithmetic (ϕ in the core on both sides, the drop exact); see the supplement and `src/hunt_leximin_nonexact.py`.

5 Reduction for proper nonsingleton bumps on exact games

Fix an exact v, a proper coalition S with $|S| \geq 2$, and $d > 0$ with $v' = v + d\,\mathbf{1}_S$ exact. Since $|S| \geq 2$ the floors are unchanged, and

$$\mathrm{Core}(v') = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\} \subseteq \mathrm{Core}(v).$$

For $\tau \in [v(S), v(S)+d]$ let v_τ agree with v except $v_\tau(S) = \tau$, so $\mathrm{Core}(v_\tau) = \mathrm{Core}(v) \cap \{x(S) \geq \tau\}$ interpolates $\mathrm{Core}(v)$ and $\mathrm{Core}(v')$.

Lemma 5.1 (exact interval)**.** *Every v_τ, $\tau \in [v(S), v(S) + d]$, is exact.*

Proof. $\mathrm{Core}(v_\tau) \supseteq \mathrm{Core}(v') \neq \emptyset$, so v_τ is balanced. Exactness needs $\min_{\mathrm{Core}(v_\tau)} x(T) = v_\tau(T)$ for all T. For $T = S$, exactness of v supplies $x^0 \in \mathrm{Core}(v)$ with $x^0(S) = v(S)$, while any $x^1 \in \mathrm{Core}(v')$ satisfies $x^1(S) \geq v(S) + d$. The core of v is convex, so the segment $[x^0, x^1]$ contains a point $x^\tau \in \mathrm{Core}(v)$ with $x^\tau(S) = \tau$. This point belongs to $\mathrm{Core}(v_\tau)$, proving $\min_{\mathrm{Core}(v_\tau)} x(S) = \tau$. For $T \neq S$: put $f_T(\tau) = \min_{\mathrm{Core}(v_\tau)} x(T)$. As τ grows $\mathrm{Core}(v_\tau)$ shrinks, so f_T is nondecreasing; and $\mathrm{Core}(v_\tau) \subseteq \mathrm{Core}(v)$ gives $f_T(\tau) \geq \min_{\mathrm{Core}(v)} x(T) = v(T)$. At the endpoints $f_T = v(T)$ because v and v' are exact. A nondecreasing function equal to $v(T)$ at both endpoints and $\geq v(T)$ between is $\equiv v(T)$. Hence $f_T(\tau) = v(T) = v_\tau(T)$. □

The entire exact-bump range can be read from exposed core faces. For $T \subseteq N$, write $F_T = \{x \in \mathrm{Core}(v) : x(T) = v(T)\}$ (with $F_N = \mathrm{Core}(v)$), and define

$$d_v^*(S) = \min_{T \neq S} \left(\max_{x \in F_T} x(S) - v(S) \right).$$

Proposition 5.2 (exact bump capacity)**.** *For an exact game v and a proper coalition S, the bumped game $v + d\mathbf{1}_S$ is exact if and only if $0 \leq d \leq d_v^*(S)$. If $\sigma_S = \phi(v)(S) - v(S)$, then every exact bump at S is non-binding if and only if $d_v^*(S) \leq \sigma_S$; a binding exact bump exists if and only if $d_v^*(S) > \sigma_S$.*

Proof. The bumped core is $K_d = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\}$. For $T \neq S$, exactness at T is equivalent to $F_T \cap K_d \neq \emptyset$, which is equivalent to

$$d \leq \max_{x \in F_T} x(S) - v(S).$$

Taking the minimum over $T \neq S$ gives all the higher-coalition exactness conditions; the term $T = N$ also gives nonemptiness of K_d. Exactness at S follows by interpolating between a point of F_S and a point of K_d, as in Lemma 5.1. This proves the first statement. The last two statements compare the exact capacity with the largest bump still satisfied by the old selection, namely σ_S. □

The capacity implication can be proved for every star-failure dual with at most two residual levels. This includes every two-player coalition and also explains the same-stratum pair certificates found in the computation.

Proposition 5.3 (two-level dual certificate)**.** *Let v be exact, S proper, $\phi = \phi(v)$, and $\sigma_S = \phi(S) - v(S)$. Suppose $(\star_S)$ fails and its dual maximal face can be represented by weights taking at most two values on S: after normalization, for some $\emptyset \neq C \subsetneq S$ and $R > 1$,*

$$\phi \in \arg \max_{x \in \mathrm{Core}(v)} \left(x(C) + R x(S \setminus C) \right).$$

Then

$$\max_{x \in F_C} x(S) \le \phi(S) \quad \textit{and hence} \quad d_v^*(S) \le \sigma_S.$$

The same conclusion holds in the one-level case where ϕ maximizes $x(S)$.

Proof. Put $H = S \setminus C$ and $h = \phi(C) - v(C) \ge 0$. Exactness makes $F_C = \{x \in \mathrm{Core}(v) : x(C) = v(C)\}$ nonempty. For $x \in F_C$, maximality of ϕ gives

$$(x(C) - \phi(C)) + R(x(H) - \phi(H)) \le 0.$$

The first parenthesis is $-h$, so $x(H) - \phi(H) \le h/R$. Therefore

$$x(S) - \phi(S) = -h + x(H) - \phi(H) \le -h\left(1 - \frac{1}{R}\right) \le 0.$$

Thus F_C certifies the required bound:

$$d_v^*(S) \le \phi(S) - v(S) = \sigma_S.$$

In the one-level case, $\phi(S) = \max_{\mathrm{Core}(v)} x(S)$, so the conclusion is immediate. □

Corollary 5.4 (the capacity implication for pairs)**.** *If v is exact and S is a proper two-player coalition, then*

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Proof. When $(\star_S)$ fails, its positive dual weights on the two members of S have either one value or two. Normalize the smaller value to one and apply Proposition 5.3. □

There is a second exact certificate, independent of the number of residual levels. It compares the slack of a coalition carrying the whole complement of S with the slack of its trace inside S.

Proposition 5.5 (off-side slack-domination certificate)**.** *Let v be exact, S proper, $\phi = \phi(v)$, and $P = N \setminus S$. Suppose C is a nonempty proper subset of S, put $T = P \cup C$, and assume*

$$\phi(T) - v(T) \le \phi(C) - v(C).$$

Then

$$\max_{x \in F_C} x(S) \le \phi(S), \qquad \textit{so} \qquad d_v^*(S) \le \phi(S) - v(S).$$

In particular, the conclusion holds whenever T is ϕ-tight.

Proof. Because $S \cup T = N$ and $S \cap T = C$, modularity gives $x(S) + x(T) = x(N) + x(C)$ for every allocation x. If $x \in F_C$, then core feasibility at T and efficiency imply

$$x(S) = v(N) + v(C) - x(T) \le v(N) + v(C) - v(T).$$

Moreover,

$$v(N) + v(C) - v(T) = \phi(S) + \big(\phi(T) - v(T)\big) - \big(\phi(C) - v(C)\big) \le \phi(S).$$

Exactness makes F_C nonempty, so this face occurs in the minimum defining $d_v^*(S)$ and proves the claim. □

The carrier comparison has a useful normal form that removes both slack expressions. It also converts failure of every candidate into a single finite-dimensional separation problem.

Proposition 5.6 (carrier marginal and face separation)**.** *Let v be exact, $\phi = \phi(v)$, S proper, and $P = N \setminus S$. For every $k \in S$,*

$$\begin{aligned}[\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] \\ = \phi(P) - [v(P \cup \{k\}) - v(\{k\})].\end{aligned}$$

Consequently k satisfies the carrier inequality if and only if

$$v(P \cup \{k\}) - v(\{k\}) \geq \phi(P). \tag{M}$$

Let $K \subseteq S$ be nonempty. If (M) fails strictly for every $k \in K$, then every point of the exact face $F_{P\cup\{k\}}$ satisfies $x(S) > \phi(S)$. If, in addition, $(\star_S)$ fails, there is a nonzero $w \in \mathbb{R}^S_+$ such that

$$\max_{k\in K} \max_{x\in F_{P\cup\{k\}}} \sum_{i\in S} w_i(x_i - \phi_i) < 0. \tag{Sep}$$

Proof. Because P and $\{k\}$ are disjoint, $\phi(P \cup \{k\}) = \phi(P) + \phi_k$, which proves the identity and (M). Now suppose the latter fails for k and take $x \in F_{P\cup\{k\}}$. Core feasibility at $\{k\}$ gives

$$x(P) = v(P \cup \{k\}) - x_k \leq v(P \cup \{k\}) - v(\{k\}) < \phi(P).$$

Efficiency therefore gives $x(S) > \phi(S)$.

Let Z be the convex hull of the union of the faces $F_{P\cup\{k\}}$, $k \in K$. It is a compact subset of the core, and every point of Z has S-sum strictly larger than $\phi(S)$. Hence failure of $(\star_S)$ implies that Z is disjoint from $\phi_S + \mathbb{R}^S_+$ after projection onto the S-coordinates. Strong separation gives a strict separating normal w. Its coordinates must be nonnegative, since otherwise its infimum on $\phi_S + \mathbb{R}^S_+$ would be $-\infty$. Translating by ϕ_S gives (Sep). □

The separator is in fact forced by the star-failure cover; it is not an independent normal that can be eliminated against that cover.

Proposition 5.7 (residual-gap separator)**.** *Suppose failure of $(\star_S)$ has a tight-coalition certificate*

$$r^S = \mu \mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T, \qquad \lambda_T \geq 0, \quad \phi(T) = v(T) \text{ if } \lambda_T > 0,$$

where $r_i \geq 1$ on S. Put

$$r_0 = \min_{i\in S} r_i, \qquad K = \{i \in S : r_i = r_0\}, \qquad H = S \setminus K, \qquad D = \sum_{i\in H}(r_i - r_0).$$

Assume $H \neq \emptyset$. If the carrier marginal fails at $k \in K$ by

$$\gamma_k = [\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] > 0,$$

then, for every $x \in F_{P\cup\{k\}}$,

$$\sum_{i\in H} \frac{r_i - r_0}{D}(x_i - \phi_i) \leq -\frac{r_0}{D}\gamma_k < 0. \tag{RG}$$

Consequently the normalized residual gap is a common strict separator for every failed candidate in K, supported entirely on the higher-residual class H.

Proof. For any $x \in \mathrm{Core}(v)$ put $d = x - \phi$. Efficiency gives $d(N) = 0$, and core feasibility plus tightness gives $d(T) \geq 0$ whenever $\lambda_T > 0$. Taking the scalar product of the cover identity with d therefore yields

$$\sum_{i \in S} r_i d_i = -\sum_{T} \lambda_T d(T) \leq 0.$$

Splitting off the minimum residual gives

$$r_0 d(S) + \sum_{i \in H} (r_i - r_0) d_i \leq 0.$$

For $x \in F_{P \cup \{k\}}$, the carrier calculation in Proposition 5.6 gives $d(S) \geq \gamma_k > 0$. Substitution and division by D prove (RG). □

Corollary 5.8 (residual-gap escape criterion)**.** *Under the hypotheses of Proposition 5.7, if some $k \in K$ satisfies*

$$\max_{x \in F_{P \cup \{k\}}} \sum_{i \in H} \frac{r_i - r_0}{D} (x_i - \phi_i) \geq 0, \tag{E}$$

then k satisfies the carrier inequality

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \leq [\phi_k - v(\{k\})].$$

Thus existence of a minimum-residual carrier is reduced to escape from one explicit scalar separator; no coordinatewise choice of a member of K is required.

Proof. The contrapositive is exactly inequality (RG). □

The strict separator also has a finite dual form. This removes the quantification over all points of the carrier-tight faces and exposes the precise slack identity induced by the residual cover.

Proposition 5.9 (dual form of the carrier-face separator)**.** *Use the hypotheses and notation of Proposition 5.6, and put $T_k = P \cup \{k\}$. A vector $w \in \mathbb{R}^S_+$ satisfies (Sep) if and only if, for every $k \in K$, there are multipliers $\theta^k_U \geq 0$ on the nonempty proper coalitions and $a_k, b_k \in \mathbb{R}$ such that*

$$w^S = a_k \mathbf{1}_N + b_k \mathbf{1}_{T_k} - \sum_{\emptyset \neq U \subsetneq N} \theta^k_U \mathbf{1}_U, \tag{D1}$$

where w^S is w on S and zero on P, and

$$\sum_{U} \theta^k_U \left[\phi(U) - v(U)\right] < b_k [\phi(T_k) - v(T_k)]. \tag{D2}$$

Whenever the carrier marginal fails at k, the right-hand slack in (D2) is positive and every such certificate has $b_k > 0$.

Proof. Fix k and maximize $w^S \cdot x$ over the core subject to $x(T_k) = v(T_k)$. In the convention $-x(U) \leq -v(U)$ for the core inequalities, the linear-programming dual has nonnegative multipliers θ^k_U, free multipliers a_k, b_k on efficiency and the face equality, and stationarity exactly (D1). Its objective is

$$-\sum_{U} \theta^k_U v(U) + a_k v(N) + b_k v(T_k).$$

Taking the scalar product of (D1) with ϕ shows that this objective is strictly below $w \cdot \phi_S$ exactly when (D2) holds. Strong duality therefore proves the equivalence, simultaneously for all k because the same w occurs in each stationarity identity.

If the carrier marginal fails, then $\phi(T_k) - v(T_k) > \phi_k - v(\{k\}) \geq 0$. The left side of (D2) is nonnegative, so strictness forces $b_k > 0$. □

Corollary 5.10 (aggregated separator identity)**.** *If the carrier marginal fails for every $k \in K$, then the separator in Proposition 5.6 supplies $c > 0$, $A \in \mathbb{R}$, a collection $\eta_U \geq 0$, and a nonzero $w \in \mathbb{R}^S_+$ such that*

$$\mathbf{1}_K = cw^S - A\mathbf{1}_N - |K|\mathbf{1}_P + \sum_U \eta_U \mathbf{1}_U, \tag{AD1}$$

and

$$\sum_U \eta_U[\phi(U) - v(U)] < \sum_{k\in K}[\phi(P \cup \{k\}) - v(P \cup \{k\})]. \tag{AD2}$$

More precisely, one may take

$$c = \sum_{k\in K} \frac{1}{b_k}, \qquad A = \sum_{k\in K} \frac{a_k}{b_k}, \qquad \eta_U = \sum_{k\in K} \frac{\theta^k_U}{b_k}.$$

Proof. Proposition 5.9 gives $b_k > 0$ for every $k \in K$. Divide (D1) and (D2) by b_k and sum over k. Since $\mathbf{1}_{P\cup\{k\}} = \mathbf{1}_P + e_k$ and $\sum_{k\in K} e_k = \mathbf{1}_K$, the stationarity sum rearranges to (AD1); the strict slack inequalities sum to (AD2). □

The same comparison has a quantitative converse at the level of total S-payoff. For a core point ψ, define

$$\gamma_S(\psi) = \min_{\emptyset \neq C \subsetneq S} \{[\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)]\},$$
$$\eta_S(\psi) = \max\{x(S) - \psi(S) : x \in \mathrm{Core}(v),\ x_i \geq \psi_i\ (i \in S)\}.$$

Thus $(\star_S)$ at ψ is equivalent to $\eta_S(\psi) > 0$.

Lemma 5.11 (translation and positive scaling)**.** *For $\psi \in \mathrm{Core}(v)$ define $\bar{v}(T) = v(T) - \psi(T)$, so that $0 \in \mathrm{Core}(\bar{v})$. If v is exact, then $\bar{v}$ is exact, and*

$$\gamma^{\bar{v}}_S(0) = \gamma^v_S(\psi), \qquad \eta^{\bar{v}}_S(0) = \eta^v_S(\psi).$$

For every $c > 0$, the game $c\bar{v}$ is exact and both quantities are multiplied by c. Thus every case with positive γ_S may be normalized to $\psi = 0$ and $\gamma_S = 1$ without loss.

Proof. The identity $\mathrm{Core}(\bar{v}) = \mathrm{Core}(v) - \psi$ proves preservation of exactness and the two translation identities by direct substitution. Likewise $\mathrm{Core}(c\bar{v}) = c\,\mathrm{Core}(\bar{v})$, and slacks and feasible displacements scale by c. □

Lemma 5.12 (exact carrier lift)**.** *If v is exact, then for every nonempty proper $C \subset S$ there is a $z^C \in \mathrm{Core}(v)$ such that*

$$z^C(S) - \psi(S) \geq [\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)].$$

Proof. Choose z^C tight at $P \cup C$. Core feasibility at C gives $z^C(P) \leq v(P \cup C) - v(C)$. Efficiency and the identity $\psi(P) = v(N) - \psi(S)$ now give the displayed inequality. □

The only possible obstruction is coordinatewise: the carrier-tight points raise the total payoff of S, but different members of S may fall at different points. For a pair, convexity removes that obstruction without loss.

There is a useful normalization that isolates exactly what remains to be proved and, unexpectedly, removes the leximin selection from the statement.

Proposition 5.13 (exact-core closure formulation)**.** *Let v be exact, let $\psi \in \mathrm{Core}(v)$, and put $D = \mathrm{Core}(v) - \psi$ and $s_\psi(A) = \psi(A) - v(A)$. Then*

$$s_\psi(A) = -\min_{d\in D} d(A) = \max_{d\in D} d(N \setminus A). \tag{EC}$$

Consequently

$$\gamma_S(\psi) = \min_{\emptyset \neq C \subsetneq S} \left\{ \max_{d\in D} d(S \setminus C) + \min_{d\in D} d(C) \right\}, \tag{CG}$$

whereas

$$\eta_S(\psi) = \max\{d(S) : d \in D,\ d_i \geq 0\ (i \in S)\}. \tag{CE}$$

Thus the quantitative carrier inequality is a statement about every translated exact core, not specifically about a leximin point.

Proof. Exactness gives $v(A) = \min_{x\in\mathrm{Core}(v)} x(A)$, hence $s_\psi(A) = -\min_{d\in D} d(A)$. Every $d \in D$ has $d(N) = 0$, so the last quantity equals $\max_{d\in D} d(N \setminus A)$. Substituting $A = P \cup C$ and then $A = C$ gives (CG); translating the defining program for η_S gives (CE). □

The closure qualification in this formulation is essential. For the convex hull in $\mathbb{R}^3$ of

$$(0,0,0), (4,2,-12), (-12,14,3), (18,9,-17), (10,17,-9), (4,-2,-5), (15,20,-5),$$

the right side of (CG) is 3, while the largest nonnegative total in the hull is 0. Append to each point a fourth coordinate equal to minus its total and take the lower-envelope exact game. Its full core is the coalitional closure of those generators: the same reference point still has $\gamma_S = 3$, but now $\eta_S = 27$, attained at $(10, 17, 0, -27)$. Thus arbitrary convexity is false by a gap of 3, while exact-core closure repairs this candidate by a gap of 24.

The remaining obstruction has an exact dual that is formally the slack-priced version of a failure cover. Define the common-coordinate margin

$$\kappa_S(\psi) = \max\{z : d \in D,\ d(S) \geq \gamma_S(\psi),\ d_i \geq z\ (i \in S)\}.$$

Proposition 5.14 (quantitative orthant dual)**.** *The value $\kappa_S(\psi)$ equals*

$$\min \left\{ \sum_T \lambda_T s_\psi(T) - q\gamma_S(\psi) : \begin{array}{l} \lambda_T, q, w_i \geq 0, \quad \sum_{i\in S} w_i = 1, \\ u\mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T = q\mathbf{1}_S + w^S \end{array} \right\}. \tag{OD}$$

Here u is free and w^S is w on S and zero on P. Consequently (Q) is equivalent to $\kappa_S(\psi) \geq 0$, while the stronger uniform lift asks for $\kappa_S(\psi) \geq \gamma_S(\psi)/|S|$.

Proof. Write the translated core inequalities as $-d(T) \leq s_\psi(T)$, the high-total constraint as $-d(S) \leq -\gamma_S(\psi)$, and the coordinate constraints as $-d_i + z \leq 0$. The nonnegative dual multipliers are respectively λ_T, q, w_i; efficiency has free multiplier u. Stationarity in z gives $\sum_i w_i = 1$, stationarity in d gives the displayed incidence identity, and the dual objective is $\sum_T \lambda_T s_\psi(T) - q\gamma_S(\psi)$. Strong LP duality proves (OD). □

Proposition 5.15 (two-level orthant separators are impossible)**.** *Let v be exact, $\psi \in \mathrm{Core}(v)$, and $\gamma_S(\psi) > 0$. For every nonempty $A \subseteq S$ there is a $d^A \in D$ such that*

$$d^A(S) \geq \gamma_S(\psi), \qquad d^A(A) \geq \gamma_S(\psi). \tag{OI}$$

Consequently, if $w = a\mathbf{1}_S + b\mathbf{1}_A$ with $a, b \geq 0$, then some $d \in D$ with $d(S) \geq \gamma_S(\psi)$ satisfies $w \cdot d_S \geq (a+b)\gamma_S(\psi)$. In particular, no strictly positive vector with at most two coordinate levels can strictly separate

$$\{d_S : d \in D,\ d(S) \geq \gamma_S(\psi)\} \quad \textit{from} \quad \mathbb{R}^S_+.$$

Thus any failure of the quantitative carrier inequality admits a strictly positive separator, but every such separator must have at least three distinct coordinate levels.

Proof. If $A \subsetneq S$, put $C = S \setminus A$ and choose a core point tight at $P \cup C$. For its displacement d^A from ψ, (EC) gives

$$d^A(A) = s_\psi(P \cup C) \geq \gamma_S(\psi) + s_\psi(C) \geq \gamma_S(\psi),$$

while Lemma 5.12 gives $d^A(S) \geq \gamma_S(\psi)$. If $A = S$, the latter lemma with any nonempty proper $C \subset S$ supplies the required point. Hence $w \cdot d^A_S = ad^A(S) + bd^A(A) \geq (a+b)\gamma_S(\psi) > 0$ whenever $w \neq 0$.

If the displayed convex set and the nonnegative orthant are disjoint, strong separation supplies a nonzero $w \geq 0$ whose maximum on the first set is strictly negative. Since that set is compact and every one of its points has S-sum at least the positive number $\gamma_S(\psi)$, adding a sufficiently small multiple of $\mathbf{1}_S$ makes w strictly positive while preserving strict separation. A strictly positive vector with at most two levels has the form $a\mathbf{1}_S + b\mathbf{1}_A$, contradicting the first part. □

Proposition 5.16 (codimension-one uniform carrier lift)**.** *Let v be exact, let $\psi \in \mathrm{Core}(v)$, let $|S| \geq 2$, and let $P = N \setminus S = \{p\}$ be a singleton. Put $m = |S|$. Then*

$$x_i = \psi_i + \frac{\gamma_S(\psi)}{m} \quad (i \in S), \qquad x_p = \psi_p - \gamma_S(\psi) \tag{UL}$$

defines a point of $\mathrm{Core}(v)$ whenever $\gamma_S(\psi) > 0$. Consequently

$$\eta_S(\psi) \geq \gamma_S(\psi),$$

and every member of S can be raised simultaneously by $\gamma_S(\psi)/|S|$.

Proof. Write $t = \gamma_S(\psi)/m$ and $d = x - \psi$. Efficiency is immediate. If $T \subseteq S$, then $d(T) = t|T| \geq 0 \geq -s_\psi(T)$.

Now let $T = P \cup C$ with $C \subsetneq S$ and put $A = S \setminus C$. If C is nonempty, the definition of γ_S gives

$$s_\psi(T) \geq \gamma_S(\psi) + s_\psi(C) \geq \gamma_S(\psi),$$

and hence

$$d(T) = -d(A) = -t|A| \geq -\gamma_S(\psi) \geq -s_\psi(T).$$

It remains only to check $T = P$. Exactness makes v superadditive on disjoint coalitions: a core point tight at $P \cup C$ gives $v(P \cup C) \geq v(P) + v(C)$. Therefore $s_\psi(P \cup C) \leq s_\psi(P) + s_\psi(C)$, and any nonempty proper $C \subset S$ yields

$$\gamma_S(\psi) \leq s_\psi(P \cup C) - s_\psi(C) \leq s_\psi(P).$$

Thus $d(P) = -\gamma_S(\psi) \geq -s_\psi(P)$. Every coalition constraint holds, so $x \in \mathrm{Core}(v)$ and $x(S) - \psi(S) = mt = \gamma_S(\psi)$. □

Proposition 5.17 (sharp quantitative pair bound and uniform split)**.** *If v is exact, $\psi \in \mathrm{Core}(v)$, and S is a proper two-player coalition, then there is an $x \in \mathrm{Core}(v)$ satisfying*

$$x_i - \psi_i \geq \frac{\gamma_S(\psi)}{2} \quad (i \in S).$$

In particular, $\eta_S(\psi) \geq \gamma_S(\psi)$, and the coefficient one is sharp.

Proof. The claim is immediate if $\gamma_S(\psi) \leq 0$, so write $\gamma = \gamma_S(\psi) > 0$. Let

$$K = \{d_S : d \in D,\ d(S) \geq \gamma\}.$$

By the exact carrier lift, K is nonempty; it is compact because it is a closed subset of the translated core. If K did not meet the closed upper orthant $(\gamma/2)\mathbf{1}_S + \mathbb{R}^S_+$, strong separation of a compact set from an upper set would supply a nonzero normal $w \geq 0$ (a negative coordinate could be driven to $-\infty$ along the upper set) such that

$$\max_{d_S \in K} w \cdot d_S < \frac{\gamma}{2} \sum_{i \in S} w_i. \tag{PS}$$

Every nonnegative two-dimensional w can be written $a\mathbf{1}_S + b\mathbf{1}_A$ for $a, b \geq 0$ and a nonempty $A \subseteq S$. Proposition 5.15 gives a point of K with weighted value at least $(a+b)\gamma$, whereas the right side of (PS) is $\gamma(2a + b|A|)/2 \leq (a+b)\gamma$, a contradiction. Hence the uniform split exists. The exact lower-envelope equality fixtures show sharpness. □

Corollary 5.18 (quantitative carrier bound through four players)**.** *For every exact game with $|N| \leq 4$, every core point ψ, and every proper nonsingleton S, one has*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

More generally, the inequality holds in every dimension whenever $|S| = 2$ or $|N \setminus S| = 1$. Hence any counterexample requires $|S| \geq 3$, $|N \setminus S| \geq 2$, and therefore at least five players.

The first ostensibly open dimension can also be closed, although our present proof is a finite exact computation.

Proposition 5.19 (computer-assisted five-player carrier theorem)**.** *For every exact game with $|N| \leq 5$, every core point ψ, and every proper nonsingleton S,*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

In the sole new case $|N| = 5$, $|S| = 3$, the exactness assumption can be weakened to total balancedness (nonemptiness of every subgame core).

Proof. The preceding results leave only $|N| = 5$, $|S| = 3$, and $P = N \setminus S = \{p, q\}$. The case $\gamma_S(\psi) \leq 0$ is immediate. Otherwise translate ψ to zero, scale so that $\gamma_S(\psi) = 1$, and write $s(A) = s_\psi(A)$. In fact, total balancedness suffices: a core point of each subgame T, translated by $\psi|_T$ and extended arbitrarily outside T to have zero grand total, gives a vector d^T such that

$$d^T(N) = 0, \qquad d^T(T) = -s(T), \qquad d^T(U) \geq -s(U) \quad (\emptyset \neq U \subseteq T). \tag{X5}$$

Together with $s \geq 0$, $s(\emptyset) = s(N) = 0$, and the six carrier inequalities

$$s(P \cup C) - s(C) \geq 1 \quad (\emptyset \neq C \subsetneq S), \tag{C5}$$

these constraints define a rational polyhedron containing every normalized totally balanced instance under consideration.

The LP defining η_S has the slack-linear dual

$$\min_{\lambda, w, \rho \geq 0} \sum_{\emptyset \neq T \subsetneq N} \lambda_T s(T) \tag{E5}$$

subject to

$$\sum_{T \ni p} \lambda_T = \sum_{T \ni q} \lambda_T = 1 + \rho, \qquad \sum_{T \ni i} \lambda_T + w_i = \rho \quad (i \in S).$$

This is a five-equation standard-form polyhedron with 34 nonnegative variables. The verifier enumerates all $\binom{34}{5} = 278{,}256$ candidate bases by exact elimination. They give 263 distinct vertices, or 43 orbits under the $S_3 \times S_2$ player symmetry. For each orbit representative it then minimizes the objective in (E5) over (X5)–(C5). Independently replayed rational dual certificates give only the lower bounds

$$1, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad 2, \quad 3, \quad 4;$$

the minimum is 1 and is attained. Since a slack-nonnegative objective has an optimal vertex in (E5), every normalized instance has $\eta_S \geq 1 = \gamma_S$. The exact enumeration and all 43 rational certificates are in `src/audit_n5_exact_cone_eta_vertices.py` and `results/n5_totally_balanced_cone_eta_vertices.json`. □

The exact-total uniform displacement used in the codimension-one proof does not extend to this case. An exact integer five-player game recorded in the supplementary counterexample artifact has $\gamma_S = 1$, $\eta_S = 2$, and $\kappa_S = 1/2$, but no displacement with $d(S) = 1$ and $d_i = 1/3$ for all $i \in S$. Thus the five-player theorem genuinely needs the freedom to gain more than γ_S in total.

The eta dual has a finite combinatorial interpretation that removes the apparently unbounded parameter ρ.

Minimal balanced collections and their generation are classical; complete computational catalogs are now available through seven players (Laplace Mermoud et al. 2025). The point here is their target-conditioned appearance as the vertices of the eta dual, followed by optimization over the exact-game cone. The latter cone has an independent facet description in terms of indecomposable minimal semi-balanced systems (Studený and Kratochvíl 2022).

Lemma 5.20 (eta-dual vertices are minimal balanced collections)**.** *Fix a proper nonsingleton $S \subset N$ and put $P = N \setminus S$. On every nonnegative slack system, the eta dual in (E5) (with the evident degree equation for every member of P) has an optimal vertex with*

$$\lambda_S = 0, \qquad \lambda_{\{i\}} = 0 \quad (i \in S). \tag{DV}$$

After imposing (DV), its vertices are in bijection with the minimal balanced collections on N that contain S. Explicitly, adjoining S with coefficient one to a dual point gives

$$\mathbf{1}_S + \sum_T \lambda_T \mathbf{1}_T + \sum_{i \in S} w_i \mathbf{1}_{\{i\}} = (1 + \rho)\mathbf{1}_N. \tag{MB}$$

Conversely, rescale a minimal balanced collection containing S so that its S-coefficient is one, interpret its S-singleton coefficients as the w_i, and interpret all remaining coefficients except that of S as the λ_T. In particular, every relevant dual vertex uses at most $|N|$ coalitions after S is adjoined.

Proof. If $a = \lambda_S > 0$, deleting a from λ_S and adding a to every w_i, $i \in S$, preserves all degree equations and weakly decreases the objective by $as(S)$. Similarly, $\lambda_{\{i\}}$ can be moved into w_i at weakly lower cost. Hence an optimum exists on the face (DV), and a vertex optimum can be chosen there.

On that face the dual degree equations are exactly (MB). Thus the face is affinely isomorphic to the slice, at S-coefficient one, of the nonnegative cone of balanced collections containing S. A point of this slice is a vertex precisely when it lies on an extreme ray with positive S-coefficient: a decomposition into two such rays gives a convex decomposition after normalization, while a nonzero ray with zero S-coefficient gives a recession direction and again precludes extremality. The extreme rays of the balanced-collection cone are the minimal balanced collections. Their positive incidence columns are linearly independent: otherwise a nonzero dependence permits perturbing the positive balancing weights in one of its two signs until a coefficient first reaches zero, preserving balance and contradicting minimality. Hence there are at most $|N|$ columns. Finally, in the reverse construction the normalized coefficient of S is at most one because every member of S has total balanced degree one; hence $\rho \geq 0$ as required. □

The totally balanced strengthening is dimension-sharp. Its first six-player escape nevertheless exposes a reusable exactness identity.

Lemma 5.21 (three-block exact-closure repair)**.** *Let $N = \{1, \dots, 6\}$, $S = \{1, 2, 3\}$, $P = \{4, 5, 6\}$, and abbreviate $s = s_\psi$. For every exact game and every $\psi \in \mathrm{Core}(v)$,*

$$s(2345) + s(146) + s(56) \geq s(1456) - s(1). \tag{TB}$$

Consequently the eta-dual supported with unit weight on $2345, 146, 56$ has cost at least $\gamma_S(\psi)$.

Proof. Translate ψ to zero. Choose core displacements a, b, c tight at $5, 45, 1456$, respectively. The incidence identities

$$\begin{aligned} \mathbf{1}_1 + \mathbf{1}_{2345} + \mathbf{1}_{56} &= \mathbf{1}_N + \mathbf{1}_5, \\ \mathbf{1}_5 + \mathbf{1}_{2345} + \mathbf{1}_{146} &= \mathbf{1}_N + \mathbf{1}_{45}, \\ \mathbf{1}_1 + \mathbf{1}_{45} + \mathbf{1}_{146} + \mathbf{1}_{56} &= 2\,\mathbf{1}_{1456} \end{aligned}$$

and core feasibility give

$$\begin{aligned} s(5) &\leq s(1) + s(2345) + s(56), \\ s(45) &\leq s(5) + s(2345) + s(146), \\ 2s(1456) &\leq s(1) + s(45) + s(146) + s(56). \end{aligned}$$

Substituting the first two bounds into the third and dividing by two proves (TB). The final assertion follows from $s(1456) - s(1) \geq \gamma_S(\psi)$, the carrier inequality for $C = \{1\}$. □

There is an integer totally balanced game for this split with $\gamma_S = 1$, $\eta_S = 0$, and zero-cost eta dual $\{2345, 146, 56\}$. It is not exact; its exact-core closure is the zero game, so its

carrier gap also collapses to zero. Thus Lemma 5.21 identifies exactly the crossing constraint supplied by exactness. The game, closure, and incidence proof are independently checked in the supplementary three-block verifier.

The repair is not isolated: it closes the complete first multilevel layer of the six-player eta dual.

Proposition 5.22 (complete six-player unit-triple layer)**.** *In the split $|S| = |P| = 3$, consider eta-dual solutions with $\rho = 1$, $w = 0$, and three unit-weight supporting coalitions. Equivalently, every target player has support degree one and every outside player has degree two. For every such dual and every exact normalized slack system satisfying the carrier inequalities, its slack cost is at least one. Hence this entire dual layer satisfies the quantitative carrier bound.*

Proof. Each target player chooses its unique supporting slot, while each outside player chooses the unique slot from which it is absent. The implementation deliberately retains a harmless superset before quotienting by $S_3 \times S_3$: thirteen orbits literally contain three proper distinct coalitions, while one has a repeated coalition and two contain a dominated target-singleton column. Bounding those three degeneracies as well is stronger and gives 16 support orbits in total. Minimization over the global exact-witness cone gives exact values one for seven orbits and two for nine. Every value is certified by a replayed rational dual lower bound; the saved rational primal points record matching attainment. The complete enumeration is in `src/audit_n6_unit_triple_eta_duals.py`, with the certificates in `results/n6_unit_triple_eta_duals.json`. □

The preceding layer is the sharp boundary of a complete six-player theorem. The full census below was the discovery proof, but the separator geometry reduces it to a much smaller rank classification.

Lemma 5.23 (rank-six three-level classification)**.** *Consider an undominated eta-dual vertex for one of the two unresolved six-player splits. If its target-floor multipliers w_i have at least three distinct values, then, up to permutations preserving S and P, necessarily $|S| = 4$ and the associated normalized minimal balanced collection is*

$$\mathbf{1}_S + e_1 + 2e_2 + \mathbf{1}_{35} + \mathbf{1}_{136} + 2\,\mathbf{1}_{456} = 3\,\mathbf{1}_N. \tag{TL}$$

In particular its eta objective is

$$s(35) + s(136) + 2s(456) \geq 2\gamma_S. \tag{TC}$$

There is no three-level vertex in the $3 + 3$ split.

Proof. By Lemma 5.20, adjoining S and the positive target-singleton columns gives a minimal balanced collection with at most six members. Let q be the number of positive w_i and r the number of remaining coalition columns. The collection cannot contain S and all target singletons, since those incidence columns are dependent. Thus at least one w_i is zero, and three levels require

$$2 \leq q \leq |S| - 1, \qquad r \leq 5 - q \leq 3. \tag{R}$$

There is a compact way to classify these residual columns. Write their positive coefficients as $\lambda_1, \ldots, \lambda_r$, let β be the common balanced degree, and put $z_j = \lambda_j/\beta$. If $a_i \in \{0,1\}^r$ is the trace row of target player i, balance gives

$$a_i \cdot z = h - \frac{w_i}{\beta}, \qquad h := 1 - \frac{1}{\beta}. \tag{H}$$

Every zero-floor target row therefore has subset-sum height $h < 1$, every positive-floor row has smaller height, and every outside trace row has height one. Three floor levels require at least two distinct subset-sum heights strictly below h.

First take the $3 + 3$ split. Necessarily $q = 2$. Modulo $\mathbf{1}_S, e_1, e_2$, the residual columns are exactly their three outside traces and remain independent. For $r = 1, 2, 3$, the full-column-rank binary matrices whose rows all have z-height one reduce, up to row and column permutations, to

r	z
1	(1)
2	$(1, 1)$
3	$(1, 1, 1)$ or $(1/2, 1/2, 1/2)$.

(M3)

Indeed, for $r = 3$, an invertible binary matrix R with $Rz = \mathbf{1}$ either has a row of weight one, which reduces inductively to a permutation matrix, or every row has weight two, which forces $R = J - I$; a weight-three row would make every row identical. The cases $r < 3$ are the same argument with fewer columns. In every row of (M3), a subset sum $h < 1$ has at most one distinct subset-sum value below it. Hence the $3 + 3$ split cannot have three floor levels.

Now take the $4+2$ split. If $q = 3$, quotienting by $\mathbf{1}_S, e_1, e_2, e_3$ leaves at most two independent outside columns. Their normalized weights are (1) or $(1, 1)$, so a zero-floor height below one has no smaller nonnegative subset sum; this is impossible with positive floors. Thus $q = 2$. Quotient by $\mathbf{1}_S, e_1, e_2$ using the coordinates

$$x \longmapsto (x_3 - x_4, x_5, x_6). \tag{Q}$$

If $r \leq 2$, equal heights for the two zero-floor rows either make their traces equal, forcing the outside traces to have full column rank and weights (1) or $(1, 1)$, or force the two weights to be equal. In the latter case outside balance makes that common weight 1 or $1/2$. Again there is at most one subset-sum height below h. Therefore $r = 3$.

Let a_3, a_4 be the zero-floor trace rows and b_5, b_6 the outside trace rows. The three quotient columns in (Q) are independent, so

$$(a_3 - a_4) \cdot z = 0, \qquad (b_5 - b_6) \cdot z = 0 \tag{E}$$

are independent relations. After common coordinates are cancelled, every equality between distinct subsets of three positive numbers has one of only two forms:

$$z_i = z_j \quad \text{or} \quad z_i + z_j = z_k. \tag{ES}$$

Two independent sum-type relations are incompatible with positivity. Two equality-type relations make all three weights equal; outside degree then sets the common weight to 1 or $1/2$, leaving at most one height below $h < 1$. In the mixed case write the weights as $(t, t, 2t)$. If the zero-floor relation is the equality type, the outside sum relation gives $t = 1/2$ and $h = t$ or $3t$, again impossible. Hence the zero-floor relation is $2t = t + t$, so $h = 2t$. The outside equality relation either has no common coordinate, giving $t = 1$ and $h > 1$, or shares the coordinate of weight $2t$, giving

$$3t = 1, \qquad z = (1/3, 1/3, 2/3), \qquad h = 2/3. \tag{U}$$

The only two lower heights are 0 and $1/3$. Equation (H) now gives $\beta = 3$, residual coefficients $(1, 1, 2)$, and positive floor coefficients 2 and 1. Reconstructing the traces, up to target and outside permutations, gives exactly (TL).

Thus the classification is a three-weight subset-sum argument rather than a coalition census. A solver-free exact replay of the reduced matrices is in `src/verify_n6_three_level_matrix_classification.py`, with output `results/n6_three_level_matrix_classification.json`. The earlier 65,133-support determinant scan remains as independent redundancy and is indexed in the supplement.

For the exceptional orbit, the carrier inequality with $C = \{4\}$ is $s(456) - s(4) \geq \gamma_S$. Slack nonnegativity therefore gives

$$s(35) + s(136) + 2s(456) \geq 2s(456) \geq 2\gamma_S + 2s(4) \geq 2\gamma_S,$$

which proves (TC). □

Proposition 5.24 (rank-reduced six-player carrier theorem)**.** *For every exact game with $|N| \leq 6$, every core point ψ, and every proper nonsingleton S,*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

Proof. Corollary 5.18, Proposition 5.19, and the general cases $|S| = 2$ and $|P| = 1$ leave only the six-player splits $(|S|, |P|) = (3, 3)$ and $(4, 2)$. Suppose, to the contrary, that $\eta_S(\psi) < \gamma_S(\psi)$. Since ψ itself is feasible in the eta program, $\eta_S(\psi) \geq 0$, so necessarily $\gamma = \gamma_S(\psi) > 0$. Translate ψ to zero using Lemma 5.11 and choose an optimal eta-dual vertex on the undominated face from Lemma 5.20; denote its target-floor multipliers by w_i and its value by $L = \sum_T \lambda_T s(T) = \eta_S(\psi)$. The adjoined S column and target-singleton columns represented by the w_i have zero slack cost; the eta objective is precisely the displayed sum over the remaining coalition columns.

Put

$$K = \{d_S : d \in D,\ d(S) \geq \gamma\}.$$

The dual degree equations, dotted with any $d \in D$, give

$$d(S) + \sum_{i \in S} w_i d_i = -\sum_T \lambda_T d(T) \leq \sum_T \lambda_T s(T) = L.$$

Hence for every $d_S \in K$,

$$\sum_{i \in S} w_i d_i \leq L - d(S) < 0.$$

Thus $w \geq 0$ strictly separates K from the nonnegative orthant. If w had at most two coordinate levels, it would have the form $a\mathbf{1}_S + b\mathbf{1}_A$ with $a, b \geq 0$, contradicting Proposition 5.15. Therefore w has at least three levels.

Lemma 5.23 now excludes the $3 + 3$ split and leaves only the exceptional $4 + 2$ orbit, whose dual value is at least 2γ, again contradicting $L < \gamma$. This proves the theorem.

For independent redundancy, the original exhaustive audit remains available. Its C++ census checks 6,508,883 supports per split, finds 21,695 and 19,501 target-conditioned minimal balanced collections, and reduces them to 870 and 619 symmetry orbits. All 1,489 exact-cone optima are at least one by replayable rational certificates. The C++ census is `src/enumerate_n6_eta_dual_vertices.cpp`; its exact output and independent structural verification are `results/n6_eta_dual_vertex_census.json` and `results/n6_eta_dual_vertex_census_verification.json`. The latter rechecks the support count, balancing equations, incidence rank, orbit uniqueness, and recovery of all 505 previously sampled orbits. The exact LP verifier and its full rational certificates are `src/audit_n6_eta_dual_vertices.py` and `results/n6_eta_dual_vertex_audit.`

`json`. Finally, `src/verify_n6_eta_dual_vertex_audit.py` replays all 1,489 saved lower-bound identities in exact arithmetic without calling an optimizer; its summary is `results/n6_eta_dual_vertex_audit_verification.json`. None of this larger audit is needed for the rank-reduced proof above. □

The six-player threshold is exact.

Proposition 5.25 (sharp seven-player exact counterexample)**.** *There is an exact seven-player game whose leximin point ϕ and a proper coalition S satisfy*

$$\gamma_S(\phi) = \frac{3}{4}, \qquad \eta_S(\phi) = 0.$$

For the same game the core reference $\psi = 0$ satisfies $(\gamma_S(\psi), \eta_S(\psi)) = (1, 0)$. Consequently the quantitative carrier inequality is false in unrestricted dimension, and Proposition 5.24 is sharp in the number of players.

Proof. Let $N = \{1, \dots, 7\}$, $S = \{1, 2, 3\}$, $P = \{4, 5, 6, 7\}$, and put

$$a = (-3, 1, 3, -1, 2, -1, -1), \qquad b = (3, -2, 0, -1, -1, -1, 2).$$

Both vectors have total zero. Define the lower-envelope game

$$v(T) = \min\{0, a(T), b(T)\} \qquad (T \subseteq N),$$

and take $\psi = 0$. Each of $0, a, b$ belongs to $\mathrm{Core}(v)$, and at least one of them is tight at every coalition. Hence v is exact and $\psi \in \mathrm{Core}(v)$.

Writing $s(T) = -v(T)$, direct evaluation gives

C	1	2	3	12	13	23
$s(C)$	3	2	0	2	0	2
$s(P \cup C)$	4	3	1	3	1	3

and therefore every one of the six carrier differences equals one. Thus $\gamma_S(0) = 1$.

It remains to calculate $\eta_S(0)$. The four coalitions

$$Q_1 = 13456, \quad Q_2 = 457, \quad Q_3 = 3467, \quad Q_4 = 567$$

all have value zero, and their incidence vectors satisfy

$$\mathbf{1}_S + \sum_{j=1}^{4} \mathbf{1}_{Q_j} + e_1 + 2e_2 = 3\,\mathbf{1}_N. \tag{N7}$$

If $x \in \mathrm{Core}(v)$ and $x_i \geq 0$ for $i \in S$, then $x(Q_j) \geq v(Q_j) = 0$ for every j. Taking the scalar product of (N7) with x yields

$$x(S) + \sum_{j=1}^{4} x(Q_j) + x_1 + 2x_2 = 3x(N) = 0.$$

Every term on the left is nonnegative, so $x(S) = 0$. Since $x = 0$ is feasible, $\eta_S(0) = 0$.

The leximin point can be certified in three elementary stages, without an optimization oracle. The singleton floors are

$$m = (-3, -2, 0, -1, -1, -1, -1).$$

At the first stage, $v(125) = 0$ and efficiency imply $x(3467) \leq 0$ for every $x \in \mathrm{Core}(v)$. If all seven gains $x_i - m_i$ are at least t, then

$$x(3467) \geq m_3 + m_4 + m_6 + m_7 + 4t = -3 + 4t,$$

so $t \leq 3/4$. Equality fixes players $3, 4, 6, 7$ at $(3/4, -1/4, -1/4, -1/4)$.

On this first-stage face their coordinates sum to zero. Since $v(13) = v(27) = 0$, the core constraints give $x_1 \geq -3/4$ and $x_2 \geq 1/4$. Efficiency therefore gives $x_5 \leq 1/2$, so the second common gain among the free players is at most $x_5 - m_5 \leq 3/2$; equality fixes player 5 at $1/2$. Finally $x_1 + x_2 = -1/2$. If their remaining two gains are at least u, then

$$-\frac{1}{2} = x_1 + x_2 \geq m_1 + m_2 + 2u = -5 + 2u,$$

so $u \leq 9/4$. All three bounds are attained by

$$\phi = \frac{1}{4}a = \left(-\frac{3}{4}, \frac{1}{4}, \frac{3}{4}, -\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}\right).$$

Thus the successive Maschler strata are

$$\{3, 4, 6, 7\} \text{ at } \frac{3}{4}, \qquad \{5\} \text{ at } \frac{3}{2}, \qquad \{1, 2\} \text{ at } \frac{9}{4},$$

which proves directly that this point is $\phi(v)$. Direct evaluation of $s_\phi(T) = \phi(T) - v(T)$ gives $s_\phi(P \cup C) - s_\phi(C) = 3/4$ for all six nonempty proper $C \subset S$, so $\gamma_S(\phi) = 3/4$. Moreover each Q_j above is ϕ-tight. If $x \in \mathrm{Core}(v)$ dominates ϕ on S, put $d = x - \phi$. Taking the scalar product of (N7) with d gives

$$d(S) + \sum_{j=1}^{4} d(Q_j) + d_1 + 2d_2 = 3d(N) = 0.$$

Every term on the left is nonnegative: the coordinate terms by domination, and the coalition terms because $x(Q_j) \geq v(Q_j) = \phi(Q_j)$. Hence $d(S) = 0$, while $x = \phi$ is feasible, proving $\eta_S(\phi) = 0$.

The construction and all 128 lower-envelope, core, carrier, degree, and zero-cost identities are replayed in integer arithmetic by `src/verify_n7_exact_carrier_counterexample.py`. The exact rational leximin calculation and the six $3/4$ differences are independently replayed by `src/verify_n7_exact_carrier_counterexample_leximin.py`. □

The preceding example minimizes target size. A complementary leximin counterexample minimizes outside-set size.

Proposition 5.26 (sharp outside-triple leximin obstruction)**.** *There is an exact seven-player game whose leximin point ϕ and a target S with $|S| = 4$ and $|N \setminus S| = 3$ satisfy*

$$\gamma_S(\phi) = \frac{2}{3}, \qquad \eta_S(\phi) = 0.$$

At the core reference zero the same game has $(\gamma_S(0), \eta_S(0)) = (1, 0)$.

Proof. Let $S = 1234$, $P = 567$, and define

$$A = (-4, 1, 2, 2, -1, 1, -1), \qquad B = (2, -1, 0, 0, 1, -1, -1),$$

and $v(T) = \min\{0, A(T), B(T)\}$. As before, the three generators are efficient core allocations and one is tight at every coalition, so v is exact and $\psi = 0$ lies in its core. Direct evaluation gives

$$s(P \cup C) - s(C) = 1 \qquad (\emptyset \neq C \subsetneq S),$$

and hence $\gamma_S(0) = 1$.

The leximin point has a short three-stage certificate. The singleton floors are

$$m = (-4, -1, 0, 0, -1, -1, -1).$$

The coalitions 1246 and 1236 have value zero. Efficiency therefore gives $x(357) \leq 0$ and $x(457) \leq 0$ for every core point. If every gain is at least t, then both of these three-player sums are at least $-2 + 3t$, so $t \leq 2/3$. Equality in the two overlapping triples fixes players $3, 4, 5, 7$ at $(2/3, 2/3, -1/3, -1/3)$.

These four fixed coordinates sum to $2/3$. Since $v(134) = 0$, core feasibility gives $x_1 \geq -4/3$, while efficiency gives $x_1 + x_2 + x_6 = -2/3$ and hence $x_2 + x_6 \leq 2/3$. If the gains of players 2 and 6 are both at least u, then

$$x_2 + x_6 \geq -2 + 2u,$$

so $u \leq 4/3$. Equality fixes both coordinates at $1/3$, after which efficiency fixes player 1 at $-4/3$, with gain $8/3$. All bounds are attained by

$$\phi = \frac{1}{3}A = (-4/3, 1/3, 2/3, 2/3, -1/3, 1/3, -1/3).$$

Thus the successive gain strata are

$$\{3, 4, 5, 7\} \text{ at } \frac{2}{3}, \qquad \{2, 6\} \text{ at } \frac{4}{3}, \qquad \{1\} \text{ at } \frac{8}{3},$$

which proves the leximin assertion without an optimization oracle. Direct evaluation gives

$$s_\phi(P \cup C) - s_\phi(C) = \frac{2}{3} \qquad (\emptyset \neq C \subsetneq S),$$

so $\gamma_S(\phi) = 2/3$.

The four coalitions $56, 357, 457, 13467$ have value zero and are ϕ-tight. Their incidence vectors satisfy

$$\mathbf{1}_S + e_1 + 3e_2 + 2\,\mathbf{1}_{56} + \mathbf{1}_{357} + \mathbf{1}_{457} + 2\,\mathbf{1}_{13467} = 4\,\mathbf{1}_N. \tag{OT}$$

If $x \in \mathrm{Core}(v)$ dominates ϕ on S, put $d = x - \phi$. Pairing the identity with d gives a sum of nonnegative terms equal to $4d(N) = 0$: the coordinate terms are nonnegative by domination and the four coalition terms by ϕ-tightness. Hence $d(S) = 0$, and $x = \phi$ is feasible, proving $\eta_S(\phi) = 0$. Applying the same identity directly to a core point nonnegative on S proves $\eta_S(0) = 0$ as well.

All lower-envelope values, the fourteen unit carrier differences, the degree-four identity, and its rank-seven support are checked without an optimizer by `src/verify_n7_exact_carrier_outside_triple_counterexample.py`. The hand Maschler stages and all fourteen $2/3$ leximin carrier differences are independently replayed in exact rational arithmetic by the companion leximin verifier. □

Proposition 5.27 (outside-triple open family)**.** *For*

$$\kappa \geq 0, \qquad \theta - 3\kappa > \frac{1}{3}, \qquad \kappa + \theta \leq 1,$$

define

$$v_{\kappa,\theta}(T) = \min\{0, A(T), (\kappa A + \theta B)(T)\},$$

where A, B are the vectors in Proposition 5.26. Then $v_{\kappa,\theta}$ is exact, has the same leximin point $\phi = A/3$, and for $S = 1234$ satisfies

$$\gamma_S(\phi) = \theta - 3\kappa - \frac{1}{3} > 0, \qquad \eta_S(\phi) = 0.$$

In particular, the sharp outside-triple obstruction persists on a relatively open two-dimensional wedge and is not an isolated game.

Proof. All three generators have total zero, so the lower-envelope game is exact. Its singleton floors are

$$(-4, \kappa - \theta, 0, 0, -1, \kappa - \theta, -1).$$

The zero-valued constraints at 1246 and 1236 give the same first-stage bound $t \leq 2/3$ and pin $3, 4, 5, 7$ as before. On that face, the constraint at 134 and efficiency give $x_2 + x_6 \leq 2/3$. Requiring their two gains to exceed u gives

$$2(\kappa - \theta) + 2u \leq \frac{2}{3},$$

so $u \leq \theta - \kappa + 1/3$, attained by $A/3$. The parameter restrictions put this strictly above the first level and below player 1's gain $8/3$; efficiency then pins player 1. Thus $A/3$ is leximin.

Direct evaluation of the fourteen carrier differences has minimum $\theta - 3\kappa - 1/3$. The four faces in (OT) remain zero-valued and ϕ-tight, so the same balanced identity proves $\eta_S(\phi) = 0$. Exact rational checks, including interior parameter points, are in `src/verify_n7_exact_carrier_outside_triple_family.py`. □

Corollary 5.28 (the outside-triple family is intrinsically unbumpable)**.** *Every game in Proposition 5.27 has*

$$d^*_{v_{\kappa,\theta}}(1234) = 0.$$

Consequently this robust carrier failure cannot itself generate a proper-coalition monotonicity counterexample.

Proof. Throughout the parameter wedge, the coalitions $S = 1234$, 35, 56, and 357 all have value zero. Their incidence vectors satisfy the elementary identity

$$\mathbf{1}_N + \mathbf{1}_{35} = \mathbf{1}_S + \mathbf{1}_{56} + \mathbf{1}_{357}. \tag{UB}$$

If $x \in F_{35}$, efficiency and (UB) give

$$0 = x(N) + x(35) = x(S) + x(56) + x(357).$$

The three terms on the right are nonnegative by core feasibility, so $x(S) = 0$. Since $v_{\kappa,\theta}(S) = 0$, the face F_{35} contributes zero to the minimum defining exact bump capacity. Nonnegativity of capacity proves equality. □

This blocking phenomenon is not peculiar to the chosen representative. A floating discovery audit of all 34 residual $4+3$ exact-cone orbit games finds no canonical representative with $d_v^*(S) > \phi(S) - v(S)$. Exactly three representatives violate the carrier inequality at their leximin point, and all three have $d_v^*(S) = 0$. This finite screen is diagnostic rather than a universal capacity theorem, but it shows that a monotonicity counterexample must leave the canonical residual layer or use a different target geometry.

That boundary can be crossed. The resulting game cleanly separates the quantitative carrier inequality from the monotonicity conclusion it was designed to prove.

Proposition 5.29 (bumpable carrier failure with a monotone exact path)**.** *Let $S = 1234$ and let v be the exact lower-envelope game generated by*

$$0, \qquad A = (-215, 53, 104, 103, -56, 51, -40), \qquad B = (108, -55, -4, -1, 50, -60, -38).$$

Its leximin point is

$$\phi = (-75, 17, 32, 35, -20, 15, -4),$$

and

$$\gamma_S(\phi) = 36 > 35 = \eta_S(\phi), \qquad \sigma_S = 9 < 10 = d_v^*(S). \tag{BM}$$

Thus the quantitative carrier inequality fails although binding exact bumps exist. Nevertheless, if v^τ agrees with v off S and $v^\tau(S) = \tau$, then the entire exact path is

$$\phi(v^\tau) = \begin{cases} \phi, & 0 \le \tau \le 9, \\ \phi + (\tau - 9)(e_1 - e_6), & 9 \le \tau \le 10. \end{cases} \tag{MP}$$

In particular every admissible bump is coalitionally monotone on S: player 1 weakly gains and players $2, 3, 4$ are unchanged.

Proof. The three displayed generators have common total zero and attain every coalition worth, proving exactness. The singleton floors are

$$m = (-215, -55, -4, -1, -56, -60, -40).$$

There is a four-stage hand certificate for the leximin point. The constraints $v(1236) = -11$ and $v(1246) = -8$, together with efficiency, give $x(457) \le 11$ and $x(357) \le 8$. The corresponding floor sums are -97 and -100, so the first common gain is at most 36; equality pins $3, 4, 5, 7$. On that face, $v(13456) = -13$ gives $x(27) \le 13$ and hence $x_2 \le 17$, pinning player 2 at gain 72. Next $v(134) = -8$ gives $x(2567) \le 8$ and hence $x_6 \le 15$, pinning player 6 at gain 75. Efficiency pins player 1 at gain 140. All bounds are attained by ϕ, so the Maschler strata are

$$\{3, 4, 5, 7\} \text{ at } 36, \qquad \{2\} \text{ at } 72, \qquad \{6\} \text{ at } 75, \qquad \{1\} \text{ at } 140.$$

Direct evaluation of the fourteen carrier differences gives 36 when $1 \in C$ and 39 otherwise, proving $\gamma_S(\phi) = 36$. For eta, use the same degree-four incidence identity as in (OT). The four coalition slacks at $56, 357, 457, 13467$ are respectively $5, 8, 11, 3$, so its weighted cost is

$$2(5) + 8 + 11 + 2(3) = 35.$$

It therefore bounds eta above by 35. The core point

$$y = (-75, 17, 51, 51, -17, 7, -34)$$

dominates ϕ on S and satisfies $y(S) - \phi(S) = 35$, proving the matching lower bound. Finally $v(S) = 0$ and $\phi(S) = 9$, so $\sigma_S = 9$.

The capacity has an equally short upper certificate. The incidence identity (UB) still holds, but now

$$v(35) = v(357) = 0, \qquad v(56) = -10.$$

Thus every $x \in F_{35}$ satisfies $x(S) = x(35) - x(56) - x(357) \leq 10$, and hence $d_v^*(S) \leq 10$. For the reverse inequality, the endpoint game v^{10} is itself the lower envelope of the four common-total allocations

$$A, \quad B, \quad (10, 0, 0, 0, 0, -10, 0), \quad (-72, 18, 36, 28, -18, 18, -10).$$

Direct evaluation shows that this lower envelope agrees with v off S and has value 10 at S. Hence v^{10} is exact, proving $d_v^*(S) = 10$ by Proposition 5.2.

It remains to certify the path. Up to $\tau = 9$, the old leximin point remains feasible, so the bump is nonbinding. Put $\phi^{10} = \phi + e_1 - e_6$. Direct substitution places ϕ^{10} in $\mathrm{Core}(v^{10})$; convexity then places $\phi + (\tau - 9)(e_1 - e_6)$ in $\mathrm{Core}(v^\tau)$ for every $9 \leq \tau \leq 10$. The first two Maschler stages above are unchanged. After they are fixed, $x_2 + x_3 + x_4 = 84$ and efficiency gives $x_1 + x_6 = -60$. The target constraint $x(S) \geq \tau$ therefore yields

$$x_1 \geq \tau - 84, \qquad x_6 \leq 24 - \tau.$$

The candidate attains equality. Player 6's gain $84 - \tau$ remains strictly above player 2's preceding level 72, while player 1's gain is larger still; thus the last two stages pin 6 and then 1. This proves (MP). Every lower-envelope, eta, capacity, endpoint-exactness, and path identity is replayed without an optimizer by `src/verify_n7_binding_carrier_monotone_path.py`. □

The preceding construction is minimal in outside-set size among seven-player obstructions, including those evaluated at the leximin point.

Proposition 5.30 (seven-player outside-pair theorem)**.** *For every exact seven-player game, every core point ψ, and every proper nonsingleton S with $|N \setminus S| \leq 2$,*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

Consequently three is the minimum number of outside players in a seven-player obstruction, both for arbitrary core references and for the leximin reference.

Proof. The codimension-one case is Proposition 5.16, so only the split $(|S|, |P|) = (5, 2)$ remains. If a violation existed, the proof of Proposition 5.24 would produce an undominated eta-dual vertex whose target-floor multiplier has at least three levels. After adjoining S, its support is a minimal balanced collection of at most seven columns. If q positive target-singleton columns occur, three levels force $2 \leq q \leq 4$, leaving at most $6 - q$ other coalition columns.

Exact determinant enumeration of precisely these supports checks 8,798,030 candidates. It finds 2,196 three-level supports. The carrier inequalities and slack nonnegativity directly bound all but 96 of them; quotienting the latter by $S_5 \times S_2$ leaves five orbits. For each orbit, minimization over the global exact-witness slack cone has exact value 3, with independently replayable rational dual certificates. Hence no three-level vertex has cost below the normalization $\gamma_S(\psi) = 1$, a contradiction.

For comparison, the same focused census in the $4+3$ split checks 9,086,132 supports and leaves 34 orbits after the direct carrier bound. Their exact-value distribution is

$$14 \text{ at } 0, \qquad 12 \text{ at } 1, \qquad 8 \text{ at } 2.$$

Thus the outside-triple example reflects several inequivalent rank-seven obstruction mechanisms rather than a numerical accident. The accompanying integer census, exact cone audit, and optimizer-free replay are all listed in the supplementary artifact; the replay checks every saved rational lower-bound identity without calling an optimizer. Proposition 5.26 supplies the matching failure at outside size three. □

The sharp counterexample propagates to every higher dimension.

Corollary 5.31 (complete dimension threshold)**.** *The quantitative carrier inequality holds for every exact game if $|N| \le 6$, and fails for an exact game at every dimension $|N| \ge 7$.*

Proof. The positive statement is Proposition 5.24. Starting from the game in Proposition 5.25, append any number of null players: extend each of the three generators by zero and put $v^+(T) = v(T \cap N)$. This remains an exact lower-envelope game.

Its core is exactly $\mathrm{Core}(v) \times \{0\}^k$. Indeed, singleton rationality makes each new coordinate nonnegative, while the old-grand-coalition constraint and efficiency make their sum nonpositive; hence all are zero. The converse inclusion is immediate from $v^+(T) = v(T \cap N)$. The new players therefore contribute fixed zero gains before the three positive leximin strata, so the old leximin coordinates are unchanged. For the same target S, the enlarged outside set merely adds null players, and hence all six carrier differences remain $3/4$. The feasible core displacements on S are also unchanged, so $\eta_S = 0$. The identities are replayed through arbitrary prescribed finite dimension by `src/verify_n7_exact_carrier_dummy_extensions.py`. □

The example has a rigid two-dimensional core but is not an isolated game.

Proposition 5.32 (triangular core and generator minimality)**.** *For the seven-player game in Proposition 5.25,*

$$\mathrm{Core}(v) = \mathrm{conv}\{0, a, b\}.$$

Moreover every representation of v as the lower envelope of core allocations requires at least three allocations. Thus the displayed three-generator representation is minimal.

Proof. The eight coalitions

$$\mathcal{R} = \{1234, 125, 1236, 13456, 27, 457, 3467, 567\}$$

all have value zero and satisfy $\sum_{R \in \mathcal{R}} \mathbf{1}_R = 4\,\mathbf{1}_N$. Hence every core point is tight on every member of $\mathcal{R}$: all eight core inequalities are nonnegative, while their sum is four times efficiency. Their incidence matrix has rank five. Both a and b lie in its kernel and are linearly independent, so every $x \in \mathrm{Core}(v)$ has the form $x = \alpha a + \beta b$.

The core inequalities at 3, 13, and 4 now read, respectively, $3\alpha \ge 0$, $3\beta \ge 0$, and $-\alpha - \beta \ge -1$. Thus $\alpha, \beta \ge 0$ and $\alpha + \beta \le 1$. The reverse inclusion holds because $0, a, b \in \mathrm{Core}(v)$ and the core is convex.

Finally, among the three vertices, a uniquely attains $v(1)$, b uniquely attains $v(2)$, and zero uniquely attains $v(123)$. Any lower-envelope representation by core allocations must therefore contain all three. The rank, degree, core, and unique-attainment calculations are replayed by `src/verify_n7_exact_carrier_structure.py`. □

Proposition 5.33 (nonisolated two-parameter family)**.** *Let*

$$\kappa \geq 0, \qquad \theta - 2\kappa > \frac{1}{4}, \qquad \kappa + \theta \leq 1.$$

Then the exact game

$$v_{\kappa,\theta}(T) = \min\{0, a(T), (\kappa a + \theta b)(T)\}$$

has leximin point $\phi = a/4$ and, for $S = 123$,

$$\gamma_S(\phi) = \theta - 2\kappa - \frac{1}{4} > 0, \qquad \eta_S(\phi) = 0.$$

Consequently the seven-player counterexample is neither isolated nor unique up to permutation and common positive scaling.

Proof. Exactness again follows from the equal-total lower-envelope construction, and $a/4$ is a core point. The singleton floors are

$$(-3, \kappa - 2\theta, 0, -1, 2\kappa - \theta, -1, -1).$$

The first leximin stage is unchanged: $v_\theta(125) = 0$ forces $x(3467) \leq 0$, so the first common gain is at most $3/4$ and fixes $3, 4, 6, 7$ at the coordinates of $a/4$. On that face the constraints at 13 and 27 give $x_1 \geq -3/4$ and $x_2 \geq 1/4$, whence efficiency gives $x_5 \leq 1/2$. The next gain is therefore at most $1/2 - 2\kappa + \theta$, attained by $a/4$. It is strictly below the gain of player 2 because $\kappa + \theta > 1/4$, and below the gain of player 1 because $\kappa + \theta \leq 1$.

After fixing player 5, efficiency gives $x_1 + x_2 = -1/2$. The preceding bound on x_1 gives $x_2 \leq 1/4$, so the next gain is at most $1/4 - \kappa + 2\theta$, again attained by $a/4$ and at most $9/4$; the remaining coordinate is then fixed. This proves the leximin assertion directly. Evaluation of the six carrier differences gives $3/4$ for the three C containing player 1, $\kappa + \theta - 1/4$ for $C = 2, 23$, and $\theta - 2\kappa - 1/4$ for $C = 3$. The last quantity is the minimum under the parameter restrictions. The four coalitions in (N7) remain zero-valued and ϕ-tight, so the same identity proves $\eta_S(\phi) = 0$.

The region contains a relatively open two-dimensional wedge (take $\kappa > 0$ and both remaining inequalities strict), giving genuine two-parameter robustness. The slice $\kappa = 0$ includes the earlier $\theta = 1$ and $\theta = 4/5$ examples, whose singleton-floor multisets are not proportional under any permutation. Exact family checks are in `src/verify_n7_exact_carrier_family.py`. □

Corollary 5.34 (the carrier failure is not bumpable)**.** *For every game in Proposition 5.33, the exact bump capacity at $S = 123$ is zero:*

$$d^*_{v_\theta}(123) = 0.$$

Hence this sharp failure of the quantitative carrier inequality cannot itself produce a counterexample to proper-coalition monotonicity.

Proof. This is forced by the same zero-cost face geometry, not merely by unique attainment at one vertex. Three of the four coalitions in (N7) satisfy the incidence identity

$$\mathbf{1}_{135} - 2\,\mathbf{1}_{123} = \mathbf{1}_{13456} + \mathbf{1}_{457} + \mathbf{1}_{567} - 2\,\mathbf{1}_N. \tag{NB}$$

All three coalitions on the right have value zero throughout the family. Hence every core point x obeys

$$x(135) - 2x(123) = x(13456) + x(457) + x(567) \geq 0.$$

The coalition 135 also has value zero (its three generator payoffs are $0, 2, 2(\kappa+\theta)$). Thus every $x \in F_{135}$ has $x(123) \le 0$, while core feasibility at 123 gives the reverse inequality. Consequently

$$\max_{x \in F_{135}} x(123) - v_\theta(123) = 0.$$

This term occurs in the minimum defining $d^*_{v_\theta}(123)$, while bump capacity is nonnegative, proving equality. Thus no positive exact bump at this target exists. In particular, the unbumpability is intrinsic to the tight balanced certificate: any deformation preserving these three zero-valued faces and $v(123) = v(135) = 0$ remains blocked. □

The negative mechanism also exposes the rank threshold. After adjoining S, the zero-cost eta support in (N7) is a minimal balanced collection of seven linearly independent incidence columns. The outside-triple identity (OT) has the same rank-seven feature. Six players permit at most six independent columns; after the target and its positive singleton columns are removed, at most three positive residual weights remain. The two-level separator theorem forces two independent equal-subset-sum relations among those weights, and the elementary alternatives $z_i = z_j$ and $z_i + z_j = z_k$ leave only the single rank-six three-level orbit in (TL), which one carrier inequality defeats. Seven players are the first dimension permitting a fourth residual weight; the two three-weight relation types no longer exhaust the geometry. Multiple three-level orbits survive, and fourteen have zero cost on the exact-game cone. This is the conceptual six/seven transition; the support censuses and the larger 1,489-orbit audit are independent redundancy.

The slack certificate is automatic on the most important thin-core model class used by the searches.

Proposition 5.35 (two-generator lower envelopes)**.** *Let $p, q \in \mathbb{R}^N$ have the same total, and define the exact game*

$$v(T) = \min\{p(T), q(T)\} \quad (T \subsetneq N), \qquad v(N) = p(N) = q(N).$$

Let S be proper and nonsingleton. If $\max_{x \in \mathrm{Core}(v)} x(S) > \phi(S)$ and $(\star_S)$ fails at ϕ, then there is a nonempty proper $C \subset S$ satisfying the hypothesis of Proposition 5.5. Consequently $d^*_v(S) \le \phi(S) - v(S)$.

Proof. Both p and q belong to $\mathrm{Core}(v)$, and one of them attains $v(T)$ at every proper T, so v is exact. Put $P = N \setminus S$. Exactness at P gives

$$\max_{x \in \mathrm{Core}(v)} x(S) = v(N) - v(P) = \max\{p(S), q(S)\}.$$

After interchanging p and q, binding therefore gives $p(S) > \phi(S)$. The two generator sums on S cannot tie, since then $v(S) = \max_{x \in \mathrm{Core}(v)} x(S)$; hence $p(S) > q(S)$ and $p(P) < q(P)$. Define

$$C = \{i \in S : p_i > \phi_i\}.$$

This set is nonempty because $p(S) > \phi(S)$, and it is proper because otherwise $p \in \mathrm{Core}(v)$ would itself witness $(\star_S)$. For $i \in C$, core feasibility at the singleton gives

$$p_i > \phi_i \ge v(\{i\}) = \min\{p_i, q_i\},$$

so $q_i \le \phi_i < p_i$. Hence

$$p(C) - q(C) \ge p(C) - \phi(C) \ge p(S) - \phi(S) = \phi(P) - p(P).$$

In particular $q(C) < p(C)$, and therefore

$$\begin{aligned} v(P \cup C) - v(C) &= \min\{p(P) + p(C), q(P) + q(C)\} - q(C) \\ &= \min\{p(P) + p(C) - q(C), q(P)\} \\ &\geq \phi(P). \end{aligned}$$

The last inequality also uses $\phi(P) \leq v(N) - v(S) = q(P)$, since $q(S) < p(S)$. Equivalently, $\phi(P \cup C) - v(P \cup C) \leq \phi(C) - v(C)$, which is precisely the off-side slack-domination hypothesis. □

Lemma 5.36 (exact-face compensation identity)**.** *Let v be exact, let $P = N \setminus S$, and let $\psi \in \mathrm{Core}(v)$. For nonempty $C \subsetneq S$, choose any $a \in \mathrm{Core}(v)$ with $a(P \cup C) = v(P \cup C)$; such an a exists by exactness. Then*

$$v(P \cup C) - v(C) - \psi(P) = [a(P) - \psi(P)] + [a(C) - v(C)]. \tag{GC}$$

In particular, $a(P) \geq \psi(P)$ is a sufficient one-way screen for $v(P \cup C) - v(C) \geq \psi(P)$. When $\psi = \phi$, this is the off-side hypothesis used in Proposition 5.5; it is not an equivalent characterization, because the nonnegative trace compensation $a(C) - v(C)$ may rescue cases with $a(P) < \psi(P)$. The algebraic identity itself is not restricted to a finite lower-envelope representation.

Proof. Since $a(P \cup C) = v(P \cup C)$ and $P \cap C = \varnothing$, the identity is algebraic. Core feasibility gives $a(C) - v(C) \geq 0$. □

The exact interval does not rescue proper-coalition monotonicity. In fact the failure already occurs with five players, zero singleton floors, and a two-generator endpoint.

Theorem 5.37 (proper-coalition monotonicity fails on exact games)**.** *Let $N = \{1, 2, 3, 4, 5\}$, $S = \{1, 3, 4, 5\}$, and let v and w be the lower-envelope games generated respectively by*

$$A = (2, 0, 12, 6, 0), \qquad B = (0, 0, 0, 0, 20), \qquad P = (0, 2, 10, 4, 4)$$

and by A, B alone. Then both games are exact, all singleton values are zero, and they differ only at S, where

$$v(S) = 18 < 20 = w(S).$$

Nevertheless

$$\phi(v) = \left(1, 1, \frac{13}{2}, 5, \frac{13}{2}\right), \qquad \phi(w) = (2, 0, 6, 6, 6). \tag{PM}$$

Thus target players 3 and 5 each lose $1/2$. Because both games have zero singleton values, this is also a direct failure of proper-coalition monotonicity for the ordinary Lmin solution.

More precisely, if v^τ agrees with v off S and $v^\tau(S) = \tau$, then every v^τ, $18 \leq \tau \leq 20$, is exact and

$$\phi(v^\tau) = \begin{cases} \phi(v), & 18 \leq \tau \leq 19, \\ (\tau - 18,\ 20 - \tau,\ 16 - \tau/2,\ \tau - 14,\ 16 - \tau/2), & 19 \leq \tau \leq 20. \end{cases} \tag{PP}$$

The losses therefore begin immediately when the bump becomes binding.

Proof. The three displayed allocations have common total 20. The additional generator P lowers only S. Indeed, if $5 \notin T$, then $B(T) = 0 \le P(T)$. If $5 \in T$, then

$$P(T) - A(T) = 4 + 2\mathbf{1}_{\{2\in T\}} - 2|T \cap \{1, 3, 4\}|,$$

which is negative only when $T = S$. At that coalition $P(S) = 18$ while $A(S) = B(S) = 20$. Thus v and w agree off S. Every generator belongs to the corresponding core and attains the coalitions on which it is minimal, so both lower-envelope games are exact. Directly, their common singleton floors are zero.

Here is a hand certificate for the first point in (PM). In the base game $v(345) = 18$, so efficiency gives $x_1 + x_2 \le 2$. Hence the first common gain is at most one, and equality pins players $1, 2$. On that face the remaining total is 18, while $v(135) = 14$ gives $x_3 + x_5 \ge 13$ and therefore $x_4 \le 5$. This pins player 4; efficiency leaves $x_3 + x_5 = 13$, which the last stage equalizes. The resulting point is the displayed $\phi(v)$ and direct substitution places it in the core.

For the endpoint, $w(S) = 20$ and efficiency pin player 2 at zero. Next $w(345) = 18$ gives $x_1 \le 2$. After player 1 is pinned at two, the remaining total is 18 and $w(35) = 12$ gives $x_4 \le 6$; the last pair is equalized. This gives $\phi(w) = (2, 0, 6, 6, 6)$, again core-feasible by direct substitution.

For the path, $\phi(v)(S) = 19$, so the nonbinding part is constant. Suppose $19 \le \tau \le 20$. The target constraint and efficiency first give $x_2 \le 20 - \tau$. Once player 2 is pinned, $v(345) = 18$ gives $x_1 \le \tau - 18$. On the resulting face $x(345) = 18$; using $v(135) = 14$ gives $x_4 \le \tau - 14$. Finally $x_3 + x_5 = 32 - \tau$, so the last stage equalizes them. These four bounds give (PP). Every coordinate is affine, and direct core feasibility at $\tau = 19, 20$ proves feasibility throughout the segment. Exactness throughout is Lemma 5.1.

The exact rational verifier recomputes both endpoint Lmin allocations by a staged Maschler LP, rather than merely checking the displayed stage bounds:

`src/verify_n5_exact_leximin_monotonicity_counterexample.py`,
`results/n5_exact_leximin_monotonicity_counterexample.json`.

□

Proposition 5.38 (a large core without a Lorenz-dominating element)**.** *Both five-player games v, w in Theorem 5.37 have large cores, and*

$$\mathrm{Lmax}(v) = \mathrm{Lmax}(w) = (2, 0, 6, 6, 6).$$

Thus Lmax is coalitionally monotonic across this bump while Lmin is not. In the base game the richest Lmin stratum is $\{3, 5\}$ and

$$\mathrm{Lmin}(v)(\{3, 5\}) = 13 > 12 = v(\{3, 5\}).$$

The base game is not an exact partition game at any core allocation and its core has no Lorenz-dominating element. The endpoint game is an exact partition game, witnessed by its common Lmin–Lmax allocation with ordered strata $\{2\} \prec \{1\} \prec \{3, 4, 5\}$. Consequently largeness alone implies neither the exact-partition property nor existence of a Lorenz-dominating core element, and the through-four-player classification below is sharp at five.

Computer-assisted proof. For each game, enumerate every vertex of Sharkey's acceptable-payoff polyhedron

$$U(v) = \{y \in \mathbb{R}^N : y(T) \ge v(T) \text{ for every nonempty } T \subseteq N\}.$$

There are $\binom{31}{5} = 169{,}911$ candidate tight bases, of which 104,286 are independent. Exact rational elimination gives 12 distinct acceptable vertices for v and 8 for w. An exact core LP finds, for every such vertex y, an $x \in \mathrm{Core}(v)$ or $x \in \mathrm{Core}(w)$ with $x \leq y$. Since $U(v)$ is pointed with nonnegative recession cone, convexity extends vertex domination to all of $U(v)$; this is the large-core property. Separate exact staged LPs compute the two Lmax allocations, and direct substitution checks the displayed strict suffix slack.

To test the exact-partition property independently of Lmin, enumerate all 541 ordered set partitions $B_1 \prec \cdots \prec B_k$ of five players. For each chain, an exact rational linear-arithmetic query imposes core feasibility, equal payoffs within each block, strict payoff inequalities between adjacent blocks, and tightness of every richer suffix $B_{j+1} \cup \cdots \cup B_k$. All 541 base-game queries are infeasible. For the endpoint, the displayed three-block chain is feasible exactly at $(2, 0, 6, 6, 6)$. Finally, any core allocation that Lorenz-dominated the whole core would be both its leximin and leximax point. The displayed inequality $\mathrm{Lmin}(v) \neq \mathrm{Lmax}(v)$ therefore excludes such an allocation in the base game. The verifier and output are

`src/verify_n5_exact_lmin_lmax_large_core.py`,
`results/n5_exact_lmin_lmax_large_core.json`.

□

The example also closes the proposed cone route. For every $19 \leq \tau < 20$, the core point

$$y^\tau = (2, 0, 16 - \tau/2, \tau - 14, 16 - \tau/2)$$

weakly dominates $\phi(v^\tau)$ on S and has $y^\tau(S) = 20 > \tau$. Thus $(\star_S)$ holds along the entire binding interior, even while players 3 and 5 decrease. The implication "$(\star_S)$ along an exact path implies coordinatewise monotonicity" is therefore false; the carrier program controls exact bump capacity but cannot prove monotonicity of this selector.

This negative result should be read against the classical impossibility literature, not as an unrestricted first impossibility theorem. Arin and Iñarra (2001) introduced Lmin, and Dietzenbacher (2021) already shows that Lmin can fail aggregate monotonicity on the broader egalitarian-stable domain while Lmax is coalitionally monotonic there. Moreover, Housman and Clark (1998) prove that no core allocation method can be coalitionally monotonic on the full balanced-game domain once there are at least four players. None of these results itself supplies an exact-game proper-coalition counterexample for Lmin. Theorem 5.37 does so with a five-player rational certificate. The next result shows that five is the minimum possible dimension. The condition needed next is established terminology. Following Llerena and Mauri (2017), a balanced game is an *exact partition game* if it has a core allocation at which every cumulative union of richest payoff strata is tight. On this class the Dutta–Ray solution is the unique such allocation and Lorenz-dominates the core (Llerena and Mauri 2017; Dietzenbacher and Yanovskaya 2020). The new contribution is the following sharp low-dimensional classification, not the definition or its Lorenz consequence.

We will use one further established description of exactness. An *exact-balanced* weight vector ρ satisfies $\sum_S \rho_S \mathbf{1}_S = \mathbf{1}_N$, with every nonzero coefficient positive except possibly one. A game is exact if and only if $\sum_S \rho_S v(S) \leq v(N)$ for every such vector (Csóka et al. 2011; Lohmann et al. 2012). At a core allocation ϕ, write $s_\phi(S) = \phi(S) - v(S)$.

Lemma 5.39 (exact-balanced slack budget)**.** *If v is exact, $\phi \in \mathrm{Core}(v)$, and ρ is exact-balanced, then*

$$\sum_S \rho_S s_\phi(S) \geq 0. \tag{SB}$$

If H is the unique negatively weighted coalition, then

$$(-\rho_H)s_\phi(H) \le \sum_{S\neq H} \rho_S s_\phi(S). \tag{SB-H}$$

Proof. The incidence identity and efficiency give $\sum_S \rho_S \phi(S) = \phi(N) = v(N)$. Subtract the exact-balancedness inequality $\sum_S \rho_S v(S) \le v(N)$. □

The next quantitative form exposes how tight transfer-blockers can be completed by singleton coalitions into an exact-balanced collection.

Lemma 5.40 (boundary-cover inequality)**.** *Let v be a singleton-normalized exact game, let $\phi \in \mathrm{Core}(v)$, and fix $\emptyset \neq H \subsetneq N$, with $L = N \setminus H$ and $s = \phi(H) - v(H)$. Give coalitions tight at ϕ arbitrary weights $\lambda_T \ge 0$, and put*

$$c_j = \sum_{T \ni j} \lambda_T.$$

If $\alpha > \beta \ge 0$, $c_h \le \alpha$ for $h \in H$, and $c_i \le \beta$ for $i \in L$, then

$$(\alpha - \beta)s \le \sum_{h\in H} (\alpha - c_h)\phi_h + \sum_{i\in L} (\beta - c_i)\phi_i. \tag{BC}$$

Proof. Exactness of H provides $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$. For $d = z - \phi$,

$$d(H) = -s, \qquad d(L) = s, \qquad d_j \ge -\phi_j, \qquad d(T) \ge 0$$

for every coalition T tight at ϕ. Therefore

$$0 \le \sum_T \lambda_T d(T) = \sum_j c_j d_j \le -(\alpha - \beta)s + \sum_{h\in H} (\alpha - c_h)\phi_h + \sum_{i\in L} (\beta - c_i)\phi_i,$$

which is (BC). □

For its exact-balanced interpretation, add singleton weight $\alpha - c_h$ at each $h \in H$ and $\beta - c_i$ at each $i \in L$, and give H weight $-(\alpha - \beta)$. The resulting coverage is $\beta \mathbf{1}_N$; after adding a common grand-coalition weight when $\beta = 0$ and rescaling, Lemma 5.39 gives the same inequality. Thus (BC) is a quantitative protector-completion form of published exact balancedness, not a new characterization of exact games.

Singleton completion can be sharpened to an intrinsic coalition-slack cost. This form needs no normalization once the tight cover is given.

Theorem 5.41 (tight deficit completion)**.** *Let v be exact, $\phi \in \mathrm{Core}(v)$, and $N = H \dot\cup L$. Suppose weights $\lambda_T \ge 0$ on ϕ-tight coalitions have common load α on H and loads $c_i \le \beta < \alpha$ on $i \in L$. Put $d_i = \beta - c_i$. If weights $\nu_Q \ge 0$ on nonempty coalitions $Q \subseteq L$ satisfy*

$$\sum_{Q\subseteq L} \nu_Q \mathbf{1}_Q = d,$$

then

$$(\alpha - \beta)s_\phi(H) \le \sum_{Q\subseteq L} \nu_Q s_\phi(Q). \tag{DC}$$

Consequently, if d lies in the incidence cone of ϕ-tight coalitions contained in L, then H is tight.

Proof. The cover and completion give

$$\sum_T \lambda_T \mathbf{1}_T + \sum_Q \nu_Q \mathbf{1}_Q - (\alpha - \beta)\mathbf{1}_H = \beta \mathbf{1}_N.$$

After rescaling when $\beta > 0$, this is exact-balanced. If $\beta = 0$, first add any positive common grand-coalition coefficient; its slack is zero. Lemma 5.39, tightness of every T, and $s_\phi(N) = 0$ now give (DC). □

The displayed incidence system is a semi-balanced system with exceptional set H in the terminology of Studený and Kratochvíl (2022); its associated exact-game inequality is prior art. The candidate new object below is the restricted optimization selected by an Lmin tight cover, not the semi-balanced inequality itself.

Minimizing the right side over all completions defines a canonical LP value $\kappa_{\phi,L}(d)$. Singleton coalitions make the LP feasible, while nonnegative slack makes its optimum nonnegative and attained. Thus

$$(\alpha - \beta)s_\phi(H) \le \kappa_{\phi,L}(d), \qquad \kappa_{\phi,L}(d) = 0 \iff d \in \operatorname{cone}\{\mathbf{1}_Q : Q \subseteq L,\ s_\phi(Q) = 0\}. \tag{DC-K}$$

This turns every slack boundary into a positive-cost ray of a poorer-side tight-set cone. The five-player two-column family and the seven-player three-trace family both attain equality: their respective parameters are $(\alpha, \beta) = (2, 1)$ with completion $\mathbf{1}_R$, and $(1, 1/2)$ with completion $\frac{1}{2}\mathbf{1}_{\{a\}}$. A standalone proof and exact audit are

`paper3/TIGHT_DEFICIT_COMPLETION_THEOREM.md`,
`src/verify_tight_deficit_completion.py`,
`results/tight_deficit_completion.json`.

Equivalently, define the fractional exact-cover worth

$$\omega_{v,L}(d) := \max\left\{ \sum_Q \nu_Q v(Q) : \nu_Q \ge 0,\ \sum_Q \nu_Q \mathbf{1}_Q = d,\ \emptyset \ne Q \subseteq L \right\}.$$

Modularity of ϕ gives the exact gap identity

$$\kappa_{\phi,L}(d) = d \cdot \phi_L - \omega_{v,L}(d). \tag{DC-G}$$

Thus coalitional completion is never more expensive than singleton completion in a normalized game, and it is strictly cheaper whenever an optimal completion uses positive total coalition worth. The invariant is a fractional exact-cover value gap, not merely a renamed uncovered-payoff sum.

The cover can in fact be reduced to one signed incidence circuit. This removes arbitrary support size from the obstruction theory.

Theorem 5.42 (atomic Lmin obstruction reduction)**.** *Let v be exact, $\phi \in \operatorname{Core}(v)$, and $N = H \dot\cup L$, with $\varnothing \ne H \subsetneq N$. Suppose nonempty proper ϕ-tight coalitions with weights $\lambda_T \ge 0$ have common load α on H and loads $c_i < \alpha$ on $i \in L$. If $s_\phi(H) > 0$, then among the positive-support tight coalitions, H, and the deficit singletons*

$$\{\{i\} : i \in L,\ c_i < \beta\}, \qquad \beta := \max_{i \in L} c_i,$$

there is a pure minimal semi-balanced subsystem $\mathcal{A}$ with exceptional coalition H. It has coefficients $a_S > 0$ for $S \neq H$, $\gamma > 0$, and $r \in \mathbb{R}$ satisfying

$$\sum_{S \in \mathcal{A} \setminus \{H\}} a_S \mathbf{1}_S - \gamma \mathbf{1}_H = r \mathbf{1}_N, \tag{AC}$$

and $|\mathcal{A}| \leq |N|$. Moreover $\mathcal{A}$ contains a deficit singleton whose ϕ-slack is strictly positive.

Consequently every slack richer-payoff suffix at Lmin has such an atomic certificate.

Proof. Put $d_i = \beta - c_i$. Singleton completion gives

$$\sum_T \lambda_T \mathbf{1}_T + \sum_{i \in L} d_i \mathbf{1}_{\{i\}} - (\alpha - \beta)\mathbf{1}_H = \beta \mathbf{1}_N. \tag{AC0}$$

Project incidence vectors to $\mathbb{R}^N / \operatorname{span}\{\mathbf{1}_N\}$. This is a dependence whose only negative coordinate is the one indexed by H. The conformal circuit decomposition theorem writes it as a positive sum of circuit dependences without introducing a new sign (see, e.g., Björner et al. 1999). Because the coefficient of H in (AC) is negative, one constituent circuit has H as its unique negative element. Lifting that circuit from the quotient gives (AC).

Circuit minimality makes $\mathcal{A}$ a minimal semi-balanced system; its unique negative sign makes it pure and makes H exceptional. The quotient has dimension at most $|N| - 1$, so a circuit has at most $|N|$ elements. Finally exact balancedness applied to (AC) gives

$$\gamma s_\phi(H) \leq \sum_{\substack{\{i\} \in \mathcal{A} \\ c_i < \beta}} a_{\{i\}} s_\phi(\{i\}),$$

because every other positive-support coalition is ϕ-tight. The left side is positive, so at least one displayed singleton has positive slack. The Lmin conclusion follows from the strict tight-cover lemma. □

The conformal decomposition and minimal semi-balanced terminology are prior art; the candidate new point is their Lmin-selected orientation and support restriction. In fact these atoms systematically avoid the exact-cone facets.

Corollary 5.43 (Lmin obstruction atoms are decomposable)**.** *Every atom supplied by Theorem 5.42 is decomposable in the sense of Studený and Kratochvíl (2022). Consequently its associated exact-game inequality is not facet-defining.*

Proof. Choose a positive-slack deficit singleton $\{i\}$ in the atom. There must be another player in L: if $L = \{i\}$, then $\beta = \max_{j \in L} c_j = c_i$, so no deficit singleton exists. Hence $E := H \cup \{i\}$ is proper. It is not already in the atom, since otherwise

$$\mathbf{1}_H + \mathbf{1}_{\{i\}} - \mathbf{1}_E = 0$$

would be a signed dependence on a proper subsupport, contrary to circuit minimality. The same identity shows that E is exceptional in $\mathcal{A} \cup \{E\}$. Thus E is a decomposition of $\mathcal{A}$ by the published definition. The facet conclusion follows from the indecomposability criterion of Studený and Kratochvíl (2022). □

The rooted construction has an intrinsic game-theoretic meaning that does not depend on Lmin. It turns exactness of a face into balancedness on the complement.

Theorem 5.44 (marginal-core factorization)**.** *Let v be exact, let $\phi \in \mathrm{Core}(v)$, fix $\varnothing \neq H \subsetneq N$, and put $L = N \setminus H$. Define the H-marginal game on L by*

$$w_H(Q) := v(H \cup Q) - v(H), \qquad Q \subseteq L.$$

Then w_H is balanced. For every $Q \subseteq L$,

$$\phi(Q) - w_H(Q) = s_\phi(H \cup Q) - s_\phi(H), \tag{MF1}$$

and

$$\phi(L) = w_H(L) - s_\phi(H). \tag{MF2}$$

Consequently

$$H \text{ is tight at } \phi \iff \phi_L \in \mathrm{Core}(w_H). \tag{MF3}$$

Proof. Exactify H: choose $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$. Then $z(L) = v(N) - v(H) = w_H(L)$, while for $Q \subseteq L$,

$$z(Q) = z(H \cup Q) - z(H) \geq v(H \cup Q) - v(H) = w_H(Q).$$

Thus $z_L \in \mathrm{Core}(w_H)$ and w_H is balanced. Equations (MF1) and (MF2) follow by expansion and efficiency. If H is tight, core feasibility of ϕ gives $\phi(Q) \geq w_H(Q)$ and (MF2) gives efficiency, so $\phi_L \in \mathrm{Core}(w_H)$. Conversely, efficiency in that marginal core and (MF2) force $s_\phi(H) = 0$. □

Thus exact partition at a core point is equivalent to compatibility of its restrictions with a canonical family of balanced marginal games. At a slack Lmin boundary, ϕ_L lies exactly $s_\phi(H)$ below the efficiency hyperplane of the balanced game w_H. The obstruction is therefore not an unstructured failure of an exact face: it is a weighted fractional-cover gap inside a balanced marginal game.

This factorization is not a recursive identity for the selector. In both the five-player and seven-player obstruction families, adding the missing total $s_\phi(H)$ uniformly to ϕ_L gives neither ordinary Lmin nor normalized Lmin of w_H. Thus the valid recursion is at the level of core compatibility and fractional-cover geometry, not at the level of the selected allocation itself; the exact counterchecks are recorded in `results/rooted_deficit_completion.json`.

The forced decomposition can be absorbed rather than discarded. Rooting a poorer coalition Q at H replaces it by $H \cup Q$ and charges its marginal slack. This strengthens the completion invariant.

Theorem 5.45 (rooted deficit completion)**.** *Under the hypotheses and notation of Theorem 5.41, let $\tau := \sum_Q \nu_Q$. Then*

$$(\alpha - \beta + \tau) s_\phi(H) \leq \sum_Q \nu_Q s_\phi(H \cup Q), \tag{RDC}$$

or equivalently

$$(\alpha - \beta) s_\phi(H) \leq \sum_Q \nu_Q \{s_\phi(H \cup Q) - s_\phi(H)\}. \tag{RMC}$$

For disjoint H, Q, the rooted marginal cost is

$$s_\phi(H \cup Q) - s_\phi(H) = s_\phi(Q) - \{v(H \cup Q) - v(H) - v(Q)\}. \tag{RMS}$$

Thus (RMC) improves the unrooted bound by the weighted superadditivity surplus.

Proof. For every $Q \subseteq L$, $\mathbf{1}_Q = \mathbf{1}_{H\cup Q} - \mathbf{1}_H$. Substitute this into the deficit-completion identity to obtain

$$\sum_T \lambda_T \mathbf{1}_T + \sum_Q \nu_Q \mathbf{1}_{H\cup Q} - (\alpha - \beta + \tau)\mathbf{1}_H = \beta \mathbf{1}_N.$$

Exact balancedness and tightness of the protectors give (RDC); rearrangement gives (RMC). Expanding the three slacks proves (RMS). Exact games are totally balanced and hence superadditive on disjoint coalitions, so the surplus in braces is nonnegative. □

Minimizing the right side of (RMC) over exact covers of d defines the finite rooted functional

$$\kappa^H_{\phi,L}(d) := min_{\nu\geq 0} \left\{ \sum_Q \nu_Q [s_\phi(H \cup Q) - s_\phi(H)] : \sum_Q \nu_Q \mathbf{1}_Q = d \right\}.$$

It satisfies

$$(\alpha - \beta) s_\phi(H) \leq \kappa^H_{\phi,L}(d) \leq \kappa_{\phi,L}(d), \tag{RK}$$

and has the marginal-worth gap form

$$\kappa^H_{\phi,L}(d) = d \cdot \phi_L - \max_{\nu\geq 0} \left\{ \sum_Q \nu_Q [v(H \cup Q) - v(H)] : \sum_Q \nu_Q \mathbf{1}_Q = d \right\}. \tag{RG}$$

Hence a nonpositive rooted completion cost forces H tight. Unlike the individual marginal costs, their optimum is nonnegative whenever it is selected by a strict tight cover, by (RK).

Corollary 5.46 (rooted atomic reduction)**.** *Every slack richer Lmin suffix H selects a pure minimal semi-balanced circuit with exceptional set H, at most $|N|$ coalitions, and positive support consisting only of Lmin-tight protectors and proper rooted singletons $H\cup\{i\}$. At least one rooted singleton is not tight.*

Proof. Use singleton completion in the rooted identity above. The existence of a deficit singleton implies $|L| \geq 2$, so every $H\cup\{i\}$ used is proper. Apply the same conformal circuit decomposition as in Theorem 5.42 and choose a constituent with H as its negative element. Circuit minimality gives the support bound. If all its positive coalitions were tight, exact balancedness would force H tight, a contradiction. □

The rooted two-column atom, the universal three-poorer atom, and one factor of the three-trace atom share a single dimension-free exterior facet.

Theorem 5.47 (block-star exact-cone facet)**.** *Let*

$$N = H \dot\cup P_1 \dot\cup \cdots \dot\cup P_k, \qquad k \geq 2,$$

with every displayed block nonempty, and put $S_j = H \cup P_j$. Then

$$\sum_{j=1}^k \mathbf{1}_{S_j} - (k-1)\mathbf{1}_H = \mathbf{1}_N. \tag{BS}$$

The system with positive coalitions $S_1, \ldots, S_k$ and exceptional coalition H is a pure indecomposable minimal semi-balanced system. Hence, by the exact-cone criterion of Studený and Kratochvíl (2022), it defines a facet of the cone of exact games.

Proof. The incidence rows are constant on every displayed block. The row of P_j occurs only in S_j, so any relation modulo $\mathbf{1}_N$ forces all spoke coefficients to be equal; the H row then forces the coefficient of H. Thus (BS) is the unique relation up to scale, and deleting any supported coalition destroys it. The system is pure and minimal.

For indecomposability, suppose that a proper coalition E outside the displayed support could be a new exceptional set. Normalize its coefficient to -1, let $a_j \geq 0$ be the coefficient on S_j, let $b \geq 0$ be that on H, and let r be the common load. Equal supported incidence rows first force E to be a union of entire blocks. Write $e_H, e_j \in \{0,1\}$ for membership of H, P_j in E, and put $q = \sum_j e_j$. The peripheral and root rows give

$$a_j = r + e_j, \qquad (k-1)r + q + b - e_H = 0. \tag{BSD}$$

If $e_H = 0$ and some peripheral block is excluded, then $r \geq 0$, so (BSD) forces $r = q = b = 0$ and $E = \varnothing$. If all peripheral blocks are included, $r \geq -1$ and the left side is at least one. If $e_H = 1$ and some peripheral block is excluded, then $r \geq 0$; if $q = 0$ then $E = H$, if $q = 1$ then the equation forces $b = r = 0$ and $E = S_j$, and $q \geq 2$ is impossible. The first two sets are already supported. If all peripheral blocks are included, $E = N$. No new proper exceptional set exists, proving indecomposability. □

The block star is the zero-kernel case of a general two-sided normal form. It exposes the correct object for classifying rooted atoms without enumerating games.

Theorem 5.48 (two-sided trace-coupling normal form)**.** *Fix $N = H \dot\cup L$. Let $\mathcal{P}$ be a family of positive coalitions other than the rooted spokes $H \cup \{i\}$, and write $R_T = T \cap H$, $Q_T = T \cap L$. For nonnegative λ_T, u_i and positive γ, the rooted identity*

$$\sum_{T \in \mathcal{P}} \lambda_T \mathbf{1}_T + \sum_{i \in L} u_i \mathbf{1}_{H \cup \{i\}} - \gamma \mathbf{1}_H = \beta \mathbf{1}_N \tag{TC}$$

holds if and only if, with $\tau = \sum_i u_i$, there is $\alpha \geq 0$ such that

$$\sum_{T \in \mathcal{P}} \lambda_T \mathbf{1}_{R_T} = \alpha \mathbf{1}_H, \tag{TC-H}$$

$$\sum_{T \in \mathcal{P}} \lambda_T \mathbf{1}_{Q_T} + \sum_{i \in L} u_i e_i = \beta \mathbf{1}_L, \tag{TC-L}$$

$$\gamma = \alpha - \beta + \tau. \tag{TC-S}$$

This is a linear isomorphism between the corresponding nonnegative rooted relation and coupling cones, and therefore preserves extreme rays. In particular, every rooted circuit maps to an extreme coupling ray.

If $\mathcal{P} = \varnothing$, then $u_i = \beta$ for every $i \in L$ and $\gamma = (|L| - 1)\beta$, so the identity is precisely the block star. If $\mathcal{P} \neq \varnothing$ and $\alpha > 0$, its normalized rich traces form a balanced packet on H; if $\alpha = 0$, every member of $\mathcal{P}$ with positive coefficient is disjoint from H and the residual geometry lies entirely on L.

Proof. Restrict (TC) to the coordinates in L to obtain (TC-L). Every rooted spoke contains all of H, so restriction to H gives

$$\sum_T \lambda_T \mathbf{1}_{R_T} + (\tau - \gamma)\mathbf{1}_H = \beta \mathbf{1}_H.$$

This is (TC-H) with $\alpha = \beta + \gamma - \tau \geq 0$ and is equivalent to (TC-S). Adding the two block equations proves the converse. The correspondence and its inverse are linear. A ray of the nonnegative relation cone is nonextreme exactly when its coupling is a nontrivial sum of two such couplings, proving the extreme-ray assertion. A support-minimal fixed-sign dependence is extreme, which gives the circuit statement. The final alternatives follow directly from the two cover equations. □

Corollary 5.49 (single-packet star-or-trace reduction)**.** *Every slack richer Lmin suffix selects a rooted circuit with at most $|N|$ coalitions of one of two types:*

(i) a pure block star on the poorer singleton blocks;

(ii) a trace-coupled atom with $\alpha > 0$, whose normalized non-spoke rich traces are exactly one minimal balanced collection on H.

Thus neither a poorer-side-only coupling nor a mixture of rich packets is needed.

Proof. In the strict Lmin cover, first delete every positive-weight tight coalition disjoint from H. This leaves the common load on H unchanged and can only decrease the loads on L, so strictness is preserved. Normalize its common rich load to one. Its weights lie in the bounded exact-cover polytope of the nonempty rich traces, and hence decompose into extreme points q with coefficients θ_q. Each extreme point is a normalized minimal balanced packet. Lift it using its original tight coalitions and denote its poorer load vector by $r^q \in \mathbb{R}^L_+$. Their average $c = \sum_q \theta_q r^q$ satisfies $c_i < 1$ for every $i \in L$.

For $m = |L|$ define the canonical rooted excess

$$\delta(r) := 1 + (m-1)\max_i r_i - \sum_i r_i. \tag{CRE}$$

This function is convex. With $\beta = \max_i c_i < 1$,

$$\delta(c) = 1 + (m-1)\beta - \sum_i c_i \geq 1 - \beta > 0.$$

Jensen's inequality therefore gives $\delta(r^q) > 0$ for some packet q. Put $b = \max_i r^q_i$ and $d_i = b - r^q_i$. The packet and its canonical singleton completion satisfy

$$\sum_T \lambda^q_T \mathbf{1}_T + \sum_{i \in L} d_i \mathbf{1}_{H \cup \{i\}} - \delta(r^q)\mathbf{1}_H = b\mathbf{1}_N. \tag{CP}$$

This follows directly from Theorem 5.48: the two cover loads are $\alpha = 1$ and $\beta = b$, while $\sum_i d_i = mb - \sum_i r^q_i$.

Apply conformal circuit decomposition to (CP) and choose a constituent containing H negatively. If it has no non-spoke term, it is the block star. Otherwise its non-spoke rich traces form a nonempty balanced subcollection of the minimal balanced support of packet q, and therefore equal that full support. This proves (ii). Exact balancedness and $s_\phi(H) > 0$ also force at least one completion spoke in the circuit to have positive slack. The circuit bound is Corollary 5.46. □

The existence mechanism is more general than the preceding cover theory. Let $K \subseteq \{x \in \mathbb{R}^N : x(N) = c\}$ be nonempty, compact, and convex, and for $\phi \in K$ define the face deficit

$$\sigma_{K,\phi}(A) := \phi(A) - \min_{x \in K} x(A).$$

Theorem 5.50 (leximin face ascent and positive variation)**.** *Fix a split $N = H \dot\cup L$. For every $\phi \in K$,*

$$\max_{H \subseteq C \subseteq N} \sigma_{K,\phi}(C) = \max_{z \in K} \sum_{i \in L} (z_i - \phi_i)_+. \tag{PV}$$

Let M denote this common value. A point z maximizes the right side if and only if, with

$$D_-(z) = \{i \in L : z_i < \phi_i\}, \qquad Z(z) = \{i \in L : z_i = \phi_i\},$$

every $C = H \cup D$ satisfying $D_-(z) \subseteq D \subseteq D_-(z) \cup Z(z)$ maximizes the left side and $z(C) = \min_{x \in K} x(C)$. Conversely, every face minimizer of every left-side maximizer has this sign-cell form and maximizes the right side.

If, in addition, ϕ is the leximin point of K, every coordinate in H is strictly larger than every coordinate in L, and $\sigma_{K,\phi}(H) > 0$, then there is an $i \in L$ such that

$$\sigma_{K,\phi}(H \cup \{i\}) > \sigma_{K,\phi}(H), \tag{FA}$$

and $H \cup \{i\}$ is proper. Equivalently,

$$\max_{z \in K} \sum_{i \in L} (z_i - \phi_i)_+ > \sigma_{K,\phi}(H) = \max_{z \in K} \sum_{i \in L} (z_i - \phi_i) > 0. \tag{VG}$$

If K is a polytope, a positive-variation maximizer, and hence an exactifier of some interval-maximizing coalition, can be chosen at a vertex.

Proof. Write $C = H \cup D$ with $D \subseteq L$ and put $a = \phi - z$. Efficiency gives $a(N) = 0$. Compactness and finiteness allow the maxima to be exchanged, so

$$\begin{aligned} \max_{D} \sigma_{K,\phi}(H \cup D) &= \max_{z \in K} \max_{D \subseteq L} a(H \cup D) \\ &= \max_{z \in K} \max_{D \subseteq L} \{-a(L) + a(D)\} \\ &= \max_{z \in K} \sum_{i \in L} (-a_i)_+ = \max_{z \in K} \sum_{i \in L} (z_i - \phi_i)_+. \end{aligned}$$

For fixed z, the inner maximum is attained exactly when D contains every index with $a_i > 0$ and may choose freely among the zero indices. If z also maximizes the outer problem, the resulting coalition attains M and z attains its face minimum. Conversely, let $C = H \cup D$ attain M and let z minimize $z(C)$. For $e \in L \setminus D$, maximality and feasibility give

$$M \geq \sigma_{K,\phi}(C \cup \{e\}) \geq \phi(C \cup \{e\}) - z(C \cup \{e\}) = M + \phi_e - z_e,$$

so $z_e \geq \phi_e$; deletion gives $z_d \leq \phi_d$ for $d \in D$. Consequently D is a maximizing sign set for this z, proving the reverse correspondence and (PV).

For ascent, choose $z \in K$ minimizing $z(H)$, put $\Delta = z - \phi$, and write $s = \sigma_{K,\phi}(H) > 0$. Then $\Delta(H) = -s$ and $\Delta(L) = s$. If every Δ_i on L were nonnegative, at least one would be positive. For all sufficiently small $t > 0$, convexity keeps $\phi + t\Delta$ in K and the strict L–H payoff gap is preserved. Every poorer coordinate weakly increases and one strictly increases, so the increasingly sorted payoff vector lexicographically improves, contrary to leximin optimality. Thus $\Delta_i < 0$ for some $i \in L$, and

$$\sigma_{K,\phi}(H \cup \{i\}) \geq \phi(H \cup \{i\}) - z(H \cup \{i\}) = s - \Delta_i > s.$$

Since $\Delta(L) = s > 0$, a distinct poorer coordinate is positive, making the extension proper. The strict variation gap follows from (PV) and the ascent. Finally, a convex function on a polytope attains its maximum at some vertex: decompose any maximizer into vertices and use convexity of the positive-variation objective. □

This theorem is a support-function statement about arbitrary convex efficient allocation sets. Exactness is precisely what identifies its face deficit with ordinary coalition slack. The proof uses no balanced collection, strict cover, or finite classification. The next result retains the strict cover only because it gives a sharper cover-dependent factor.

Theorem 5.51 (strict-cover slack transfer)**.** *Let v be exact, let $\phi \in \mathrm{Core}(v)$, and write $N = H \dot\cup L$. Suppose nonnegative weights on proper ϕ-tight coalitions have common load one on H and loads $c_i < 1$ on $i \in L$. Put*

$$b = \max_{i \in L} c_i, \qquad d_i = b - c_i, \qquad \delta = 1 - b + \sum_{i \in L} d_i.$$

Then

$$\delta s_\phi(H) \le \sum_{i \in L} d_i s_\phi(H \cup \{i\}). \tag{ST}$$

If $s_\phi(H) > 0$, then $\sum_i d_i > 0$ and

$$\max_{i : d_i > 0} \frac{s_\phi(H \cup \{i\})}{s_\phi(H)} \ge \frac{\delta}{\sum_i d_i} = 1 + \frac{1 - b}{\sum_i d_i} > 1. \tag{SA}$$

Proof. With the notation of the theorem, the canonical completion identity is

$$\sum_T \lambda_T \mathbf{1}_T + \sum_{i \in L} d_i \mathbf{1}_{H \cup \{i\}} - \delta \mathbf{1}_H = b \mathbf{1}_N;$$

indeed both sides load every player by b. Apply exact balancedness to this relation; if $b = 0$, first add any positive common grand-coalition coefficient and rescale, which changes no slack. Every cover coalition T is tight at ϕ, so subtracting the worth inequality from the identical incidence equality at ϕ gives (ST). Since $b < 1$, $\delta = 1 - b + \sum_i d_i$. When H is slack, the slack-transfer inequality rules out $\sum_i d_i = 0$. Dividing by the positive sum and taking the largest completion slack proves (SA). For the selected i, the inequality $d_i > 0$ leaves a distinct maximally loaded poorer player outside $H \cup \{i\}$, so that completion is proper. □

Corollary 5.52 (universal Lmin slack ascent)**.** *Let v be exact and $\phi = \mathrm{Lmin}(v)$. At every strict payoff boundary with poorer side L and richer suffix H, either H is tight or there is an $i \in L$ such that*

$$s_\phi(H \cup \{i\}) > s_\phi(H), \qquad v(H \cup \{i\}) - v(H) < \phi_i. \tag{UA}$$

Thus every exact-partition failure has a strict local marginal-deficit witness on an adjacent edge of the coalition lattice. In particular, no slack Lmin suffix is an upper local maximum of the slack function.

Proof. Take $K = \mathrm{Core}(v)$. Exactness gives $v(A) = \min_{x \in K} x(A)$ for every coalition A, so $s_\phi(A) = \sigma_{K,\phi}(A)$. Theorem 5.50 therefore gives a proper one-player extension with strictly larger slack. Finally, the slack identity

$$s_\phi(H \cup \{i\}) - s_\phi(H) = \phi_i - \big(v(H \cup \{i\}) - v(H)\big)$$

gives the equivalent marginal inequality. Theorem 5.51 supplies the separate quantitative cover-dependent strengthening. □

The ascent cannot simply be iterated: after adjoining one poorer player the coalition need not remain a payoff suffix, so Lmin duality need not supply a new strict cover. Maximizing over the whole Boolean interval above the failed suffix bypasses that obstruction and gives a structural terminal object.

Theorem 5.53 (terminal punctured-suffix theorem)**.** *Under the hypotheses of Corollary 5.52, suppose H is slack. Choose $C = H \dot\cup D$ maximizing s_ϕ over $[H, N] = \{H \cup Q : Q \subseteq L\}$, and put $E = L \setminus D$. Then $\varnothing \neq D \subsetneq L$, C is proper, and*

$$s_\phi(C) > s_\phi(H).$$

For every $z \in \mathrm{Core}(v)$ exact on C, put $\Delta = z - \phi$ and $S = s_\phi(C)$. Then

$$S = \max_{y \in \mathrm{Core}(v)} \sum_{i \in L} (y_i - \phi_i)_+, \qquad \{i \in L : \Delta_i < 0\} \subseteq D \subseteq \{i \in L : \Delta_i \leq 0\}. \tag{TC-0}$$

At least one terminal coalition and exactifier can be chosen with the exactifier a core vertex. Moreover,

$$\Delta_d \leq 0 \quad (d \in D), \qquad \Delta_e \geq 0 \quad (e \in E), \quad \Delta(E) = S, \tag{TC-1}$$

$$v(C) - v(C \setminus \{d\}) \leq \phi_d \quad (d \in D), \quad v(C \cup \{e\}) - v(C) \geq \phi_e \quad (e \in E). \tag{TC-2}$$

Normalize any strict Lmin cover of H to rich load one, write its poorer loads as $c_i < 1$, and put $q_i = 1 - c_i$. Every such exactifier satisfies

$$\sum_{d \in D} q_d(-\Delta_d) \geq \sum_{e \in E} q_e \Delta_e > 0. \tag{TC-3}$$

Writing $q_E = \min_{e \in E} q_e$, one consequently has the terminal singleton-sacrifice budget

$$q_E s_\phi(C) \leq \sum_{d \in D} q_d s_\phi(\{d\}). \tag{TC-4}$$

In particular, there are $d \in D$ and $e \in E$ such that

$$\Delta_d < 0 < \Delta_e, \qquad \phi_d \leq \phi_e. \tag{TC-5}$$

Thus the terminal slack continuation of a failed suffix is payoff-inverted: it contains a losing poorer player while excluding a weakly richer gaining poorer player.

Proof. Corollary 5.52 gives an $i \in L$ with $s_\phi(H \cup \{i\}) > s_\phi(H)$. Since $s_\phi(N) = 0$, every interval maximizer has the asserted form with nonempty proper $D \subset L$, and its slack S is strictly larger than the root slack. Theorem 5.50, applied to $K = \mathrm{Core}(v)$, gives (TC-0) and the core-vertex choice because exactness identifies face deficit with coalition slack.

For $d \in D$ and $e \in E$, interval maximality gives $s_\phi(C \setminus \{d\}) \leq S$ and $s_\phi(C \cup \{e\}) \leq S$. Expanding the two slack differences proves (TC-2). Exactness supplies a core point z with $z(C) = v(C)$. Core feasibility on the two neighboring coalitions gives

$$z_d = z(C) - z(C \setminus \{d\}) \leq v(C) - v(C \setminus \{d\}) \leq \phi_d,$$

and

$$z_e = z(C \cup \{e\}) - z(C) \geq v(C \cup \{e\}) - v(C) \geq \phi_e.$$

Also $\Delta(C) = -S$ and efficiency gives $\Delta(E) = S > 0$, proving (TC-1) and a strict gain in E.

Let λ be any normalized strict cover of H. Every supported coalition is tight at ϕ, so $\Delta(T) \geq 0$. Summing and using efficiency gives

$$0 \leq \sum_T \lambda_T \Delta(T) = \Delta(H) + \sum_{i \in L} c_i \Delta_i = -\sum_{i \in L} q_i \Delta_i.$$

Separating D and E proves (TC-3); its right side is positive because every $q_e > 0$ and some $\Delta_e > 0$. Hence some member of D loses strictly. Moreover, $\Delta(E) = S$ makes the weighted gain at least $q_E S$, while singleton core feasibility gives $-\Delta_d \leq \phi_d - v(\{d\}) = s_\phi(\{d\})$. Substitution in (TC-3) proves (TC-4) without a normalization assumption.

Finally, if every losing member of D had strictly larger Lmin payoff than every gaining member of E, then along a sufficiently short segment from ϕ toward z all strict payoff-stratum gaps would be preserved. The first affected payoff stratum would contain a gain and no loss, so its sorted payoff vector would lexicographically improve, contrary to $\phi = \mathrm{Lmin}(v)$. This proves (TC-5). □

The theorem does not classify all terminal cuts, but it replaces an invalid iteration heuristic by a dimension-free exact-face object with a forced sign pattern, a cover-deficit budget, and a payoff inversion. The pure-star, three-trace, and four-co-singleton mechanisms saturate (TC-3). A standalone proof, hostile-review prompt, and exact audit are

`paper3/TERMINAL_SLACK_CUT_THEOREM.md`,
`paper3/TERMINAL_SLACK_CUT_EXTERNAL_REVIEW_PROMPT.md`,
`src/verify_terminal_slack_cut.py`,
`results/terminal_slack_cut.json`.

Corollary 5.54 (two-poorer canonical slack amplifier)**.** *At a slack Lmin boundary with exactly two poorer players, label them so that balanced-trace transport selects a lifted minimal packet with $0 \leq r_a < r_b < 1$. Then*

$$s_\phi(H \cup \{a\}) \geq \frac{1 - r_a}{r_b - r_a} s_\phi(H) > s_\phi(H). \tag{2P}$$

The amplified coalition adjoins the losing poorer player and omits the unique gaining poorer player.

Proof. Here $d_a = r_b - r_a$, $d_b = 0$, and $\delta = 1 - r_a$, so the claim is the specialization of Theorem 5.51. Equivalently, exactify H at $z \in \mathrm{Core}(v)$ and write $\Delta = z - \phi$. Balanced-trace transport gives $\Delta_a < 0 < \Delta_b$ and $\Delta_a + \Delta_b = s_\phi(H)$. Hence

$$z(H \cup \{a\}) = \phi(H \cup \{a\}) - \Delta_b,$$

so core feasibility already yields $s_\phi(H \cup \{a\}) \geq \Delta_b > s_\phi(H)$; the packet relation gives the displayed quantitative factor. □

Corollary 5.55 (local marginal-deficit witness)**.** *Under the hypotheses of Corollary 5.54,*

$$v(H \cup \{a\}) - v(H) \leq \phi_a - \frac{1 - r_b}{r_b - r_a} s_\phi(H) < \phi_a. \tag{MD}$$

Thus the universal local witness has the displayed sharp packet-dependent margin when there are exactly two poorer players.

Proof. The slack difference on that edge is

$$s_\phi(H \cup \{a\}) - s_\phi(H) = \phi_a - \big(v(H \cup \{a\}) - v(H)\big).$$

Subtracting one from the amplification factor in (2P) gives $(1 - r_b)/(r_b - r_a)$ and proves the claim. □

Thus the general object is not a bare list of exact-cone facets. It is the polyhedral cone coupling a balanced rich-trace packet to a singleton-completed poor-trace packet. The three-trace atom is rich-active: its non-spoke traces form the internal triangle. The rooted broken diamond and the universal block-star identities have empty coupling kernel. A dimension-free classification or forbidden-minor theorem for extreme coupling rays is the remaining general-theory target. A standalone proof and exact audit are

`paper3/TWO_SIDED_TRACE_COUPLING.md`,
`paper3/UNIVERSAL_LMIN_SLACK_ASCENT.md`,
`paper3/UNIVERSAL_LMIN_SLACK_ASCENT_EXTERNAL_REVIEW_PROMPT.md`,
`src/verify_two_sided_trace_coupling.py`,
`results/two_sided_trace_coupling.json`.

Thus raw exact-cone facets are not the right atoms for Lmin failure: the egalitarian orientation first forces a decomposable singleton completion. Rooting quotients out that forced triangle while preserving the distinguished suffix. In the five-player broken diamond the resulting rooted circuit is indecomposable; in the seven-player three-trace family it has exactly one further decomposition. Thus rooting is a genuine bridge to the facet theory but not a universal facet correspondence. The later Theorem 5.108 shows that the residual is exactly a sum of two saturated indecomposable facets. The standalone proof, scope calibration, and exact probes are in

`paper3/ATOMIC_LMIN_OBSTRUCTION_REDUCTION.md`,
`paper3/MARGINAL_CORE_FACTORIZATION.md`,
`paper3/ROOTED_DEFICIT_COMPLETION.md`,
`src/verify_atomic_lmin_obstruction.py`,
`results/atomic_lmin_obstruction.json`,
`src/probe_lmin_atom_indecomposability.py`,
`results/lmin_atom_indecomposability_probe.json`,
`src/verify_rooted_deficit_completion.py`,
`results/rooted_deficit_completion.json`,
`paper3/BLOCK_STAR_FACET_THEOREM.md`,
`src/verify_block_star_facet.py`,
`results/block_star_facet.json`.

The same inequality has a useful internal-face specialization. It is the common budget law behind both the unique-richest threshold and the later trace-rounding exclusions.

Corollary 5.56 (exact-face overload budget)**.** *Let v be a singleton-normalized exact game, let $\phi \in \mathrm{Core}(v)$, and let $\varnothing \neq D \subseteq H \subsetneq N$. Suppose nonnegative real weights on proper ϕ-tight coalitions have incidence loads*

$$c_h = 1 + q\mathbf{1}_{\{h \in D\}} \quad (h \in H), \qquad 0 \le c_i \le 1 \quad (i \in L := N \setminus H) \tag{OF}$$

for some $q > 0$. Then

$$q\, s_\phi(D) \le \sum_{i \in L} (1 - c_i)\phi_i. \tag{OB}$$

In particular, if the cover loads every player outside H once, then D is tight at ϕ.

Proof. Apply Lemma 5.40 with target coalition D, $\alpha = 1 + q$, and $\beta = 1$. The players in $H \setminus D$ have load one, the players in L have load at most one, and the completion terms on D and $H \setminus D$ vanish. The resulting inequality is (OB). If every poorer load is one, its right side is zero; core feasibility gives $s_\phi(D) \geq 0$. □

Equivalently, exactify D at a core point x. Summing the tight-coalition inequalities gives $x(H) + qv(D) + \sum_{i \in L} c_i x_i$ on the left. Singleton normalization makes $x_i \geq 0$, so replacing the poorer loads by one can only increase that side; efficiency then gives (OB). This direct proof emphasizes that the overloaded set may be any rich coalition, not merely one player, and that the cover weights need not be integral.

For a fixed family $\mathcal{T}$ of ϕ-tight coalitions, normalize $\alpha - \beta = 1$ in (BC) and minimize its right side over $\lambda_T \geq 0$ and $\beta \geq 0$, subject to $c_h \leq \beta + 1$ on H and $c_i \leq \beta$ on L. Denote this finite linear-program value by $\Delta_{\phi,\mathcal{T}}(H)$; taking all weights zero shows $\Delta_{\phi,\mathcal{T}}(H) \leq \phi(H)$.

Corollary 5.57 (boundary-defect bound)**.** *Under the hypotheses of Lemma 5.40,*

$$0 \leq s_\phi(H) \leq \Delta_{\phi,\mathcal{T}}(H) \leq \phi(H). \tag{BD}$$

In particular, a zero-defect tight-coalition cover forces H to be tight. If $\phi = \mathrm{Lmin}(v)$ and such a cover exists at every strict payoff boundary, then v is an exact partition game witnessed at ϕ.

Proof. The middle inequality is Lemma 5.40 for every feasible cover, followed by minimization. Core feasibility gives the first inequality, and the zero-weight cover gives the last. The final assertion applies the zero-defect conclusion to every cumulative richer suffix. □

The following transfer-blocking property is not new: it is the egalitarian-core characterization of Arin and Iñarra (2001), applied to Lmin. We use "protector coalition" only as a descriptive label. The new use below is to combine this published property with exact-game singleton witnesses, replacing the finite stratum enumeration used in an earlier version.

Lemma 5.58 (protector coalition)**.** *Let $\phi = \mathrm{Lmin}(v)$, and order its payoff strata $B_1, \dots, B_k$ from lowest to highest. If $i \in B_j$ and $h \in B_{j+1} \cup \cdots \cup B_k$, then some coalition T is tight at ϕ and satisfies*

$$h \in T, \qquad i \notin T. \tag{PC}$$

Proof. The claim follows directly from the transfer-blocking characterization of the egalitarian core in Arin and Iñarra (2001), since Lmin belongs to that set. For completeness, the following LP-dual derivation also checks the paper's inequality conventions. The face is indexed by final payoff strata rather than by possibly finer algorithmic pinning substeps; tied players that pin at different degenerate substeps therefore cause no ambiguity. Put $\alpha = \phi_i$ and $P = B_1 \cup \cdots \cup B_{j-1}$. On the stratum face

$$F = \{x \in \mathrm{Core}(v) : x_p = \phi_p \ (p \in P), \quad x_\ell \geq \alpha \ (\ell \notin P)\},$$

the maximum of x_i is α. Indeed, if $x_i > \alpha$ at some $x \in F$, then a sufficiently short segment from ϕ toward x keeps every richer coordinate strictly above α, keeps all coordinates in B_j at least α, and raises coordinate i. This lexicographically improves ϕ.

Apply LP duality to $\max_{x \in F} x_i$. Let $\lambda_S \geq 0$ multiply $v(S) - x(S) \leq 0$, let $\mu_\ell \geq 0$ multiply $\alpha - x_\ell \leq 0$, and write $w_q = \sum_{S \ni q} \lambda_S$. Stationarity at a coordinate outside P is

$$-\mathbf{1}_{\{q=i\}} - w_q - \mu_q + \nu = 0.$$

Coordinates in P instead carry free equality multipliers and impose no additional restriction on this coverage comparison. The floor of the richer player h is strictly inactive, so $\mu_h = 0$. Consequently

$$w_h - w_i = 1 + \mu_i > 0.$$

Some positive-multiplier coalition contains h and omits i; complementary slackness makes it tight at ϕ. □

Lemma 5.59 (fractional protector-packing obstruction)**.** *Let v be a normalized exact game and let $\phi = \mathrm{Lmin}(v)$ have a unique richest player r, with $\phi_r = \gamma$. Put $L = N \setminus \{r\}$ and $q_i = \phi_i$ for $i \in L$. For each $i \in L$, Lemma 5.58 supplies a tight protector $\{r\} \cup Q_i$ omitting i. It has $\emptyset \neq Q_i \subseteq L \setminus \{i\}$ because $\gamma > 0 = v(\{r\})$, so the singleton $\{r\}$ is slack. Fix any such choices. If $\lambda_i \geq 0$ and*

$$c_j := \sum_{i : j \in Q_i} \lambda_i \leq 1 \quad (j \in L), \qquad \Lambda := \sum_{i \in L} \lambda_i,$$

then

$$\gamma(\Lambda - 1) \leq \sum_{j \in L} (1 - c_j) q_j. \tag{FP}$$

Proof. If $\Lambda \leq 1$, the left side of (FP) is nonpositive and the right side is nonnegative. If $\Lambda > 1$, apply Corollary 5.56 with $H = D = \{r\}$, $q = \Lambda - 1$, and the weighted tight protectors $T_i = \{r\} \cup Q_i$. Their poorer loads are the c_j, while $s_\phi(\{r\}) = \phi_r - v(\{r\}) = \gamma$. The overload budget is exactly (FP). □

The packing inequality isolates a sharp sacrifice-budget threshold that is already visible before the full exact-partition classification.

Theorem 5.60 (indecomposable separation family)**.** *In a singleton-normalized exact game with $2 \leq |N| \leq 4$, Lmin cannot have a unique richest player. More generally, fix integers $h \geq 1$ and $m \geq h + 3$. There is a four-generator integral exact game on a poorer block P of size m and a richer block R of size h with*

$$\mathrm{Lmin}(v)_i = \begin{cases} h, & i \in P, \\ m - 2, & i \in R, \end{cases} \qquad v(R) = 0.$$

For the same game, putting

$$a = \frac{h^2(m-1)}{(h+1)(m-2)}, \qquad c = \frac{h(m-1)}{h+1},$$

one has

$$\mathrm{Lmax}(v)_i = \begin{cases} c, & i \in R \cup \{3, 4\}, \\ a, & i \in P \setminus \{3, 4\}. \end{cases}$$

Their ℓ^1 distance is

$$\|\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1 = \frac{4h(m-h-2)}{h+1}.$$

Every game in the family is indecomposable, meaning that it admits no nontrivial direct-sum partition. Normalizing the distance by the common total $v(N) = 2h(m-1)$ gives

$$\frac{\|\mathrm{Lmin}(v) - \mathrm{Lmax}(v)\|_1}{v(N)} = \frac{2(m-h-2)}{(h+1)(m-1)}.$$

In particular, at $h = 1$ the normalized distance is $(m-3)/(m-1) \to 1$. Thus five is the sharp unique-richest threshold, and an indecomposable four-generator family has growing Lmin–Lmax separation rather than merely preserving a fixed example under null-player padding.

Proof. Suppose r is uniquely richest, put $L = N \setminus \{r\}$, $\gamma = \phi_r$, and $q_i = \phi_i < \gamma$. Choose the protectors in Lemma 5.59. If $|L| = 1$, no nonempty Q_i exists. If $|L| = 2$, both Q_i are the opposite singleton; weight both by one in (FP), giving $\gamma \le 0$.

Let $|L| = 3$. If some $Q_i = \{j\}$, weight the protectors indexed by i and j by one. The sets Q_i and Q_j are nonempty and disjoint, so their union leaves at most one player u uncovered. Inequality (FP) gives either $\gamma \le 0$ or $\gamma \le q_u < \gamma$. If no Q_i is a singleton, then $Q_i = L \setminus \{i\}$ for all i; weight all three by $1/2$. Every poorer load is one and $\Lambda = 3/2$, forcing $\gamma/2 \le 0$. This proves the four-player bound.

For the family, let $P = \{1, \dots, m\}$ and let R be a disjoint set of h players. Write e_i for unit vectors and $\mathbf{1}_R = \sum_{r \in R} e_r$. Take the lower envelope of

$$\begin{aligned} g^1 &= h(m-1)e_4 + (m-1)\mathbf{1}_R, \\ g^2 &= h(m-1)(e_3 + e_4), \\ g^3 &= h(e_1 + e_2 + e_3) + h(m-4)e_5 + (m-1)\mathbf{1}_R, \\ g^4 &= h\mathbf{1}_P + (m-2)\mathbf{1}_R, \qquad v(S) := \min_{1 \le k \le 4} g^k(S). \end{aligned} \tag{UF}$$

At the endpoint $m = 4$, the zero term $h(m-4)e_5$ is simply omitted; player 5 then belongs to R. Every row has total $2h(m-1)$ and is a core allocation of the lower-envelope game; every coalition attains its worth at a row. Hence v is exact directly from the definition. Put $\phi = g^4$. The four rows give

$$v(N \setminus \{1\}) = v(N \setminus \{2\}) = 2h(m-1) - h, \qquad v(R \cup \{3\}) = v(R \cup \{4\}) = h(m-1). \tag{UF-T}$$

The point ϕ gives every player at least h. At any core point with all coordinates at least h, the first two displayed constraints and efficiency force $x_1 = x_2 = h$. Write $y = x - h\mathbf{1}_N$. Then $y \ge 0$, $y_1 = y_2 = 0$, and $y(N) = h(m-h-2)$. The last two constraints give

$$y_3 + y(R) \ge h(m-h-2), \qquad y_4 + y(R) \ge h(m-h-2).$$

Comparing each with the total forces $y_i = 0$ for every $i \in P$. Thus the poorer block is pinned at the first Lmin level h. The richer block has fixed total $h(m-2)$, so its next common level is at most $m-2$, attained by ϕ. This proves the Lmin formula. Since $g^2(R) = 0$, the richest suffix R has slack $h(m-2)$.

It remains to identify Lmax. Every core point satisfies $x_3 + x(R) \ge h(m-1)$ and $x_4 + x(R) \ge h(m-1)$ by (UF-T). Hence its largest coordinate is at least $c = h(m-1)/(h+1)$. Put

$$t = \frac{h(m-1)}{(h+1)(m-2)}.$$

The hypothesis $m \ge h+3$ gives $0 < t < 1$. The core point

$$z = tg^4 + (1-t)g^2$$

has coordinates $z_i = c$ on $R \cup \{3,4\}$ and $z_i = a$ otherwise. Thus c is the minimum possible largest coordinate. On the face where every coordinate is at most c, the two displayed inequalities force

every coordinate in $R \cup \{3,4\}$ to equal c. The remaining $m-2$ coordinates have fixed total and average a; the displayed point attains that common ceiling. This proves the Lmax formula and its uniqueness. Direct subtraction gives the displayed ℓ^1 distance.

It remains to prove indecomposability. In a singleton-normalized direct sum, two players in different summands have pair worth zero. Directly from (UF),

$$v(\{1,4\}) = v(\{2,4\}) = v(\{3,4\}) = h > 0,$$
$$v(\{3,r\}) = v(\{4,r\}) = m-1 > 0 \quad (r \in R),$$

and, when $m \geq 5$, $v(\{4,5\}) > 0$. Thus players $1,\dots,5$ and all of R lie in one summand (with the evident $m = 4$ interpretation). If $m = 6$, put $T = \{3,4,5\} \cup R$; then $v(T) = 7h$ but $v(T \cup \{6\}) = 8h$. If $m \geq 7$, put $T = \{4,5\}$; for every $j \in P \setminus \{1,\dots,5\}$,

$$v(T) = 2h, \qquad v(T \cup \{j\}) = 3h.$$

Since every such T is already contained in the first summand and $v(\{j\}) = 0$, these strict interactions rule out placing any remaining poorer player in another summand. Hence the game is indecomposable.

The escape saturates the exact-balanced circuit

$$\mathbf{1}_{R\cup\{3\}} + \mathbf{1}_{R\cup\{4\}} + \sum_{i \in P\setminus\{3,4\}} \mathbf{1}_{\{i\}} - \mathbf{1}_R = \mathbf{1}_N. \tag{UF-C}$$

The two tight protector slacks are zero, the $m-2$ uncovered poorer singleton slacks are h each, and the negative richest-suffix slack is $h(m-2)$. When $h = 1$, this is equivalently equality in (FP). In that case, only three poorer players leave at most one uncovered payoff, strictly less than the unique richest payoff; from five players onward, the family supplies exactly the sacrifice budget needed for equality. The two-parameter audit, unique-richest verifier, and fixed five-player replay are recorded in the following paired scripts and artifacts:

- `src/verify_rich_block_family.py` and `results/rich_block_family.json`;
- `src/verify_unique_richest_family.py` and `results/unique_richest_family.json`;

The fixed five-player replay is indexed in the accompanying structural note. □

The qualifier *indecomposable* is essential to the two-parameter statement. A direct sum of h copies of the five-player $h = 1$ game trivially produces h tied rich players and keeps normalized distance $1/3$; null-player padding likewise preserves dimensional existence. Neither operation yields the $h = 1$ family's growing normalized distance $(m-3)/(m-1) \to 1$. For fixed h, the displayed family instead tends to $2/(h+1)$; no uniform claim is made as $h \to \infty$.

The closest published low-dimensional antecedent is that exact games with at most four players are PES stable (Dietzenbacher 2021); see also the large-core route in Biswas et al. (1999). That solution-stability statement does not imply exact partition at Lmin, because PES stability is strictly weaker than the exact-partition tightness condition in general. The next lemma supplies the stronger coalitional statement and its sharp structural explanation.

Lemma 5.61 (exact-partition classification through four players)**.** *Let v be an exact game with $|N| \leq 4$ and $v(\{i\}) = 0$ for every player. Order the payoff strata of* $\mathrm{Lmin}(v)$ *as $B_1,\dots,B_k$ from lowest to highest. At every strict gap, the richer suffix is tight:*

$$H_j := B_{j+1} \cup \cdots \cup B_k, \qquad \mathrm{Lmin}(v)(H_j) = v(H_j) \quad (j < k). \tag{RST}$$

Equivalently, v is an exact partition game witnessed at its Lmin allocation. Consequently $\mathrm{Lmin}(v)$ *is the Lorenz-dominating core allocation and equals* $\mathrm{Lmax}(v)$.

Proof. Put $\phi = \mathrm{Lmin}(v)$ and consider a strict payoff boundary with richer suffix H and poorer side $L = N \setminus H$. If $|H| = 1$, its member is the unique richest player, impossible through four players by Theorem 5.60. Hence $|H| \geq 2$, so $|L| \leq 2$. The human two-poorer boundary theorem proved below, Theorem 5.105, now makes H tight (and also covers $|L| = 1$). This proves (RST); the one-stratum case is immediate. Thus the dimension-free balanced-trace transport obstruction, rather than a four-player payoff-stratum case split, explains the exact-partition threshold.

For completeness, here is the connection with the published exact-partition result in the present ordering convention. Take any $x \in \mathrm{Core}(v)$ and let H consist of the r richest coordinates of ϕ, at a stratum boundary. Then

$$\sum_{i=1}^{r} x_i^{\downarrow} \geq x(H) \geq v(H) = \phi(H) = \sum_{i=1}^{r} \phi_i^{\downarrow}.$$

Write $U_x(r) = \sum_{i=1}^{r} x_i^{\downarrow}$. Between consecutive boundaries, U_x is concave in r, whereas U_ϕ is affine because the intervening coordinates of ϕ are tied. Endpoint dominance therefore gives $U_x(r) \geq U_\phi(r)$ throughout the block. Thus ϕ Lorenz-dominates the core, as in Llerena and Mauri (2017). If y denotes Lmax, Lorenz dominance and leximax minimality force $y^{\downarrow} = \phi^{\downarrow}$; uniqueness of the leximax-minimal point of the compact convex core then gives $y = \phi$. □

The five-player certificate also identifies where this mechanism stops. Its failed boundary has three poorer and two richer players, with richer pair $\{3, 5\}$ slack. The only tight nontrivial proper coalitions at the base Lmin point are $\{3, 4, 5\}$ and $\{1, 3, 5\}$: they protect the three poorer players while retaining both richer players. More precisely, their incidence vectors satisfy the exact-balanced identity

$$\mathbf{1}_{\{1,3,5\}} + \mathbf{1}_{\{3,4,5\}} + \mathbf{1}_{\{2\}} - \mathbf{1}_{\{3,5\}} = \mathbf{1}_N. \tag{E5}$$

At the base Lmin allocation the two protector slacks are zero, the singleton $\{2\}$ has slack one, and the richer suffix $\{3, 5\}$ has slack one. Thus (SB) is attained at equality: the fifth player supplies exactly the unit of positive singleton slack that funds the unit of richer-suffix slack. Equivalently, (BC) is tight with $\alpha = 2$ and $\beta = 1$; the sole poorer-side coverage deficit is player 2. This exact calculation is reproduced by `src/verify_exact_balanced_slack_budget.py`, with output in `results/exact_balanced_slack_budget.json`. The verifier also solves the boundary-defect program in exact rational arithmetic. The displayed cover has defect one, while (BD) gives the matching lower bound from the unit suffix slack, so the program is sharp in this example.

In particular, Lemma 5.59 is a unique-richest obstruction and does not apply; exactness of a singleton can zero one rich player, but not the whole rich side. This explains the sharp example's escape without claiming that every higher-dimensional obstruction has this form.

The number of players below a failed boundary has its own sharp threshold. The first step is a local dual consequence of the progressive Lmin stages.

The qualitative tight-coalition optimality criterion behind this step is prior art. Arin et al. (2008, Lemma 1) characterize Lmax by their *property I*: a hierarchical system built from the ordered payoff partition and the coalitions tight at the candidate has only the zero feasible direction. Under the standard antiduality between Lmax and Lmin, this becomes a property-I characterization of Lmin using the reversed payoff partition and complement-tight coalitions (Mauri 2018, Lemma 3). The lemma below is not claimed as a new characterization of Lmin. It records the quantitative certificate needed here: one nonnegative combination of the original tight-coalition incidence vectors that is regular on an entire richer suffix and simultaneously

strictly deficient at every poorer player. The explicit LP dual and backward combination show how that certificate follows from the same lexicographic optimality structure; the subsequent projection to rich-side traces is the additional step used in this paper.

Lemma 5.62 (strict richer-side cover)**.** *Let $\phi = \mathrm{Lmin}(v)$ and let a strict payoff boundary have nonempty poorer side L and richer suffix H. There are nonnegative weights λ_T on proper coalitions tight at ϕ and a number $\alpha > 0$ such that, with*

$$c_i := \sum_{T \ni i} \lambda_T,$$

one has

$$c_h = \alpha \quad (h \in H), \qquad c_i < \alpha \quad (i \in L). \tag{SC}$$

Proof. At a progressive Lmin stage, hold all earlier coordinates fixed, retain the common floor on the current and later players, and maximize one player i in the current stratum. Because i is finalized at that level, the optimum is ϕ_i: if a feasible point raised i, a short segment from ϕ toward it would keep every later stratum strictly above the current level and would lexicographically improve the current tied block.

Here is the dual calculation explicitly. Put P for the earlier players, $\ell = \phi_i$, and minimize $-x_i$ subject to

$$v(S) - x(S) \le 0 \quad (\emptyset \ne S \subsetneq N), \qquad \ell - x_q \le 0 \quad (q \notin P),$$

together with $x_p = \phi_p$ for $p \in P$ and $x(N) = v(N)$. Let $\lambda_S, \mu_q \ge 0$ be the two families of inequality multipliers, let ν_i be the free efficiency multiplier, and put $w_q = \sum_{S \ni q} \lambda_S$. Stationarity at $q \notin P$ is

$$-\mathbf{1}_{\{q=i\}} - w_q - \mu_q + \nu_i = 0. \tag{SC-D}$$

Every later player's floor is strictly inactive at ϕ, hence $w_q = \nu_i$ there. At the target, $w_i = \nu_i - 1 - \mu_i \le \nu_i - 1$, and every other player in the current stratum has $w_q = \nu_i - \mu_q \le \nu_i$. Complementary slackness restricts positive λ_S to ϕ-tight coalitions. Earlier coordinates carry free equality multipliers and impose no load restriction. Finally $w_i \ge 0$ implies $\nu_i \ge 1$.

Write the poorer payoff strata as $B_1, \dots, B_j$, from lowest to highest. For each t, add the individual-pinning duals of all $i \in B_t$ and call the resulting cover D_t. It has a common load η_t on every later player, and every $i \in B_t$ has load at most $\eta_t - 1$: its own dual contributes the unit deficit and every other dual in the stratum contributes at most its common later load. Loads on earlier strata are finite but unrestricted.

Combine the D_t backwards. Start with any positive multiple of D_j. Suppose a combination of $D_{t+1}, \dots, D_j$ has common load A on H and finite loads r_i on B_t. Adding MD_t changes the common rich load to $A + M\eta_t$, while the load on $i \in B_t$ is at most $r_i + M(\eta_t - 1)$. Its rich-side deficit is at least $A - r_i + M > 0$ for all sufficiently large M. Players in later poorer strata receive the same added load as H, so their already strict deficits are preserved. Backward induction gives one cover satisfying (SC). Since every target load is nonnegative and at most $\nu_i - 1$, each $\nu_i \ge 1$, so the final α is positive. □

Theorem 5.63 (laminar strict-cover domain)**.** *Let v be exact, let $\phi \in \mathrm{Core}(v)$, and let $\varnothing \ne H \subsetneq N$. Suppose nonnegative weights on proper ϕ-tight coalitions have a common positive load on H, strictly smaller load on every player in $N \setminus H$, and laminar positive support. Then H is tight at ϕ.*

Proof. Let $\mathcal{F}$ be the positive support of the cover. For each $h \in H$, choose an inclusion-minimal $M_h \in \mathcal{F}$ containing h. If $i \in M_h \cap L$, every supported coalition containing h also contains i: it intersects M_h, so laminarity makes the two sets nested, and minimality rules out a strict supported subset of M_h containing h. The load of i would therefore be at least the common load of h, contrary to the strict poorer-side deficit. Hence $M_h \subseteq H$.

The inclusion-maximal distinct members of $\{M_h : h \in H\}$ are pairwise disjoint and cover H. They form a partition $\mathcal{P}$ of H by ϕ-tight coalitions. If $\mathcal{P} = \{H\}$, the conclusion is immediate. Otherwise, give each $B \in \mathcal{P}$ weight one, give H weight minus one, and give N weight one. The incidence sum is $\mathbf{1}_N$, with H the only negative term, so exact balancedness yields

$$\sum_{B \in \mathcal{P}} v(B) - v(H) + v(N) \leq v(N).$$

Tightness of the blocks gives $\phi(H) \leq v(H)$, while core feasibility gives the reverse inequality. Thus H is tight. □

Corollary 5.64 (laminar Lmin domain)**.** *If a singleton-normalized exact game admits a laminar strict richer-side cover at every strict Lmin boundary, it is exact partition witnessed at Lmin. In particular, this holds when its complete proper Lmin-tight family is laminar.*

Proof. Apply Lemma 5.62 and Theorem 5.63 at every boundary. If the complete tight family is laminar, the lemma's cover has laminar support. □

Thus every slack Lmin suffix forces crossing coalitions in the support of *every* strict richer-side cover. This is a boundary-local, dimension-free domain theorem, not merely a cardinality exclusion. The known five-player failure escapes exactly through its two crossing protector coalitions. A standalone proof and finite audit are

`paper3/LAMINAR_LMIN_DOMAIN_THEOREM.md`,
`src/verify_laminar_lmin_domain.py`,
`results/laminar_lmin_domain.json`.

The laminar condition admits a substantially broader uncrossing form. Say that v is *Lmin-locally supermodular* if, for $\phi = \mathrm{Lmin}(v)$, every pair A, B tight at ϕ satisfies

$$v(A) + v(B) \leq v(A \cap B) + v(A \cup B). \tag{LSM}$$

Unlike convexity, this condition is imposed only on coalitions binding at one canonical core allocation.

Theorem 5.65 (exact-face uncrossing)**.** *Let v be exact, let $\phi \in \mathrm{Core}(v)$, and let $\varnothing \neq H \subsetneq N$. Suppose nonnegative weights on proper ϕ-tight coalitions have a common positive load on H and strictly smaller load on every player outside H. If the ϕ-tight family is closed under union and intersection, then H is tight at ϕ. Conversely, if H is slack, there are crossing ϕ-tight coalitions A, B with*

$$v(A) + v(B) - v(A \cap B) - v(A \cup B) = s_\phi(A \cap B) + s_\phi(A \cup B) > 0. \tag{UD}$$

The pair may be chosen from a minimum-energy representation of a strict cover with the prescribed player loads.

Proof. First observe that if tight A, B satisfy (LSM), then both $A \cap B$ and $A \cup B$ are tight. Indeed,

$$v(A) + v(B) \le v(A \cap B) + v(A \cup B) \le \phi(A \cap B) + \phi(A \cup B) = \phi(A) + \phi(B) = v(A) + v(B),$$

so all inequalities are equalities and both nonnegative slacks vanish.

Retain the complete player-load vector c of the assumed cover. Among all nonnegative representations of c by nonempty ϕ-tight coalitions, now allowing N, minimize

$$\mathcal{E}(\lambda) := \sum_S \lambda_S |S|(n - |S|). \tag{UE}$$

The representation set is nonempty and compact: after the empty coalition is discarded, every coefficient is bounded by a coordinate of c.

If positive-support coalitions A, B cross, put $\delta = \min\{\lambda_A, \lambda_B\}$. Subtract δ from their weights and add it to $A \cap B$ and $A \cup B$, discarding an empty intersection. The incidence identity $\mathbf{1}_A + \mathbf{1}_B = \mathbf{1}_{A\cap B} + \mathbf{1}_{A\cup B}$ preserves c, while tight-pair closure preserves feasibility. But the energy falls by

$$2\delta |A \setminus B|\, |B \setminus A| > 0, \tag{UE-D}$$

a contradiction. Thus the minimizing support is laminar. If it contains N with weight γ, remove that term. Every load is at least γ; the residual rich load is positive because each poorer load was strictly below the original rich load, and all rich–poor gaps are unchanged. The remaining proper support is a laminar strict cover, so Theorem 5.63 makes H tight. Applying this at every boundary proves the first assertion.

For the converse, minimize (UE) without assuming uncrossing closure. If H is slack, the minimizing support cannot be laminar, by Theorem 5.63; choose a crossing positive pair A, B. Their intersection and union cannot both be tight, or the same uncrossing move would lower the energy. Modularity of ϕ and tightness of A, B give the equality in (UD); at least one derived-set slack is positive. □

The broken-diamond obstruction has a quantitative conservation law. For a core point ϕ, write

$$D_\phi(A, B) := v(A) + v(B) - v(A \cap B) - v(A \cup B).$$

Theorem 5.66 (uncrossing-defect budget)**.** *Let v be exact, let $\phi \in \mathrm{Core}(v)$, and let $\varnothing \neq H \subsetneq N$. Suppose an integer multiset of proper ϕ-tight coalitions has common load $k > 0$ on H and load at most $k - 1$ on $N \setminus H$. Repeatedly uncross a positive pair A_t, B_t, with δ_t equal to the smaller multiplicity, until the support is laminar. From the final multiplicities μ, extract the partition $\mathcal{P}$ of H formed by the maximal minimal supported sets containing its players, and put $m = \min_{B\in\mathcal{P}} \mu_B$. Then*

$$0 \le m s_\phi(H) \le \sum_S \mu_S s_\phi(S) = \sum_t \delta_t D_\phi(A_t, B_t). \tag{UDB}$$

The procedure terminates after finitely many steps.

Proof. Each uncrossing preserves player loads and lowers the integer energy $\sum_S \lambda_S |S|(n - |S|)$ by $2\delta_t |A_t \setminus B_t|\, |B_t \setminus A_t| > 0$. Thus the procedure terminates with laminar support. The minimal-set argument from Theorem 5.63 applies without tightness of the final sets: if a minimal set

containing $h \in H$ also contained $i \notin H$, laminarity would make the final load of i at least k. Hence the maximal minimal sets form a partition $\mathcal{P}$ of H.

Initially the weighted slack is zero. One uncrossing changes it by

$$\delta_t\{s_\phi(A_t \cap B_t) + s_\phi(A_t \cup B_t) - s_\phi(A_t) - s_\phi(B_t)\} = \delta_t D_\phi(A_t, B_t),$$

so telescoping proves the equality in (UDB). Final slacks are nonnegative and every block of $\mathcal{P}$ has multiplicity at least m. Exact balancedness applied with weight one on the partition blocks, minus one on H, and plus one on N gives $v(H) \geq \sum_{B\in\mathcal{P}} v(B)$, hence $s_\phi(H) \leq \sum_{B\in\mathcal{P}} s_\phi(B)$. Multiplying by m proves the inequalities. □

Thus local supermodularity defects do not merely witness failure: their signed cumulative total exactly equals the weighted slack created by uncrossing and must finance the target-face slack. Any rational strict cover can be scaled to the integer form above; in particular, the strict-cover polyhedron at an Lmin boundary is rational. The sharp five-player cover has cumulative defect two and target slack one. The three-column seven-player cover uncrosses in two steps with defects 3 and 3/2, again funding unit target slack. A standalone proof and exact audit are

`paper3/UNCROSSING_DEFECT_BUDGET.md`,
`src/verify_uncrossing_defect_budget.py`,
`results/uncrossing_defect_budget.json`.

The entire two-column obstruction tier has a normal form and a sharp slack budget.

Theorem 5.67 (two-column exterior budget)**.** *Let v be singleton-normalized and exact, let $\phi = \mathrm{Lmin}(v)$, and let H be a slack cumulative richer suffix. Suppose a strict ϕ-tight cover of H has exactly two positive-support proper coalitions. Then there are nonempty, pairwise disjoint X, Y, R such that*

$$N = H \dot\cup X \dot\cup Y \dot\cup R, \qquad A = H \cup X, \qquad B = H \cup Y,$$

where A, B are the two support coalitions. Moreover

$$0 < s_\phi(H) \leq s_\phi(R). \tag{ESB}$$

Thus every two-column failure is an oriented broken diamond, and its exterior coalition finances the failed exact face.

Proof. Write the two positive weights as λ, μ and their common load on H as α. Since H is slack, Theorem 5.65 and its laminar endpoint imply that the support coalitions cross. Their four Venn-cell loads are $\lambda + \mu, \lambda, \mu, 0$. Since α is one of these four loads, $\alpha \leq \lambda + \mu$. The nonempty intersection cannot therefore contain an outside player, whose load must be strictly below α. Hence it lies in and meets H, so $\alpha = \lambda + \mu$. Equality of the loads on H and strict inequality outside now force $A \cap B = H$; crossing gives nonempty arms X, Y outside H.

Put $s = s_\phi(H)$ and choose, by exactness, $z \in \mathrm{Core}(v)$ with $z(H) = v(H)$. For $d = z - \phi$, tightness of A, B gives $d(X), d(Y) \geq s$. Efficiency yields

$$d(R) = s - d(X) - d(Y) \leq -s.$$

Core feasibility gives $z(R) \geq v(R)$, equivalently $d(R) \geq -s_\phi(R)$, proving (ESB) and also $R \neq \emptyset$. This is the slack form of the exact-balanced identity

$$\mathbf{1}_A + \mathbf{1}_B + \mathbf{1}_R - \mathbf{1}_H = \mathbf{1}_N.$$ □

Replacing the exterior term R in the last identity by its rooted spoke $H \cup R$ gives

$$\mathbf{1}_{H\cup X} + \mathbf{1}_{H\cup Y} + \mathbf{1}_{H\cup R} - 2\mathbf{1}_H = \mathbf{1}_N.$$

Thus the entire rooted two-column obstruction, in every block size, is the $k = 3$ block-star facet of Theorem 5.47, rather than only an isolated five-player facet.

The normal form has a sharp converse. Let $N = C\dot\cup X\dot\cup Y\dot\cup R$, with all four blocks nonempty, put $A = C \cup X$, $B = C \cup Y$, and write $c = |C|$, $x = |X|$, $y = |Y|$, $r = |R|$.

Theorem 5.68 (binary broken-diamond realization)**.** *Choose payoff levels $\rho > \sigma > 0$ and suppose*

$$c \geq 2 \qquad or \qquad \sigma r \geq \rho. \tag{CAP}$$

Give members of C payoff ρ and all other players payoff σ, obtaining ϕ. Let

$$K = \{z \geq 0 : z(N) = \phi(N),\ z(A) \geq \phi(A),\ z(B) \geq \phi(B)\}, \qquad v(S) = \min_{z\in K} z(S).$$

Then v is singleton-normalized and exact, $\mathrm{Core}(v) = K$*, and* $\mathrm{Lmin}(v) = \phi$*. Coalitions A, B are tight, while*

$$s_\phi(C) = \min\{\rho c, \sigma r\} > 0, \qquad s_\phi(A \cup B) = \sigma \min\{x, y\} > 0. \tag{BDR}$$

For this lower-envelope family, condition (CAP) *is also necessary for singleton normalization when $c = 1$. Across all exact games, the first two-column Lmin failure has five players, attained at $\rho = 2, \sigma = 1$ by exactly the shapes $(c, x, y, r) = (2, 1, 1, 1)$ and $(1, 1, 1, 2)$.*

Proof. The point ϕ is feasible and makes both displayed constraints tight. The lower-envelope two-inclusion argument gives $\mathrm{Core}(v) = K$ and exactness. Every coordinate has minimum zero. The σ units on a player in X, Y, or R can be moved to C. A member of C can transfer its ρ units to another member when $c \geq 2$. If $c = 1$, zeroing the member of C, adding ρ units in total to each of X, Y, and removing ρ units in total from R is feasible exactly when $\sigma r \geq \rho$. Thus singleton values are zero.

The unit weights on A, B load C twice, $X \cup Y$ once, and R zero times. More directly, if a core point has every coordinate at least σ, write it as $\sigma\mathbf{1}_N + w$ with $w \geq 0$. The two inequalities and efficiency give

$$w(C) + w(X),\ w(C) + w(Y) \geq (\rho - \sigma)c, \qquad w(N) = (\rho - \sigma)c.$$

Hence w vanishes on $X \cup Y \cup R$. The remaining total on C is max–min equalized at ρ per player, proving $\mathrm{Lmin}(v) = \phi$.

At the block-aggregate level the constraints are $z(C) + z(X) \geq \rho c + \sigma x$ and $z(C) + z(Y) \geq \rho c + \sigma y$. Efficiency gives

$$v(C) = \max\{0, \rho c - \sigma r\}, \qquad v(C \cup X \cup Y) = \rho c + \sigma \max\{x, y\},$$

which proves (BDR).

If $c = 1$, any feasible point zeroing C must increase each arm total by at least ρ. Efficiency therefore decreases R by at least ρ, proving necessity of $\sigma r \geq \rho$. For the general lower bound, four players in Theorem 5.67 make both H and R singletons. Normalization turns (ESB) into $\phi(H) \leq \phi(R)$, contradicting the strict payoff boundary. At five players, placing the extra member in an arm leaves H, R singleton and gives the same contradiction. Hence only a second member of H or of R can occur at the threshold; the displayed levels realize both possibilities. □

The atomic reduction now makes the lower side of the threshold nearly formal and removes the earlier payoff-stratum case split.

Corollary 5.69 (circuit proof of the four/five threshold)**.** *Every singleton-normalized exact game on at most four players is an exact partition game witnessed at Lmin. The bound is sharp at five players.*

Proof. Suppose a richer Lmin suffix H were slack. By Theorem 5.42, it would have an atom containing H, at least one positive-slack poorer singleton, and some tight protector coalitions. There must be at least two protectors: a single positive protector with strictly larger common load on H than outside would have to contain every member of H and no member outside it, making it the slack coalition H itself. Hence the atom has at least four members.

For $|N| \leq 3$ this contradicts the circuit bound $|\mathcal{A}| \leq |N|$. For $|N| = 4$ equality is forced: the atom has exactly two tight protectors and one poorer singleton besides H. Removing the singleton term from (AC) leaves protector load $r+\gamma$ on H, load r on every other outside player, and load strictly below r on that singleton. It is therefore a strict two-column tight cover of H. Theorem 5.67 then gives four nonempty Venn blocks H, X, Y, R, so all are singletons, and

$$0 < s_\phi(H) \leq s_\phi(R).$$

Singleton normalization turns this into $\phi_h \leq \phi_r$, contradicting that H lies strictly above its poorer complement. Therefore every richer suffix is tight. The five-player realization in Theorem 5.68 proves sharpness. □

The classification has a direct interpretation in imprecise probability. The correspondence itself is established prior art: a coherent lower probability on a finite state space is a normalized nonnegative exact game, and its induced credal set is the game core (Studený and Kratochvíl 2017). The following sharp selection consequence is the candidate cross-field contribution.

Corollary 5.70 (sharp credal Lorenz threshold)**.** *Let $\underline{P}$ be a coherent lower probability on a state space Ω with $|\Omega| \leq 4$ and $\underline{P}(\{i\}) = 0$ for every state. Its induced credal set contains a unique probability distribution that Lorenz-dominates every other induced distribution. The cardinality bound is sharp: the conclusion can fail on five states, still with all singleton lower probabilities zero.*

Proof. Under the exact-game correspondence, Corollary 5.69 makes Lmin an exact-partition allocation. The Lorenz argument in Lemma 5.61 gives the first assertion. For sharpness, on five states take the lower envelope of the four probability distributions

$$\frac{1}{6}(0,0,0,3,3), \quad \frac{1}{6}(0,0,3,3,0), \quad \frac{1}{6}(1,1,1,0,3), \quad \frac{1}{6}(1,1,1,1,2). \tag{CLP5}$$

Every generator belongs to the induced credal set and one attains the lower probability of every event, so the lower envelope is coherent; its singleton values are zero. The envelope constraints for the complements of states $1, 2$ give $p_1, p_2 \leq 1/6$, while those for $\{3,5\}$ and $\{4,5\}$ give $p_3 + p_5, p_4 + p_5 \geq 1/2$. These inequalities and efficiency first pin the unique increasingly maximal distribution and then the unique decreasingly minimal distribution as

$$\phi = (1/6, 1/6, 1/6, 1/6, 1/3), \qquad \psi = (1/8, 1/8, 1/4, 1/4, 1/4).$$

Indeed, ϕ is the fourth generator in (CLP5); the upper bounds on p_1, p_2 cap the minimum at $1/6$, and equality plus the two pair constraints pins all five coordinates. Also $\psi = \frac{1}{4}g^2 + \frac{3}{4}g^4$ is feasible;

the two pair constraints cap the smallest possible maximum at $1/4$, pin $p_3 = p_4 = p_5 = 1/4$, and decreasing lexicographic minimization uniquely equalizes the remaining mass between p_1, p_2. Their increasing-order cumulative sums are, respectively,

$$(1/6, 1/3, 1/2, 2/3, 1), \qquad (1/8, 1/4, 1/2, 3/4, 1).$$

The Lorenz curves cross. Any Lorenz-dominating point would have to be the unique increasingly maximal point ϕ, but ϕ does not dominate ψ. Hence no Lorenz-dominating distribution exists. □

More generally, put $\ell_i = \underline{P}(\{i\})$ and $R = 1 - \sum_i \ell_i$. If $R > 0$, then

$$\underline{Q}(A) = \frac{\underline{P}(A) - \sum_{i \in A} \ell_i}{R} \quad \text{and} \quad q \longmapsto \ell + Rq \tag{AN}$$

give an atom-vacuous coherent lower probability and an affine bijection from its credal set to that of $\underline{P}$; if $R = 0$, the original credal set is the singleton $\{\ell\}$. Thus the corollary also gives a residual-Lorenz selection theorem, applied to $(p_i - \ell_i)/R$, for every coherent lower probability through four states. This residual formulation is not asserted to imply ordinary Lorenz dominance after unequal atom floors are restored.

The full affine proof, five-state arithmetic, and hostile-review questions are recorded in

`paper3/COHERENT_LOWER_PROBABILITY_THRESHOLD.md`,
`src/verify_credal_four_five_threshold.py`,
`results/credal_four_five_threshold.json`,
`paper3/COHERENT_LOWER_PROBABILITY_EXTERNAL_REVIEW_PROMPT.md`.

This threshold should not be confused with the established all-dimensional base-polyhedron theorem: increasingly maximal and decreasingly minimal bases coincide for supermodular (equivalently, 2-monotone) lower probabilities (Fujishige 1980; Frank and Murota 2022). The new candidate concerns arbitrary coherent lower probabilities. Existing exact-capacity selection work also studies Shapley, vertex-average, incenter, and contraction centroids, rather than the Lorenz selector used here (Miranda and Montes 2023). The sharp threshold still requires specialist novelty review.

Thus the four/five threshold is a slack-financing threshold: crossing needs the three cells C, X, Y, exact balancedness requires the exterior cell R, and singleton exactness requires one further unit of target-or-exterior capacity. The proof is human and dimension-free. Standalone versions and exact parameter audits are

`paper3/TWO_COLUMN_LMIN_OBSTRUCTION.md`,
`src/verify_two_column_lmin_obstruction.py`,
`results/two_column_lmin_obstruction.json`,
`paper3/BINARY_BROKEN_DIAMOND_REALIZATION.md`,
`src/verify_binary_broken_diamond_realization.py`,
`results/binary_broken_diamond_realization.json`.

Corollary 5.71 (Lmin local-supermodularity domain)**.** *Every singleton-normalized exact Lmin-locally supermodular game is an exact partition game witnessed at Lmin. Conversely, every slack cumulative richer suffix at Lmin yields a broken tight diamond satisfying* (UD)*.*

Proof. At every strict Lmin boundary, Lemma 5.62 supplies the cover required by Theorem 5.65. As observed before the theorem, local supermodularity makes the tight family closed under union and intersection. Apply the theorem at every boundary. Its converse gives the final assertion. □

The exact-face theorem does not require normalization or an egalitarian allocation; these enter only to generate its cover automatically. Its Lmin corollary subsumes the laminar domain and the established convex-game domain but is stated only in terms of local behavior on the Lmin-tight skeleton. The inclusion beyond convex games is strict. For example, take the lower envelope of the four common-total allocations

$$(0,4,6,5,0), \quad (5,0,2,7,1), \quad (3,0,0,11,1), \quad (2,2,6,0,5).$$

It is singleton-normalized and exact, has $\mathrm{Lmin}(v) = (5/2, 5/2, 10/3, 10/3, 10/3)$, and its only nonempty proper Lmin-tight coalition is the richer suffix $\{3,4,5\}$. Hence it is Lmin-locally supermodular and exact partition. It is not convex: the pair $\{1,2,5\}, \{1,3,5\}$ violates supermodularity by four.

In the sharp five-player failure, the tight protectors $\{1,3,5\}$ and $\{3,4,5\}$ have intersection $\{3,5\}$ and union $\{1,3,4,5\}$, each with slack one, so their defect in (UD) is exactly two. Their support has only two columns and is therefore balanced and totally balanced; this disproves any extension based only on balanced incidence. A standalone proof, hostile review prompt, and exact audit are

`paper3/LMIN_UNCROSSING_DOMAIN_THEOREM.md`,
`paper3/LMIN_UNCROSSING_EXTERNAL_REVIEW_PROMPT.md`,
`src/verify_lmin_uncrossing_domain.py`,
`results/lmin_uncrossing_domain.json`.

The preceding certificates also show that the entire characteristic function away from the Lmin-tight coalitions is dispensable. This gives a canonical support-independent reduction before any trace classification is attempted.

Theorem 5.72 (Lmin-tight-skeleton retraction)**.** *Let v be a singleton-normalized exact game, put $\phi = \mathrm{Lmin}(v)$, and let*

$$\mathcal{T}_\phi := \{S : \varnothing \neq S \subsetneq N,\ v(S) = \phi(S)\}.$$

Define

$$K_\phi := \{x \in \mathbb{R}^N_+ : x(N) = v(N), \quad x(S) \geq \phi(S)\ (S \in \mathcal{T}_\phi)\}, \tag{SK-1}$$

and let

$$(R_\phi v)(A) := \min_{x \in K_\phi} x(A). \tag{SK-2}$$

Then $R_\phi v$ is singleton-normalized and exact, $\mathrm{Core}(R_\phi v) = K_\phi$*, and*

$$R_\phi v \leq v, \qquad \mathrm{Lmin}(R_\phi v) = \phi, \qquad (R_\phi v)(A) = \phi(A) \Longleftrightarrow v(A) = \phi(A). \tag{SK-3}$$

Thus the complete Lmin-tight/slack pattern, including exact-partition status at every cumulative richer suffix, is preserved. The operation is idempotent. For rational input, a common positive scaling makes $R_\phi v$ integral.

Proof. Exactness and singleton normalization give $\mathrm{Core}(v) \subseteq K_\phi$. Consequently, for every coalition A,

$$(R_\phi v)(A) \leq \min_{x \in \mathrm{Core}(v)} x(A) = v(A). \tag{SK-4}$$

If $A \in \mathcal{T}_\phi$, its defining inequality and the feasible point ϕ give $(R_\phi v)(A) = \phi(A) = v(A)$. For a singleton, (SK-4) and nonnegativity give value zero; efficiency handles N. If a proper A is not in $\mathcal{T}_\phi$, then $(R_\phi v)(A) \le v(A) < \phi(A)$. This proves the last equivalence in (SK-3).

Every point of K_ϕ satisfies all inequalities of its lower envelope, so it lies in $\mathrm{Core}(R_\phi v)$. Conversely, a core point of $R_\phi v$ is efficient, is nonnegative because the new singleton values are zero, and satisfies the inequalities indexed by $\mathcal{T}_\phi$ at their just-proved values $\phi(S)$. It lies in K_ϕ. Hence the two sets coincide. Every coalition minimum is attained on the compact polytope, proving exactness.

Write the payoff strata of ϕ as $B_1, \dots, B_k$, with increasing levels. For $j < k$, put $L_j = B_1 \dot\cup \cdots \dot\cup B_j$ and $H_j = N \setminus L_j$. Lemma 5.62 supplies weights on $\mathcal{T}_\phi$ with common load $\alpha_j > 0$ on H_j and loads $d_i < \alpha_j$ on L_j. Summing their defining inequalities in K_ϕ and eliminating $x(H_j)$ by efficiency gives

$$\sum_{i \in L_j} (\alpha_j - d_i)(x_i - \phi_i) \le 0. \tag{SK-5}$$

All coefficients are positive. At the first progressive leximin stage, (SK-5) for $j = 1$ pins every member of B_1 at its value in ϕ. Inductively, after the earlier blocks are fixed, the jth inequality reduces to a positive weighted sum over B_j and pins that block. Efficiency pins the last block. Therefore ϕ remains Lmin.

The tight set and total are now unchanged, so a second application uses the same polytope and lower envelope; this proves idempotence. Rational data define a rational polytope and rational coalition minima, whose finitely many denominators can be cleared. □

Corollary 5.73 (certificate-only hypergraph realization)**.** *Let $\phi \in \mathbb{R}^N_+$, let $\mathcal{T}$ be any family of proper nonempty coalitions, and define*

$$K(\mathcal{T}, \phi) := \{x \in \mathbb{R}^N_+ : x(N) = \phi(N), \quad x(T) \ge \phi(T) \ (T \in \mathcal{T})\}.$$

Write the increasing payoff strata of ϕ as $B_1, \dots, B_k$. Suppose

(i) $\min_{x \in K(\mathcal{T},\phi)} x_i = 0$ *for every player i;*

(ii) for every $j < k$, nonnegative weights on $\mathcal{T}$ have common load $\alpha_j > 0$ on $H_j = N \setminus (B_1 \dot\cup \cdots \dot\cup B_j)$ and load strictly below α_j on every player outside H_j.

Then the lower envelope of $K(\mathcal{T}, \phi)$ is a singleton-normalized exact game with Lmin ϕ, and every member of $\mathcal{T}$ is tight at ϕ. Its proper tight set is exactly $\mathcal{T}$ if and only if

$$\min_{x \in K(\mathcal{T},\phi)} x(A) < \phi(A) \quad \text{for every proper } A \notin \mathcal{T}. \tag{SK-6}$$

Conversely, the complete Lmin-tight family of every singleton-normalized exact game satisfies (i), (ii), and (SK-6). Thus normalized exact Lmin skeletons have a certificate-only characterization involving no unspecified coalition values.

Proof. Condition (i), the listed inequalities, and the feasible point ϕ prove that the lower envelope has singleton values zero and has $w(T) = \phi(T)$ for $T \in \mathcal{T}$. The two-inclusion argument of Theorem 5.72 gives $\mathrm{Core}(w) = K(\mathcal{T}, \phi)$ and exactness. Condition (ii) gives (SK-5) at every payoff boundary, so the same progressive induction proves $\mathrm{Lmin}(w) = \phi$. Condition (SK-6) is exactly the assertion that no omitted proper coalition is tight. For the converse, apply exactness to every singleton, Lemma 5.62 to every strict payoff boundary, and the definition of the complete tight family. □

The strict-cover ingredient follows from the published Lmin optimality structure discussed above. The map sending a balanced game to the unique exact game with the same core is already known as its lower envelope or exact cover; see, for example, Schmeidler (1972); Studený and Kratochvíl (2022). The candidate new claim is their exact idempotent combination: deleting every Lmin-slack coalition constraint, then exactifying, preserves Lmin and the entire coalitionwise tightness pattern. We do not claim novelty for either ingredient separately. The standalone proof and an exact audit on three unrelated families are

`paper3/LMIN_TIGHT_SKELETON_RETRACTION.md`,
`src/verify_lmin_tight_skeleton_retraction.py`,
`results/lmin_tight_skeleton_retraction.json`.

Theorem 5.74 (balanced-trace transport obstruction)**.** *Let v be exact, let $\phi = \mathrm{Lmin}(v)$, and let H be a cumulative richer suffix at a strict payoff boundary, with $L = N \setminus H \neq \emptyset$. If*

$$s := \phi(H) - v(H) > 0,$$

then there are a core point z, a displacement $\Delta = z - \phi$, and a finite convex mixture of lifted normalized minimal balanced packets on H with weights $\theta_q > 0$ and poorer-player load vectors $r^q \in \mathbb{R}_+^L$ such that

$$\Delta(H) = -s, \qquad \Delta(L) = s, \tag{BT-1}$$

$$\min_{i \in L} \Delta_i < 0 < \max_{i \in L} \Delta_i, \tag{BT-2}$$

$$-s + \sum_{i \in L} r_i^q \Delta_i \geq 0 \qquad \text{for every packet } q, \tag{BT-3}$$

$$\sum_q \theta_q r_i^q < 1 \quad (i \in L), \qquad \sum_q \theta_q = 1. \tag{BT-4}$$

Thus every slack Lmin boundary requires a minimal balanced rich-side circuit that transports the suffix slack through opposing gains and losses on the poorer side.

Proof. Choose $z \in \mathrm{Core}(v)$ exact on H. Efficiency gives (BT-1). Apply Lemma 5.62, normalize its rich load to one, and discard cover weight on coalitions disjoint from H. The remaining cover still has unit load on every member of H and has loads $c_i < 1$ on L. Summing $z(T) - \phi(T) \geq 0$ over its tight coalitions gives

$$-s + \sum_{i \in L} c_i \Delta_i \geq 0. \tag{BT-C}$$

If every Δ_i on L were nonnegative, the left side would be strictly less than $-s + \Delta(L) = 0$. Since $\Delta(L) = s > 0$, there is also a positive entry. This proves (BT-2).

Project the remaining tight coalitions to their nonempty traces on H. Their weights form a point of the bounded polytope

$$\mathcal{P}_H = \left\{ \beta \geq 0 : \sum_{\emptyset \neq R \subseteq H} \beta_R \mathbf{1}_R = \mathbf{1}_H \right\}.$$

(Boundedness follows from $\sum_R \beta_R \leq |H|$.) Its extreme points are exactly the unique normalized balancing weights on minimal balanced collections: a nonminimal support contains a balanced subcollection and admits a nontrivial feasible perturbation, while a minimal balanced support

has unique weights. Thus the projected cover is a convex combination of normalized minimal balanced packets; see also Laplace Mermoud et al. (2025).

For each trace R in a packet, split among the original coalitions having that trace using their conditional cover weights. Taking the product distribution across the finitely many trace slots refines the decomposition into a finite convex mixture of lifted deterministic packets. Indeed, after averaging first over the product choices and then over packets, the weight of an original coalition T is its trace weight times its conditional share, hence exactly the original λ_T. The lift therefore preserves the entire average cover, not only its rich projection.

For a lifted packet q, let r_i^q be its total packet weight on coalitions containing poorer player i. Its rich trace has unit load, so summing $z(T) - \phi(T) \geq 0$ with its packet weights proves (BT-3). Averaging the lifted packets recovers the original poorer loads $c_i < 1$, which is (BT-4). □

The preceding theorem forces a fractional hypergraph obstruction whenever the poor-side transport has a unique gaining coordinate. This gives a dimension-free condition on the actual tight-coalition geometry, rather than only on the number of players.

Definition 5.75 (active trace polytope)**.** *At a strict Lmin boundary (L, H), let*

$$\mathcal{A}_\phi(H) := \{T \cap H : \ T \subsetneq N, \quad \phi(T) = v(T), \ T \cap H \neq \varnothing\}$$

and define its fractional exact-cover polytope

$$\mathcal{Q}_\phi(H) := \left\{ \beta \in \mathbb{R}_+^{\mathcal{A}_\phi(H)} : \sum_{R \in \mathcal{A}_\phi(H)} \beta_R \mathbf{1}_R = \mathbf{1}_H \right\}. \tag{AT}$$

Call the boundary trace-integral *if every vertex of $\mathcal{Q}_\phi(H)$ is integral. An integral point of* (AT) *is a set partition of H.*

Lemma 5.62 makes $\mathcal{Q}_\phi(H)$ nonempty, and it is bounded because every trace is nonempty.

Theorem 5.76 (integral-trace obstruction)**.** *Let v be exact, let $\phi = \mathrm{Lmin}(v)$, and let (L, H) be a strict payoff boundary. Suppose the exact H-face contains a point $z \in \mathrm{Core}(v)$ such that*

$$z(H) = v(H), \qquad \big|\{i \in L : z_i > \phi_i\}\big| \leq 1. \tag{IT}$$

If the boundary is trace-integral, then H is tight at ϕ.

Proof. Suppose instead that $s := \phi(H) - v(H) > 0$ and put $\Delta = z - \phi$. Normalize the cover from Lemma 5.62 to unit load on H, discard coalitions disjoint from H, and write $c_i < 1$ for its loads on L. The argument proving (BT-2), applied to this chosen z, shows that Δ_L has both signs. Since $\Delta(L) = s > 0$, condition (IT) therefore gives a unique $b \in L$ with $\Delta_b > 0$, while $\Delta_i \leq 0$ for $i \neq b$.

The projected cover lies in $\mathcal{Q}_\phi(H)$. Trace integrality decomposes it into a convex mixture of integral points, hence set partitions of H. Lift every partition to deterministic original tight coalitions by the same conditional product construction used in Theorem 5.74. In a lifted partition packet the poorer-player loads r_i are nonnegative integers, and its tight coalition inequalities at z give

$$-s + \sum_{i \in L} r_i \Delta_i \geq 0. \tag{IT-P}$$

If $r_b = 0$, the left side is at most $-s$, so every packet has $r_b \geq 1$. Averaging gives $c_b \geq 1$, contradicting $c_b < 1$. □

Equivalently, if a trace-integral boundary is slack, then every core point exact on H has at least two gaining poorer coordinates; the sign argument also gives at least one losing poorer coordinate. Thus trace-integral slack requires at least three poorer players. This sign-complexity statement is independent of $|H|$.

Corollary 5.77 (odd-trace-cycle necessity)**.** *With exactly two poorer players, every slack richer Lmin suffix has a nonintegral active trace polytope. Consequently its player-by-trace incidence matrix is not balanced and contains an odd-order square submatrix with exactly two ones in every row and column.*

Proof. Exactness supplies z exact on H. Under slack, applying the normalized strict cover to this z, exactly as in the proof of (BT-2), makes its two poor displacements have opposite signs. Thus (IT) holds. Theorem 5.76 rules out integrality. A balanced zero–one matrix has an integral set-partitioning polytope (Berge 1972); taking the contrapositive and the defining forbidden submatrix characterization of balanced matrices gives the odd trace cycle. □

Definition 5.78 (safe rooted tight cover)**.** *At a strict boundary with two poorer players $L = \{a, b\}$ and richer side H, let $\mathcal{T}$ be any finite family of proper ϕ-tight coalitions. A vector $y \in \mathbb{Z}_+^{\mathcal{T}}$ is a* safe rooted tight cover *if, for some $d \in H$ and integer $k \geq 1$, its incidence loads satisfy*

$$\sum_{T \in \mathcal{T}} y_T \mathbf{1}_{T \cap H} = \mathbf{1}_H + k e_d, \qquad p := \sum_{T \in \mathcal{T}} y_T \mathbf{1}_{T \cap L} \in \{0,1\}^L \setminus \{0\}. \tag{SRC}$$

Equivalently, the affine semigroup generated by the lifted tight-coalition columns meets a rooted exact-cover fibre while loading each poor player at most once and at least one poor player once.

Theorem 5.79 (exact-face rooted-semigroup obstruction)**.** *Let v be singleton-normalized and exact, let $\phi \in \mathrm{Core}(v)$, and let (L, H) be a strict payoff boundary with $|L| = 2$. No family of proper ϕ-tight coalitions admits a safe rooted tight cover.*

Proof. Apply Corollary 5.56 with $D = \{d\}$ and $q = k$. Singleton normalization makes $s_\phi(\{d\}) = \phi_d$. If both poor players are loaded, its right side is zero. If exactly one is loaded and j is omitted, its right side is ϕ_j. The strict boundary gives $k\phi_d > \phi_j \geq 0$ in either case, a contradiction. □

The singleton root is the smallest instance of a more general overload footprint left by alternating around an odd trace cycle.

Lemma 5.80 (odd-cycle overload footprint)**.** *Let v be singleton-normalized and exact, let $\phi \in \mathrm{Core}(v)$, and let a strict payoff boundary have richer side H and poorer side $L = \{a, b\}$. Suppose an odd family of proper ϕ-tight coalitions has, on a row set $R \subseteq H$, the cycle incidence matrix: after cyclic indexing, each cycle row lies in its two incident columns. Select an alternating near-perfect cover, so its load on R is $\mathbf{1}_R + e_d$ for the duplicated cycle row d. Suppose its load on every $h \in H \setminus R$ is either one or two, and put*

$$F := \{d\} \cup \{h \in H \setminus R : \text{the selected load on } h \text{ is two}\}. \tag{OC-F}$$

If the selected poor incidence is $p \in \{0,1\}^{\{a,b\}} \setminus \{0\}$, then

$$s_\phi(F) \leq \sum_{i \in \{a,b\} : p_i = 0} \phi_i. \tag{OC-B}$$

In particular, if both poor players are loaded then F is ϕ-tight. If $R = H$, such an alternating selection is impossible.

Proof. By construction, the selected rich incidence is $\mathbf{1}_H + \mathbf{1}_F$, while its poor incidence is zero–one and nonzero. Apply Corollary 5.56 with $q = 1$. This gives (OC-B). If $R = H$, then $F = \{d\}$ and singleton normalization gives $s_\phi(D) = \phi_d$. If both poor players are loaded, the right side is zero. If exactly one is loaded and j is omitted, the right side is ϕ_j; the strict boundary gives $\phi_d > \phi_j$. Both cases are impossible. □

Thus the first possible escape from odd-cycle rounding needs an external rich row to enlarge the overload footprint beyond a singleton. This is exactly what happens at the sharp $2 \mid 4$ co-singleton packet. On any three chosen cycle rows, two alternating co-singletons cover the fourth row twice, so F has two rich players. More explicitly, at $h = 4$ relabel the rich block as $H = \{r_1, r_2, k_1, k_2\}$. In the integral family (TF), select the poor-free coalition $H \setminus \{r_1\}$ and the p_1-carrying coalition with trace $H \setminus \{k_1\}$. Their footprint is $F = \{r_2, k_2\}$, the omitted poor player is p_2, and direct substitution gives

$$s_\phi(F) = \phi_{p_2} = D,$$

where $D = (h-2)(h-3)$ is the scale parameter of that family. Hence (OC-B) is attained at equality when $h = 4$. The three-rich obstruction fails because its footprint is a prohibitively expensive singleton; the fourth rich player supplies the first larger internal face capable of absorbing the same alternating-rounding excess.

The underloaded packet also gives a cost-sensitive bridge to the classical set-covering polyhedron, rather than only a qualitative nonintegrality statement.

Theorem 5.81 (unit-gap set-cover bridge)**.** *Let v be singleton-normalized and exact, let $\phi = \mathrm{Lmin}(v)$, and let H be a slack cumulative richer suffix at a strict boundary with poorer side $\{a, b\}$. Choose a core point z exact on H and label the poor players so that, for $\Delta = z - \phi$,*

$$\Delta_a < 0 < \Delta_b.$$

Suppose a lifted exact-cover packet of proper ϕ-tight coalitions has unit rich incidence on H and gaining-player load $r_b < 1$.

Let M be its rich trace-incidence matrix, retaining duplicate lifted columns, and give a column T the integral cost

$$c_T := |T \cap H| + \mathbf{1}_{\{b \in T\}}. \tag{UG-1}$$

Then

$$\min\{c^\top x : Mx \geq \mathbf{1}_H,\ x \geq 0\} < |H| + 1 \leq \min\{c^\top y : My \geq \mathbf{1}_H,\ y \in \mathbb{Z}_+^{\mathcal{T}}\}. \tag{UG-2}$$

In particular, M is nonideal. For every lifted exact-cover point x, its cost is exactly $|H| + r_b(x)$; hence preservation of the strict unit gap after lifting a minor point back to H with the original costs is equivalent to preservation of gaining-player load below one.

Proof. The given fractional packet has total rich incidence $|H|$, so its cost is $|H| + r_b < |H| + 1$. Now take an integral cover y and let

$$m := \sum_T |T \cap H| y_T.$$

Since every rich row is covered, $m \geq |H|$. If $m > |H|$, integrality gives $m \geq |H| + 1$, and the claimed cost bound follows. If $m = |H|$, every rich row is covered exactly once. Were its b-load zero, summing at z the tight-coalition inequalities selected by y would give

$$0 \leq \sum_T y_T \Delta(T) = \Delta(H) + r_a(y)\Delta_a = -\big(\phi(H) - v(H)\big) + r_a(y)\Delta_a < 0,$$

which is impossible. Thus its integral b-load is at least one and again $c^\top y \geq |H| + 1$. Finally, any exact-cover point has total rich incidence $|H|$, so (UG-1) gives the last assertion. □

The rich-incidence term can be separated from the poor label by a large integer scale. This makes exactness of a cost-preserving Lehman lift a consequence rather than an additional hypothesis.

Corollary 5.82 (scale-separated exactness)**.** *Under the hypotheses of Theorem 5.81, let q be the number of active lifted columns, retaining duplicates, and choose an integer $\Lambda > q$. Give each column the cost*

$$c_T^{(\Lambda)} := \Lambda |T \cap H| + \mathbf{1}_{\{b \in T\}}. \tag{UG-M1}$$

Then the fractional optimum is below $\Lambda|H|+1$, while every integral cover costs at least $\Lambda|H|+1$.

Suppose a set-cover minor has a positive Lehman core with column sum r, its uniform point is feasible for every row of the minor, and that point is lifted to a cover of H by adjoining unit-weight contracted columns. If this lifted point, evaluated with the original costs (UG-M1), still costs less than $\Lambda|H| + 1$, then every rich row has load exactly one and the gaining-player load is below one. Consequently the contracted columns have pairwise disjoint rich traces, surviving columns avoid those traces, and no contracted column contains either poor player. In particular, a reduction preserving this one scaled objective gap automatically supplies the full exact-cover, deficit-preserving, poor-label-neutral lift.

Proof. The first claim repeats the proof of Theorem 5.81. The given fractional packet costs $\Lambda|H|+r_b$. An integral cover with more than $|H|$ units of rich incidence costs at least $\Lambda(|H|+1) > \Lambda|H| + 1$; one with exactly $|H|$ units is a partition and has integral b-load at least one.

For the lifted minor point, every rich row has load at least one. Every positive excess is a sum of integers from contracted columns and multiples of $1/r$ from surviving columns, and is therefore at least $1/r$. Hence any overcovered row makes the total cost at least

$$\Lambda|H| + \frac{\Lambda}{r}.$$

The Lehman core uses at most the original q columns, so $r \leq q < \Lambda$ and this is strictly greater than $\Lambda|H| + 1$, a contradiction. Thus every rich row has unit load. This forces the unit-weight contracted traces to be pairwise disjoint and every surviving column to avoid them. The cost is now $\Lambda|H|$ plus the nonnegative b-load, which must be below one; its contracted part is an integer, hence zero. Writing the total poor loads as ρ_a, ρ_b, the packet transport inequality becomes

$$0 \leq (\rho_b - 1)\sigma + (\rho_a - \rho_b)\Delta_a, \qquad \sigma := \phi(H) - v(H) > 0.$$

Because $\rho_b < 1$ and $\Delta_a < 0$, it forces $\rho_a < \rho_b < 1$. The contracted a-load is also an integer part of a load below one, so it vanishes as well. □

The corollary is deliberately conditional on preservation of the displayed objective. Classical minimally-nonideal reduction preserves nonideality, not an arbitrarily prescribed separating cost. The minor-compatible covering/packing pairs and ideal unit-increasing set functions of Király and Pap (2016) broaden Lehman's framework, but do not state that this scale-separated objective survives. Establishing such a relative reduction, or finding a counterexample to it, is the remaining lifting problem.

There is an exact unweighted reformulation of that problem. Let $\mathcal{C}$ be the clutter on the active columns whose member C_h is the set of columns containing rich row h. For a positive integral cost vector c, form the standard parallelization $\mathcal{C}^c$ by replacing column T with c_T clones and replacing every member by all choices of one clone from each of its elements; see Cornuéjols (2001).

Proposition 5.83 (parallelized weighted-gap identity)**.** *For every positive integral c,*

$$\tau(\mathcal{C}^c) = \tau^c(\mathcal{C}), \qquad \tau_f(\mathcal{C}^c) = \tau_f^c(\mathcal{C}). \tag{PAR}$$

Consequently the scale-separated gap makes $\mathcal{C}^c$ nonideal for the ordinary all-ones objective.

Proof. In an integral cover of $\mathcal{C}^c$, every original member contains a clone class selected in its entirety: otherwise choosing one unselected clone from each class gives an uncovered parallel member. Partial clone classes can therefore be discarded, and the selected full classes are precisely a c-weighted cover of $\mathcal{C}$. This proves the first identity.

For the fractional identity, average any feasible point over all permutations within every clone class. Feasibility and objective value are preserved, and the averaged point has one common value x_T on the c_T clones of T. Its parallel covering inequalities are exactly $\sum_{T \in C_h} x_T \geq 1$, and its cardinality is $c^\top x$. □

Parallelization does not itself solve the lifting problem. An arbitrary MNI minor may fail to remain below the original global threshold. More importantly, its core rows are parallel clutter members rather than original rich players. Averaging a lifted core within clone classes introduces denominators $c_T r$, so the $1/r$ excess quantum used in Corollary 5.82 is lost. Thus the unweighted identity (PAR) is an exact reformulation of the objective gap, not a Lehman lift of its player-incidence geometry. Minor inequalities describe all nonboolean facets only in special covering classes, and generalized minor inequalities are needed even for broader circulant families. Thus ordinary Lehman reduction alone does not prove the desired relative statement; the exact-game cost and label structure must do additional work before taking the minor.

There is also a label-preserving balanced-matrix formulation. Instead of trying to carry the poor labels alongside a rich-side minor, make them rows of the matrix before taking any obstruction.

Theorem 5.84 (labeled odd-cycle necessity)**.** *In the setting of Theorem 5.81, let a lifted packet have poor loads r_a, r_b with $r_b < 1$. Form the zero–one matrix $\widehat{M}$ whose rows are $H \cup \{a, b\}$ and whose original columns are the full player-incidence vectors of the lifted tight coalitions. Adjoin two dummy columns e_a, e_b, supported only on their named poor rows. Then the exact-cover polytope*

$$\{x \geq 0 : \widehat{M}x = \mathbf{1}_{H \cup \{a,b\}}\} \tag{LO-1}$$

is nonintegral. Consequently $\widehat{M}$ is not balanced and contains an odd-order square submatrix with exactly two ones in every row and column. Neither dummy column belongs to such a submatrix,

so the odd cycle uses only original lifted tight coalitions while retaining whichever poor-label rows it meets.

If, moreover, the positive rich traces of the packet form a minimal balanced collection, then extending the packet by dummy weights $1 - r_a$ and $1 - r_b$ gives a fractional vertex of (LO-1). Its positive original columns together with e_a, e_b are linearly independent. After adjoining the right-hand-side column $\mathbf{1}_{H\cup\{a,b\}}$, the resulting dependence is a linear circuit, with every packet and dummy coefficient strictly positive.

Proof. The packet inequality and $r_b < 1$ imply $r_a < r_b$: indeed, with $\sigma = \phi(H) - v(H)$ and $\Delta_b = \sigma - \Delta_a$,

$$0 \le -\sigma + r_a\Delta_a + r_b\Delta_b = (r_b - 1)\sigma + (r_a - r_b)\Delta_a,$$

where the first term is negative and $\Delta_a < 0$. Thus both poor loads are below one. Extend the packet by assigning dummy weights $1 - r_a$ and $1 - r_b$. This gives a feasible point of (LO-1) with strictly positive e_b coordinate.

For the additional assertion, let M_S be the rich trace-incidence matrix on the positive packet support. A minimal balanced collection has unique normalized balancing weights, so the columns of M_S are linearly independent: a nonzero kernel direction would permit two sufficiently small opposite perturbations of the positive balancing vector. The lifted full columns remain independent because their restrictions to the rich rows are the columns of M_S. Adjoining e_a, e_b preserves independence, since these dummy columns vanish on every rich row. Hence the displayed feasible point is a vertex. Also $r_b > 0$, since $r_b = 0$ would make the packet inequality at most $-\sigma + r_a\Delta_a < 0$. Thus $0 < 1 - r_b < 1$, and the vertex is fractional. Finally, the unique expression of the all-ones right-hand side uses every positive packet coefficient and both positive dummy coefficients. The independent support columns together with that right-hand side therefore form a minimally dependent set, proving the circuit claim.

In any integral point of (LO-1), the selected original columns partition H and load each poor row at most once. If the e_b dummy were selected, no original selected coalition would contain b. Summing their tight-coalition inequalities at z would then give

$$0 \le \Delta(H) + q_a\Delta_a = -\sigma + q_a\Delta_a < 0,$$

where $q_a \in \{0, 1\}$, a contradiction. Hence every integral point has e_b coordinate zero. The displayed fractional point cannot lie in the convex hull of the integral points, proving nonintegrality.

Balanced zero–one matrices have integral set-partitioning polytopes (Berge 1972); the contrapositive and the forbidden-submatrix characterization give the odd cycle. A dummy column has only one nonzero entry in the entire matrix, so it cannot occur in a submatrix having two ones in every column. □

Corollary 5.85 (two one-label odd-cycle obstructions)**.** *Under the hypotheses of Theorem 5.84, form $\widehat{M}_a$ from the original lifted columns restricted to $H \cup \{a\}$ and a dummy singleton column e_a. Define $\widehat{M}_b$ analogously on $H \cup \{b\}$. The exact-cover polytope of each matrix is nonintegral. Hence each matrix contains an odd-order square submatrix with two ones in every row and column, using only original lifted tight coalitions. Thus one odd-cycle obstruction lives on rows contained in $H \cup \{a\}$ and a second lives on rows contained in $H \cup \{b\}$.*

Proof. For $\widehat{M}_b$, extend the packet by dummy weight $1 - r_b > 0$. In every integral exact cover the original columns partition H. Selecting e_b would make their b-load zero, and their displacement

sum would be $-\sigma + q_a\Delta_a < 0$, impossible. Thus every integral point has dummy coordinate zero, whereas the displayed fractional point does not.

For $\widehat{M}_a$, extend the packet by dummy weight $1 - r_a > 0$. If its exact-cover polytope were integral, this point would be a convex combination of integral exact covers. Every such cover partitions H and has a-load at most one. Its integer b-load cannot be zero, again because $-\sigma+q_a\Delta_a < 0$. Hence every integral point has b-load at least one, while their convex combination has b-load $r_b < 1$, a contradiction.

Both matrices are therefore nonbalanced. Their dummy columns have only one nonzero entry and cannot belong to the corresponding forbidden odd submatrices. □

Corollary 5.86 (unconditional labeled-circuit obstruction)**.** *Every slack cumulative richer suffix at a strict Lmin boundary with exactly two poorer players furnishes an extreme lifted packet whose rich trace matrix has the strict unit gap* (UG-2) *and whose poor-row-augmented incidence matrix contains the labeled odd cycle of Theorem 5.84; moreover, both one-label augmented matrices have the odd-cycle obstructions of Corollary 5.85. The packet extended by its two dummy columns is a fractional vertex on the original player rows, and its support together with the all-ones right-hand side is a linear circuit. The corresponding support in each one-label augmentation is likewise a fractional-vertex circuit.*

Proof. Use the packet mixture in Theorem 5.74 and label its common exact-face displacement by $\Delta_a < 0 < \Delta_b$. Its average b-load is below one, so some positive-weight extreme packet has $r_b < 1$. Apply Theorems 5.81 and 5.84, and then Corollary 5.85, to that packet. Because the packet comes from an extreme point of the rich exact-cover polytope, its traces form a minimal balanced collection. The final assertion of Theorem 5.84 therefore gives, on the original player rows, a canonical fractional labeled-cover circuit supported by that same packet and its two dummy columns. The analogous support columns in each one-label augmentation are also independent; adjoining the relevant all-ones right-hand side gives one circuit on $H \cup \{a\}$ and another on $H \cup \{b\}$. The latter vertex is fractional because $0 < 1 - r_b < 1$. The former is fractional as well: integral packet weights would partition H and give an integer b-load below one, hence zero, contradicting the packet displacement inequality. □

The exceptional minimally-nonideal family in Lehman's theorem is incompatible with this underload, independently of the positive Lehman equation.

Theorem 5.87 (degenerate-MNI trace exclusion)**.** *In the setting of Theorem 5.81, an underloaded packet cannot have the trace support of a degenerate projective plane. Explicitly, let its rich rows be $h_0, h_1, \dots, h_t$, $t \geq 2$, and its columns be*

$$R_0 = \{h_1, \dots, h_t\}, \qquad R_i = \{h_0, h_i\} \quad (1 \leq i \leq t), \tag{DP-1}$$

with their unique fractional exact-cover weights $(t-1)/t, 1/t, \dots, 1/t$. No lifting of these columns to proper ϕ-tight coalitions can have gaining-player load below one and satisfy the packet transport inequality.

Proof. Let $u_0, v_0 \in \{0, 1\}$ record whether the lift of R_0 contains a, b, and let u, v count the leaf columns R_i containing a, b. The poor loads are

$$r_a = \frac{(t-1)u_0 + u}{t}, \qquad r_b = \frac{(t-1)v_0 + v}{t}.$$

The packet inequality, $\Delta_b = \sigma - \Delta_a$, $r_b < 1$, and $\Delta_a < 0$ give

$$0 \le (r_b - 1)\sigma + (r_a - r_b)\Delta_a,$$

so necessarily $r_a < r_b$.

If $v_0 = 0$, then $1 \le v \le t - 1$. The strict load inequality forces $u_0 = 0$ and $u < v$, so some i has R_i labeled by b but not by a. Select the lifts of R_0 and R_i. If $v_0 = 1$, underload forces $v = 0$, while $r_a < r_b = (t-1)/t$ forces $u_0 = 0$ and $u \le t - 2$; choose a leaf R_i not labeled by a. In this case select the same two lifted columns.

In either case their rich incidence is $\mathbf{1}_H + e_{h_i}$, and their poor incidence is nonzero and zero–one. They form a safe rooted tight cover, contradicting Theorem 5.79. □

The first label cardinality not covered by the elementary pair count admits a dimension-free Lehman separation argument. It uses no classification of Lehman matrices.

Lemma 5.88 (three-column Lehman separation)**.** *Let A, B be a positive Lehman pair satisfying $AB^\top = J + kI$, with A r-regular and B s-regular. If $r \ge 4$, then every set V of three columns of B has a row d such that*

$$|\operatorname{row}_d(B) \cap V| = 1. \tag{LH-3}$$

Proof. Suppose instead that the three-column row-load vector $y := B\mathbf{1}_V$ has entries only in $\{0, 2, 3\}$. Let x be the number of rows on which y equals three. The transpose Lehman identity gives

$$A^\top y = (J + kI)\mathbf{1}_V = 3\mathbf{1} + k\mathbf{1}_V. \tag{LH-3a}$$

Here $A^\top B = J + kI$ is the classical transpose property of a Lehman pair (Cornuéjols et al. 2009). Hence every column of A outside V meets exactly one of the x rows of load three and no row of load two: its scalar product with y is three, and the only positive entries of y are two and three. Counting these incidences and using $n = rs - k$ gives

$$rs - k - 3 = n - 3 \le rx, \qquad x \ge s - \frac{k+3}{r}. \tag{LH-3b}$$

The diagonal Lehman intersection has size $k + 1 \le r$. Since $r \ge 4$, $(k + 3)/r \le 1 + 2/r \le 3/2$, so the integer x satisfies $x \ge s - 1$.

The three columns have total incidence $3s$. Their common x rows contribute $3x$, and every remaining nonzero row contributes two under the contrary assumption. Thus $3(s - x)$ is even. Also $x \le s$, so $s - x \in \{0, 1\}$; parity forces $x = s$. All three columns are therefore supported on the same s rows and are identical. This contradicts nonsingularity of B, which follows from $AB^\top = J + kI$. Hence (LH-3) holds. □

The same transpose argument has a parity consequence for four columns when the Lehman multiplier is odd.

Lemma 5.89 (four-column separation at odd multiplier)**.** *Let A, B be a positive Lehman pair satisfying $AB^\top = J + kI$, with A r-regular. If $r \ge 5$ and k is odd, every set V of four columns of B has a row meeting V exactly once.*

Proof. Suppose no such row exists and put $y = B\mathbf{1}_V$. Then the entries of y belong to $\{0, 2, 3, 4\}$. No entry can equal three. Indeed, if $y_d = 3$, then row d of A has an entry in some column $c \notin V$ because its row sum is $r \ge 5$. But

$$(A^\top y)_c = 4,$$

whereas the contribution three from row d cannot be completed to four using entries from $\{0,2,3,4\}$. Thus y is even entrywise. On the other hand, the transpose Lehman identity gives $(A^\top y)_c = 4+k$ for every $c \in V$, which is odd. This contradiction proves the claim. □

Theorem 5.90 (positive-Lehman trace exclusion)**.** *Let v be a singleton-normalized exact game, let $\phi = \mathrm{Lmin}(v)$, and let H be a slack cumulative richer suffix at a strict payoff boundary with exactly two poorer players. Suppose a lifted extreme packet furnished by Theorem 5.74 has gaining-poor load below one, has a square rich trace-incidence matrix A, and gives every positive trace weight $1/r$. Suppose A has a zero–one Lehman dual B satisfying*

$$AB^\top = J + kI, \qquad k \geq 1, \tag{LH}$$

where A is r-regular and B is s-regular. Let U and V be the sets of packet columns whose lifted tight coalitions contain respectively the losing and gaining poor players. For a column set Z, put

$$\Pi_B(Z) := \sum_d \binom{|\operatorname{row}_d(B) \cap Z|}{2}. \tag{LH-P}$$

The packet is impossible whenever

$$s|V| - 2\Pi_B(V) > \Pi_B(U). \tag{LH-D}$$

In particular, if any two columns of B occur together in at most one row, it is enough that

$$|V|(s - |V| + 1) > \binom{|U|}{2}. \tag{LH-L}$$

The packet is also impossible, without either displayed density hypothesis, whenever $|V| \leq 2$.

Proof. The packet transport inequality gives $0 \leq |U| < |V| < r$, because its poor loads are $|U|/r$ and $|V|/r$. For each row d of B, write $u_d = |\operatorname{row}_d(B) \cap U|$ and $w_d = |\operatorname{row}_d(B) \cap V|$, and let n_1 count the rows with $w_d = 1$. Column regularity of B gives $\sum_d w_d = s|V|$. Since $t \leq 2\binom{t}{2}$ for every integer $t \geq 2$,

$$n_1 = s|V| - \sum_{d:w_d \geq 2} w_d \geq s|V| - 2\Pi_B(V). \tag{LH-1}$$

If every row counted by n_1 had $u_d \geq 2$, each would contribute at least one to $\Pi_B(U)$, contradicting (LH-D). Hence some row d has $w_d = 1$ and $u_d \leq 1$.

Use that row of B as a zero–one selector of the lifted tight coalitions. Equation (LH) says that its rich incidence load is $\mathbf{1}_H + ke_d$; its gaining-poor load is one and its losing-poor load is at most one. It is therefore a safe rooted tight cover, forbidden by Theorem 5.79.

If B is column-linear, every pair of columns contributes at most one to (LH-P). Thus $\Pi_B(Z) \leq \binom{|Z|}{2}$, and (LH-L) implies (LH-D).

Finally, the density condition is automatic when $|V| \leq 2$. For $|V| = 1$, $\Pi_B(V) = 0$ and $|U| = 0$. For $|V| = 2$, one has $|U| \leq 1$ and $\Pi_B(U) = 0$. The two columns of B are distinct because (LH) is nonsingular, so their codegree is at most $s-1$. Hence $s|V| - 2\Pi_B(V) \geq 2 > 0$. □

The real-valued overload budget also permits interpolation with a dual row that meets the gaining labels more than once. This yields a row test strictly at the level of the original packet, without an integer safe-cover requirement.

Corollary 5.91 (two-sided convex Lehman-row exclusion)**.** *Under the hypotheses and notation of Theorem 5.90, put $p = |U|$, $\ell = |V|$, and for a row d of B write*

$$u_d = |\operatorname{row}_d(B) \cap U|, \qquad w_d = |\operatorname{row}_d(B) \cap V|.$$

Define its label residual by

$$F_d := (r-p)w_d + p - \ell - (r-\ell)u_d. \tag{LH-R}$$

The packet is impossible if some row satisfies either

$$w_d \geq 1, \qquad 0 \leq F_d \leq k(r-\ell), \tag{LH-C$_+$}$$
$$u_d \geq 1, \qquad -k(r-p) \leq F_d \leq 0. \tag{LH-C$_-$}$$

The positive band is equivalently

$$rw_d - \ell - k(r-\ell) \;\leq\; p(w_d-1) + u_d(r-\ell) \;\leq\; rw_d - \ell. \tag{LH-C}$$

For $w_d = 1, u_d \leq 1$, it contains the safe-row conclusion used in Theorem 5.90. More strongly, writing $F = (F_d)_d$, the residuals obey the local transport identity

$$A^\top F = k\big((r-p)\mathbf{1}_V - (r-\ell)\mathbf{1}_U\big). \tag{LH-T}$$

Thus every packet column $c \notin U \cup V$ has the exact trace balance

$$\sum_{d \in \operatorname{col}_c(A)} F_d = 0. \tag{LH-T$_0$}$$

If neither exclusion band applies anywhere, every such neutral column meets both a positive-residual row and a negative-residual row. Summing the local identity gives the conservation law

$$\sum_d F_d = k(\ell - p) > 0. \tag{LH-S}$$

Since $w_d = 0$ implies $F_d < 0$, some row with $w_d \geq 1$ has positive residual. Thus any packet not excluded by the corollary must have a row with $F_d > k(r-\ell)$.

Proof. The uniform packet has losing and gaining loads

$$\alpha := \frac{p}{r} < \beta := \frac{\ell}{r} < 1.$$

For a row with $w_d \geq 1$, put

$$t := \frac{1-\beta}{w_d - \beta} \in (0, 1].$$

Take weight $1-t$ on the uniform packet and weight t on the zero–one coalition selector given by row d of B. The Lehman identity makes the rich load $\mathbf{1}_H + tke_d$. The gaining-poor load is exactly $(1-t)\beta + tw_d = 1$, while the losing-poor load is

$$a_d := (1-t)\alpha + tu_d = \frac{p(w_d-1) + u_d(r-\ell)}{rw_d - \ell}. \tag{LH-C1}$$

The inequalities in (LH-C$_+$) are exactly $a_d \le 1$ and $tk \ge 1 - a_d$.

Apply Corollary 5.56 to this nonnegative real combination. It gives

$$tk\,\phi_d \le (1 - a_d)\phi_a. \tag{LH-C2}$$

If $a_d < 1$, then $tk \ge 1 - a_d > 0$ and the strict boundary $\phi_d > \phi_a$ contradicts (LH-C2). If $a_d = 1$, the right side of (LH-C2) is zero while $tk\phi_d > 0$, again a contradiction.

When $w_d = 1$, one has $t = 1$. The positive band reduces to $(1-k)(r-\ell) \le u_d(r-\ell) \le r-\ell$, which holds for $u_d \in \{0, 1\}$ because $k \ge 1$.

For the negative band, put

$$\widehat{t} := \frac{1-\alpha}{u_d - \alpha} = \frac{r-p}{ru_d - p} \in (0, 1].$$

Interpolate with the same dual-row selector using weight $\widehat{t}$. The losing-poor load is now exactly one, while the gaining-poor load is

$$b_d := (1-\widehat{t})\beta + \widehat{t}w_d = \frac{\ell(u_d - 1) + w_d(r-p)}{ru_d - p}. \tag{LH-C3}$$

The inequality $F_d \le 0$ is exactly $b_d \le 1$, and $F_d \ge -k(r-p)$ is exactly $\widehat{t}k \ge 1 - b_d$. Corollary 5.56 gives

$$\widehat{t}k\phi_d \le (1 - b_d)\phi_b, \tag{LH-C4}$$

which contradicts $\phi_d > \phi_b$ exactly as above, including the endpoint $b_d = 1$.

For the final claims, regard $u = B\mathbf{1}_U$, $w = B\mathbf{1}_V$, and $F = (r-p)w + (p-\ell)\mathbf{1} - (r-\ell)u$. The classical transpose identity $A^\top B = J + kI$ and $A^\top \mathbf{1} = r\mathbf{1}$ give

$$\begin{aligned} A^\top F &= (r-p)(\ell\mathbf{1} + k\mathbf{1}_V) + (p-\ell)r\mathbf{1} - (r-\ell)(p\mathbf{1} + k\mathbf{1}_U) \\ &= k\big((r-p)\mathbf{1}_V - (r-\ell)\mathbf{1}_U\big), \end{aligned}$$

because the common all-ones coefficient cancels. This proves (LH-T) and its neutral-column specialization. Left-multiply the transport identity by $\mathbf{1}^\top$. Since every column of A has sum r, its left side is $r\sum_d F_d$, while its right side is

$$k\big((r-p)\ell - (r-\ell)p\big) = kr(\ell - p).$$

This proves (LH-S). If $w_d = 0$, then $F_d = p - \ell - (r-\ell)u_d < 0$. Conversely, a zero residual on a row with $w_d \ge 1$ lies in the positive exclusion band. Hence a surviving packet has no zero residuals, and (LH-T$_0$) forces both signs in every neutral column. The zero-V observation also proves the final assertion of the corollary. □

The separation lemma closes the next label cardinality without a density or codegree hypothesis.

Corollary 5.92 (three-gaining-column exclusion)**.** *Under the hypotheses of Theorem 5.90, the packet is impossible whenever $|V| \le 3$. Consequently any surviving positive Lehman packet has*

$$|V| \ge 4, \qquad r \ge 5. \tag{LH-4}$$

Proof. Theorem 5.90 already handles $|V| \le 2$. Suppose $|V| = 3$. Then $r \ge 4$, and Lemma 5.88 supplies a row d with $w_d = 1$. Put $p = |U| \le 2$. Its residual is

$$F_d = (r-3)(1-u_d). \tag{LH-3c}$$

If $u_d = 0$, then $0 < F_d = r - 3 \le k(r-3)$; if $u_d = 1$, then $F_d = 0$. In both cases the positive band (LH-C$_+$) applies. If $u_d = 2$, necessarily $p = 2$, and

$$-k(r-p) = -k(r-2) \le -(r-3) = F_d < 0,$$

so the negative band (LH-C$_-$) applies instead. These cases are exhaustive because $u_d \le p \le 2$. Finally $|V| < r$ gives $r \ge 5$ for any survivor. □

Thus the first unresolved positive-Lehman branch is not merely beyond the graph-like case: it requires at least four gaining labels on a trace degree of at least five. This strengthens the dimension-free obstruction hierarchy behind the sharp low-dimensional theorem. It does not eliminate the opposite-sign co-singleton matrix $J - I$ realized by the sharp family.

The first open cardinality has a rigid local form in the thin case. The following proposition is a structural reduction, not an exclusion theorem.

Proposition 5.93 (thin four-label normal form)**.** *Under the hypotheses of Theorem 5.90, suppose $k = 1$, $|V| = 4$, and neither residual band excludes the packet. Then*

$$r \ge 6, \qquad |U| = 3, \qquad |U \cap V| \le 1. \tag{LH-4a}$$

Every row with $w_d = 1$ has $u_d = 3$, and every row with $u_d = 1$ has $w_d \ge 2$.

Moreover, let $c \notin U \cup V$ be a label-free packet column. Ignore rows $d \in \mathrm{col}_c(A)$ with $(w_d, u_d) = (0,0)$. The multiset of all remaining load pairs in that trace is exactly one of

1	$(4,3)$
2	$(1,3),(3,0)$
3	$(0,3),(4,0)$
4	$(0,3),(2,0),(2,0)$
5	$(0,2),(4,1)$
6	$(0,2),(2,0),(2,1)$.

(LH-4b)

If $|U \cap V| = 1$, only types $1,2,5,6$ are feasible.

Proof. Lemma 5.89 supplies a row with $w_d = 1$. For $\ell = 4$, such a row has

$$F_d = (r-4)(1-u_d).$$

The positive band excludes $u_d \in \{0,1\}$, and the negative band excludes $u_d = 2$. If $p := |U| \le 2$, these cases are exhaustive, so $p = 3$ and every V-simple row has $u_d = 3$. When $r = 5$, the remaining value $u_d = 3$ gives $F_d = -2 = -k(r-p)$ and is also excluded. Hence $r \ge 6$. A row containing all three U-columns and only one V-column is impossible when $|U \cap V| \ge 2$, proving the overlap bound.

Apply Lemma 5.88 to the three columns of U. More generally, on any row with $u_d = 1$ one has

$$F_d = (r-3)(w_d - 1).$$

The negative band excludes $w_d = 0$ at equality, and either band excludes $w_d = 1$. Thus every such row has $w_d \geq 2$.

Now fix $c \notin U \cup V$. The transpose Lehman identity gives

$$\sum_{d \in \mathrm{col}_c(A)} w_d = 4, \qquad \sum_{d \in \mathrm{col}_c(A)} u_d = 3. \tag{LH-4c}$$

The two residual bands leave the following nonzero row types:

w_d	possible u_d
0	$2, 3$
1	3
2	$0, 1$
3	$0, 1, 2$
4	$0, 1, 2, 3.$

(LH-4d)

Partition the total w-load four. A part four yields types $1, 3, 5$ after filling the remaining u-load; a $3+1$ partition yields type 2; and a $2+2$ partition yields types $4, 6$. Two or more parts of size one already have total u-load at least six. This proves that (LH-4b) is exhaustive. If $|U \cap V| = 1$, a row with $w_d = 0$ cannot have $u_d = 3$, eliminating types $3, 4$. □

The local load restrictions already identify a canonical column globally. This is the first point at which the six local tiles collapse to an exact Lehman incidence rather than merely a list of possibilities.

Proposition 5.94 (heavy-row column rigidity)**.** *Under the hypotheses of Proposition 5.93, let*

$$H_U := \{d : u_d \geq 2\}.$$

There is a unique column $a_* \in U$ whose A-trace contains two rows of H_U; every other A-column contains exactly one. Moreover

$$|H_U| = s, \qquad H_U = \mathrm{col}_{a_*}(B). \tag{LH-4e}$$

Equivalently, a row of B contains a_* if and only if it contains at least two columns of U.

Proof. For every column $c \notin U$, the transpose Lehman identity gives

$$\sum_{d \in \mathrm{col}_c(A)} u_d = 3.$$

If c is label-free, the six types in (LH-4b) show directly that its trace contains exactly one row of H_U. If $c \in V \setminus U$, three rows with $u_d = 1$ would have total w-load at least six by Proposition 5.93, whereas the transpose identity gives total w-load five. Thus this trace also contains exactly one heavy row.

For $c \in U$, the total u-load is four. There can be at most two heavy rows. There must be at least one: otherwise four rows have $u_d = 1$ and therefore total w-load at least eight, while the actual total is four when $c \notin V$ and five when $c \in V$. Hence every U-column contains one or two rows of H_U.

Let t be the number of U-columns containing two heavy rows. Double-count the incidences between H_U and the columns of A. Every column outside U contributes one, while the three columns of U contribute $3+t$. Therefore

$$r|H_U| = (n-3) + (3+t) = n + t = rs - 1 + t. \tag{LH-4f}$$

Here $0 \le t \le 3$ and $r \ge 6$. Divisibility by r forces $t = 1$ and $|H_U| = s$. If a_* is the unique doubled column, the trace counts say

$$A^\top \mathbf{1}_{H_U} = \mathbf{1} + \mathbf{1}_{\{a_*\}}.$$

But $A^\top B = J + I$ gives the same vector as $A^\top B \mathbf{1}_{\{a_*\}}$. Since A is nonsingular, $\mathbf{1}_{H_U} = B\mathbf{1}_{\{a_*\}}$, proving the claim. □

Corollary 5.95 (dual-cubic four-label exclusion)**.** *Under the hypotheses of Proposition 5.93, one has $s \ge 4$. Thus no positive thin Lehman pair whose dual matrix B is cubic supports a surviving four-label packet.*

Proof. Suppose $s = 3$. A row with $(w_d, u_d) = (1, 3)$ exists. Since its B-row has only three entries, $U \cap V$ must be the singleton $\{g\}$ and that row is exactly U. Every row with $u_d = 3$ is therefore identical to it; distinctness of the rows of the nonsingular matrix B shows that there is exactly one such row.

Proposition 5.94 gives $|H_U| = s = 3$, so the other two heavy rows have $u_d = 2$. The allowed row types (LH-4d) give $w_d \in \{0, 3, 4\}$ on a row of u-load two. Here $w_d = 4$ is impossible because the row has size three. The value $w_d = 3$ is also impossible: a three-entry row contained in V cannot contain two entries of U, because $U \cap V = \{g\}$. Hence both rows have $w_d = 0$. They avoid g and contain the other two columns of U. The unique $u_d = 3$ row contains those same two columns. Consequently the two noncommon columns of U have identical B-columns, namely H_U, contrary to nonsingularity of B. □

The rows that carry the negative extreme in the preceding normal form cannot overlap arbitrarily on the primal side. The classical triple-intersection property of a Lehman pair turns them into a genuine packing.

Proposition 5.96 (simple-row star packing)**.** *Under the hypotheses of Proposition 5.93, put*

$$Q := \{d : u_d = 3\}.$$

The supports $\mathrm{row}_d(A)$*, $d \in Q$, are pairwise disjoint. Consequently*

$$1 \le \#\{d : w_d = 1\} \le |Q| \le s - 1. \tag{LH-4g}$$

Moreover, every row d with $(w_d, u_d) = (1, 3)$ contains at least $r - 5$ columns $c \notin U \cup V$ in its A-support. On each such column the neutral trace is necessarily type 2 of (LH-4b): apart from zero-load rows it consists of d and a unique row e with

$$(w_e, u_e) = (3, 0). \tag{LH-4h}$$

Proof. Fix distinct columns $a, b \in U$. The column form of the classical triple-intersection property, applied to the swapped Lehman pair (B, A), says that for every column c of A,

$$\left|\mathrm{col}_a(B) \cap \mathrm{col}_b(B) \cap \mathrm{col}_c(A)\right| \le 1. \tag{LH-4i}$$

Indeed, if $c \neq a$, the displayed intersection is contained in $\mathrm{col}_a(B) \cap \mathrm{col}_c(A)$, which has size one by $B^\top A = J + I$; if $c = a$, use $b \neq c$ instead. Every row in Q belongs to both B-columns a, b. Hence two rows in Q cannot belong to the same A-column, which proves that their A-supports are disjoint. Each support has size r, while there are $n = rs - 1$ columns, so $r|Q| \le rs - 1$ and therefore $|Q| \le s - 1$. The lower bounds follow from Proposition 5.93 and Lemma 5.89.

Now let $(w_d, u_d) = (1, 3)$ and put $L = U \cup V$. If $U \cap V$ is empty, then B-row d contains four labels of L, leaving three labels of L outside that row. If $|U \cap V| = 1$, its unique V-label must be the common label, so it contains three labels of L and again leaves three outside. In either case the diagonal Lehman identity gives $|\,\mathrm{row}_d(A) \cap \mathrm{row}_d(B)| = 2$, whence

$$|\,\mathrm{row}_d(A) \cap L| \le 2 + 3 = 5.$$

Thus at least $r - 5$ columns in $\mathrm{row}_d(A)$ are label-free.

Fix one such column c. Its trace has total load $(4, 3)$ by (LH-4c). Row d already contributes $(1, 3)$, so all other rows have u-load zero and total w-load three. The surviving row types in (LH-4d) allow no positive $u = 0$ contribution of w-load one. The only possible completion is therefore one row of type $(3, 0)$, together with zero-load rows. This proves (LH-4h). □

The same interpolation works for an arbitrary set of dual rows. Unlike a general convex combination, their uniform average overloads a rich coalition by a constant amount, so exactness can evaluate that entire face.

Corollary 5.97 (subset Lehman residual face budget)**.** *Under the hypotheses and notation of Corollary 5.91, let D be a nonempty set of rows of B, identifying row indices with the corresponding rich players in H. Put $d = |D|$ and write*

$$U_D := \sum_{j \in D} u_j, \qquad W_D := \sum_{j \in D} w_j, \qquad F(D) := \sum_{j \in D} F_j.$$

If $W_D \ge d$ and $F(D) \ge 0$, then

$$k(r - \ell)s_\phi(D) \le F(D)\phi_a. \tag{LH-B$_+$}$$

If $U_D \ge d$ and $F(D) \le 0$, then

$$k(r - p)s_\phi(D) \le -F(D)\phi_b. \tag{LH-B$_-$}$$

In particular, under either average-load hypothesis, $F(D) = 0$ forces D to be tight at ϕ.

Proof. Average the zero–one selectors given by the rows in D. The Lehman identity makes its rich load

$$\mathbf{1}_H + \frac{k}{d}\mathbf{1}_D,$$

and its poor loads are U_D/d and W_D/d.

First suppose $W_D \geq d$. Interpolate this average with the uniform packet using

$$t := \frac{d(r-\ell)}{rW_D - d\ell} \in (0,1].$$

The gaining-poor load becomes one, the rich overload on D is

$$q = \frac{tk}{d} = \frac{k(r-\ell)}{rW_D - d\ell},$$

and direct simplification gives losing-poor deficit

$$1 - a_D = \frac{F(D)}{rW_D - d\ell}.$$

Thus $F(D) \geq 0$ makes both poor loads at most one. Apply Corollary 5.56 and clear the positive denominator to obtain (LH-B$_+$).

If $U_D \geq d$, interpolate instead with

$$\widehat{t} := \frac{d(r-p)}{rU_D - dp} \in (0,1].$$

The losing-poor load becomes one, the rich overload is $k(r-p)/(rU_D - dp)$ on D, and the gaining-poor deficit is $-F(D)/(rU_D - dp)$. The hypothesis $F(D) \leq 0$ and the same overload budget give (LH-B$_-$). If $F(D) = 0$, either right side is zero, so $s_\phi(D) = 0$. □

The corollary does not assert that every surviving labeled Lehman packet has a zero-residual set satisfying an average-load hypothesis, nor that a tight set obtained this way inherits a smaller positive Lehman packet. Those are the unresolved existence and recursion steps required for a dimension-free classification.

Equation (LH) and its role as the core identity for nondegenerate minimally nonideal matrices are classical (Cornuéjols et al. 2009). The new claim here is only the candidate two-poor-label exclusion and face-budget mechanism obtained by combining that identity with the exact-face obstruction. The alternating selection on an odd trace cycle is the $r = 2$ instance. For a self-dual projective-plane matrix, the linear bound (LH-L) is the coarse global density bound used below. By contrast, the surviving co-singleton support $J - I$ satisfies the opposite-sign identity $(J-I)I = J - I$, not a positive Lehman identity; this locates the sharp family on the other side of the structural divide.

The same counting argument can retain noncore rows rather than assuming that the Lehman fractional vertex covers them exactly. The key MNI fact is that rows of the blocker core are transversals of every row of the original clutter; hence a dual selector cannot miss a noncore row and need only avoid hitting it twice.

Corollary 5.98 (noncore-aware Lehman rounding)**.** *Let A, B be an n-row positive Lehman pair satisfying $AB^\top = J + kI$, with A r-regular and B s-regular. Let U, V be column sets, define Π_B by* (LH-P)*, and let $\mathcal{E}$ be a finite family of additional zero–one row supports on the same columns such that*

$$|\operatorname{row}_d(B) \cap W| \geq 1 \qquad (d \in [n],\ W \in \mathcal{E}). \tag{NC-1}$$

If

$$s|V| - 2\Pi_B(V) > \Pi_B(U) + \sum_{W \in \mathcal{E}} \big(s|W| - n\big), \tag{NC-2}$$

then some row d of B meets V exactly once, meets U at most once, and meets every $W \in \mathcal{E}$ exactly once.

Consequently, suppose these columns lift to proper ϕ-tight coalitions at a two-poorer exact-game boundary with rich suffix $\widetilde{H}$, the n core rows and the rows in $\mathcal{E}$ together represent every uncontracted rich player, and any contracted columns are poor-label-free and partition the remaining rich players, which the surviving columns avoid. If U, V are the losing- and gaining-poor label sets, then (NC-2) *is impossible.*

Proof. As in (LH-1), at least $s|V| - 2\Pi_B(V)$ dual rows meet V exactly once. At most $\Pi_B(U)$ of those rows meet U at least twice. For a fixed $W \in \mathcal{E}$, every dual row meets W by (NC-1), while each row that meets W at least twice contributes at least one to $|\operatorname{row}_d(B) \cap W| - 1$. Column regularity and $n = rs - k$ give

$$\sum_d \big(|\operatorname{row}_d(B) \cap W| - 1\big) = s|W| - n = s(|W| - r) + k.$$

Thus at most $s|W| - n$ rows fail to meet W exactly once. (This number is nonnegative under (NC-1).) The union bound and (NC-2) give the required row d.

Select the lifted coalitions indexed by that row, together with every contracted column. The Lehman identity gives load $\mathbf{1} + ke_d$ on the core rich rows, the choice of d gives load one on every additional row, and the contracted columns cover their disjoint rich rows once. The poor load is nonzero and zero–one. This is a safe rooted tight cover, forbidden by Theorem 5.79. □

For an exact-cover lift of a Lehman minor, the apparent label-preservation problem has a useful one-sided resolution. It is enough to preserve the gaining-player deficit; the transport inequality then removes both poor labels from every contracted column automatically. A general MNI minor may also have noncore rich rows overcovered by its Lehman fractional vertex; the corollary does not suppress that separate excess-row issue. If instead a minor lift preserves the scaled gap of Corollary 5.82, even one such excess is already too expensive, so exact coverage and the gaining-player deficit follow simultaneously.

Corollary 5.99 (deficit-preserving Lehman-minor lift)**.** *Let v be singleton-normalized and exact, let $\phi = \operatorname{Lmin}(v)$, and let H be a slack cumulative richer suffix at a strict boundary with poorer side $\{a, b\}$. Let z be a core point exact on H, put $\Delta = z - \phi$, and label the poor players so that $\Delta_a < 0 < \Delta_b$. Let $\mathcal{F}$ be lifted proper ϕ-tight coalitions, taken with unit weight, whose pairwise disjoint rich traces partition $H \setminus K$ for some nonempty $K \subseteq H$. On the remaining rows K, suppose lifted tight coalitions have rich traces contained in K, square trace matrix A, uniform weight $1/r$, and a positive Lehman dual B satisfying*

$$AB^\top = J + kI,$$

with A r-regular and B s-regular. Let U, V label the core columns containing respectively a, b, and let t_a, t_b count the members of $\mathcal{F}$ containing respectively a, b.

If the combined lifted packet has gaining-player load below one,

$$t_b + \frac{|V|}{r} < 1, \tag{LM-1}$$

then

$$t_a = t_b = 0, \qquad 0 \le |U| < |V| < r. \tag{LM-2}$$

Consequently the combined packet is impossible whenever (LH-D) *holds (or under either of its sufficient conditions in Theorem 5.90).*

Without a density hypothesis, put $T = s|V|$. *There is a nonempty proper* $D \subset K$ *with* $|D| = T$ *and, writing*

$$u(D) := \sum_{d \in D} |\operatorname{row}_d(B) \cap U|,$$

one has $u(D) < T$ *and*

$$\phi(H) - v(H) < s_\phi(D) \le \frac{T - u(D)}{k} \phi_a. \tag{LM-3}$$

Proof. Write $\sigma = \phi(H) - v(H) > 0$. Summing at the exact point z all tight-coalition inequalities in the combined packet gives

$$-\sigma + \left(t_a + \frac{|U|}{r}\right) \Delta_a + \left(t_b + \frac{|V|}{r}\right) \Delta_b \ge 0. \tag{LM-4}$$

Condition (LM-1) first gives $t_b = 0$ and $|V| < r$. Since $\Delta_b = \sigma - \Delta_a$, the left side of (LM-4) is

$$\left(\frac{|V|}{r} - 1\right) \sigma + \left(t_a + \frac{|U| - |V|}{r}\right) \Delta_a.$$

The first term is strictly negative and $\Delta_a < 0$. Hence the second coefficient is negative, so $rt_a + |U| < |V|$. This proves (LM-2). In particular every member of $\mathcal{F}$ is poor-label-free.

Under the density condition, the counting argument in Theorem 5.90 supplies a row d of B meeting V once and U at most once. Select that dual row together with every member of $\mathcal{F}$. The latter cover $H \setminus K$ once, and the Lehman identity gives load $\mathbf{1}_K + ke_d$ on K. The total rich load is therefore $\mathbf{1}_H + ke_d$, while the poor load is nonzero and zero–one. This is a safe rooted tight cover, contradicting Theorem 5.79.

For the final assertion, let P be the dual rows meeting V. Since $|P| \le s|V| = T$, extend P to a row set D of size T. This is a proper subset of K because $1 + k \le s$, $|V| < r$, and $T \le s(r-1) = |K| + k - s < |K|$; moreover $u(D) \le s|U| < s|V| = T$. Average those rows and add every member of $\mathcal{F}$ with unit weight. The resulting rich load is $\mathbf{1}_H + (k/T)\mathbf{1}_D$, its b-load is one, and its a-load is $u(D)/T < 1$. Corollary 5.56 gives the weak upper bound in (LM-3).

For $d \in K \setminus D$, select the dth dual row together with $\mathcal{F}$. Its displacement inequality is

$$-\sigma + k\Delta_d + u_d \Delta_a \ge 0, \qquad u_d := |\operatorname{row}_d(B) \cap U|,$$

so $\Delta_d > 0$. Every member of $\mathcal{F}$ is now a coalition contained in H and is tight at ϕ, hence its displacement at z is nonnegative. Their disjoint traces partition $H \setminus K$, so $\Delta(H \setminus K) \ge 0$. Since $K \setminus D$ is nonempty, $\Delta(H \setminus D) > 0$. Therefore $\Delta(D) = -\sigma - \Delta(H \setminus D) < -\sigma$, and core feasibility on D gives $s_\phi(D) > \sigma$, as claimed. □

Even when the density test does not produce one safe dual row, positive Lehman duality still forces a nontrivial internal exact-face certificate.

Corollary 5.100 (canonical Lehman internal face)**.** *Under the hypotheses and notation of Theorem 5.90, there is a nonempty proper set* $D \subset H$ *such that, writing*

$$u(D) := \sum_{d \in D} |\operatorname{row}_d(B) \cap U|,$$

one has $|D| = s|V|$, $u(D) < |D|$, *and the strict slack amplification*

$$\phi(H) - v(H) < s_\phi(D) \le \frac{|D| - u(D)}{k}\phi_a, \tag{LH-F}$$

where a *is the losing poor player.*

Proof. Put $T = s|V|$. The diagonal of (LH) gives $1 + k \le s$, while $|V| < r$ and regularity give $T \le s(r-1) = |H| + k - s < |H|$. Let P be the dual rows meeting V. Since $|P| \le \sum_d |\operatorname{row}_d(B) \cap V| = T$, extend P to a set D of exactly T rows. Every row outside D has zero V-load, so

$$\sum_{d \in D} |\operatorname{row}_d(B) \cap V| = s|V| = T.$$

On the other hand, $u(D) \le s|U| < s|V| = T$.

For this row set, $W_D = T = |D|$ and

$$F(D) = (r - \ell)(T - u(D)) \ge 0.$$

The positive subset budget (LH-B$_+$) therefore gives the weak upper inequality in (LH-F). Equivalently, this is the average-dual-row construction with gaining-poor load one, losing-poor load $u(D)/T$, and rich overload $(k/T)\mathbf{1}_D$.

It remains to prove the strict lower inequality. Use the core point z and displacement $\Delta = z - \phi$ attached to the lifted packet by Theorem 5.74; put $\sigma = \phi(H) - v(H) > 0$, so $\Delta(H) = -\sigma$ and, after the packet's labeling, $\Delta_a < 0 < \Delta_b$. For every $d \notin D$, the dth dual row has zero V-load. Summing at z the tight-coalition inequalities selected by that row gives

$$-\sigma + k\Delta_d + u_d\Delta_a \ge 0, \qquad u_d := |\operatorname{row}_d(B) \cap U|.$$

Hence $k\Delta_d \ge \sigma - u_d\Delta_a > 0$. Since $H \setminus D$ is nonempty, $\Delta(H \setminus D) > 0$, and therefore

$$\Delta(D) = -\sigma - \Delta(H \setminus D) < -\sigma.$$

Core feasibility gives $v(D) \le z(D)$, so $s_\phi(D) \ge \phi(D) - z(D) = -\Delta(D) > \sigma$, as required. □

Theorem 5.101 (proper-hyperedge obstruction)**.** *Let* v *be a singleton-normalized exact game, let* $\phi = \operatorname{Lmin}(v)$*, and let* H *be a cumulative richer suffix at a strict payoff boundary with exactly two poorer players* $L = \{a, b\}$*. If* H *is slack at* ϕ*, label the poor players by the exact-face displacement* $\Delta_a < 0 < \Delta_b$*. Every lifted extreme packet from Theorem 5.74 whose gaining-player load satisfies* $r_b < 1$ *contains a positive proper rich trace* R *with*

$$3 \le |R| \le |H| - 1. \tag{GH}$$

Such a packet always exists. Consequently every slack two-poorer boundary has $|H| \ge 4$*; in particular, if* $|H| \le 3$*, then* H *is tight. More generally,* H *is tight whenever every proper active rich trace has cardinality at most two.*

Proof. Suppose $s := \phi(H) - v(H) > 0$. Use the lifted extreme-point packet mixture in Theorem 5.74, relabeling the poor players so that

$$\Delta_a < 0 < \Delta_b, \qquad \Delta_a + \Delta_b = s. \tag{RT-1}$$

Fix any positive-weight lifted extreme packet with $r_b < 1$ and suppose, toward a contradiction, that it has no proper trace of cardinality at least three. Its minimal balanced trace support cannot contain H: the singleton collection $\{H\}$ is already balanced, so minimality would force it to be the whole support with weight one; then $r_b < 1$ would give $r_b = 0$, and the packet inequality $-s + r_a\Delta_a \geq 0$ would be impossible. Hence every positive trace of the fixed packet has size at most two.

Its support is therefore a disjoint union of singleton traces and pair traces of weight one, together with vertex-disjoint odd cycles whose pair traces have weight one half. Here is the elementary argument. The positive incidence columns at an extreme point are linearly independent. Consider a connected pair-graph component on n vertices, with e pair columns and q singleton columns. Independence gives $e + q \leq n$. If $q > 0$ and $n > 1$, connectedness gives $e \geq n - 1$, so $q = 1$, $e = n - 1$, and the pair graph is a tree. Choose a leaf distinct from the singleton vertex. Its equation forces its incident pair weight to one, so the equation at its neighbour forces another positive incident column to zero; this also covers the two-vertex endpoint, where the singleton itself is forced to zero. Hence a positive singleton component stands alone. With $q = 0$, one has $e \in \{n - 1, n\}$. The same leaf argument reduces a tree component to a single edge. A unicyclic component cannot have an attached tree, again by the leaf argument, so it is a cycle. An even cycle has dependent incidence columns; an odd cycle is nonsingular and its cover equations force weight one half.

An integral packet has integer load on b, and its transport inequality forces that load to be at least one. Thus the fixed packet is fractional. All its coefficients are in $\{1, 1/2\}$, so r_b is a nonnegative half-integer. Its transport inequality

$$-s + r_a\Delta_a + r_b\Delta_b \geq 0 \tag{RT-2}$$

first gives $r_b > 0$, hence $r_b = 1/2$. If $r_a > 0$, then $r_a \geq 1/2$ and

$$-s + r_a\Delta_a + r_b\Delta_b \leq -s + \tfrac{1}{2}(\Delta_a + \Delta_b) = -\tfrac{1}{2}s < 0.$$

Thus $r_a = 0$. Consequently exactly one half-weight edge coalition on one odd cycle contains b, and no supported coalition contains a.

Write that labeled odd cycle as $e_0, e_1, \ldots, e_{2k}$ in cyclic order, with the unique b-carrying coalition over e_0. Replace the half weights on this component by unit weights on the alternating edges

$$e_0, e_2, \ldots, e_{2k}.$$

These edges cover every vertex of the labeled cycle once except their common endpoint $d = e_0 \cap e_{2k}$, which they cover twice. Leave every other packet component at its original exact-cover weights, including weight $1/2$ on any other odd cycle. Because no other supported coalition carries a poor player, the resulting nonnegative *real* combination of proper ϕ-tight coalitions has rich load $\mathbf{1}_H + e_d$ and poor load e_b. Corollary 5.56, with overload set $\{d\}$ and $q = 1$, gives $\phi_d = s_\phi(\{d\}) \leq \phi_a$, contradicting the strict boundary.

The use of the real-valued overload budget is essential when the packet has additional odd-cycle components: leaving those components at half weight is not an integer semigroup fibre. Thus the argument proves the trace exclusion, but only a one-cycle packet directly yields the safe rooted *integer* cover of Theorem 5.79.

Hence every underloaded extreme packet contains a proper trace satisfying (GH). Finally, (BT-4) makes the mixture's average b-load strictly less than one, so at least one such packet

exists. Its proper trace has size at least three, forcing $|H| \geq 4$. The last assertion follows because a proper active trace of size at least three has just been shown necessary. □

Theorem 5.102 (uniqueness of the first hypergraphic obstruction)**.** *Under the hypotheses of Theorem 5.101, suppose additionally that $|H| = 4$. If H is slack at ϕ, then the active trace exact-cover polytope has the co-singleton vertex*

$$\beta_{H\setminus\{h\}} = \frac{1}{3} \quad (h \in H), \qquad \beta_R = 0 \quad \textit{otherwise.} \tag{FO}$$

In particular, every co-singleton $H \setminus \{h\}$ is the trace of a proper ϕ-tight coalition. Thus the first possible two-poorer obstruction is, up to relabeling, exactly the fractional co-singleton packet realized by the sharp six-player family.

Proof. Assume slack and use the extreme-point packet mixture from balanced-trace transport, with $\Delta_a < 0 < \Delta_b$ and $\Delta_a + \Delta_b = s$. Choose a lifted packet with $r_b < 1$. Writing $\delta = -\Delta_a > 0$, its transport inequality becomes

$$s(r_b - 1) + \delta(r_b - r_a) \geq 0, \tag{FO-1}$$

so necessarily $r_a < r_b$. The packet is fractional because an integral packet has integer b-load at least one.

We first record the elementary four-point exact-cover classification. At a vertex, the positive trace-incidence columns are linearly independent, so there are at most four. The full trace H can occur only as the integral vertex by itself. For proper traces, group the support by the number t of three-player traces. The graph argument in Theorem 5.101 handles $t = 0$. For $t = 1, 2, 3, 4$, relabel the missing vertices of the triples and subtract the four cover equations. Positivity and column independence leave the following complete list of fractional vertices, up to a permutation of $H = 1234$:

	positive traces	weights
A	$1;\ 23, 24, 34$	$1;\ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
B	$1, 23, 24, 134$	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
C	$12, 134, 234$	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
D	$12, 13, 14;\ 234$	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3};\ \frac{2}{3}$
E	$123, 124, 134, 234$	$\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}$.

(FO-2)

Indeed, the cases $t = 0, 1, 2, 3, 4$ give respectively A; B or D; C; no positive independent solution; and E. Substitution in the four cover equations verifies every displayed row and also shows directly that no omitted coefficient can be positive.

Types A–D contradict exactness. To see this without checking poor labels one by one, call a selection of supported lifted coalitions a *safe root rounding* if its rich load is $\mathbf{1}_H + e_d$ for some $d \in H$, each poor load is zero or one, and at least one poor player is loaded. In types A–C, equation (FO-1) and half-integrality imply that exactly one half-weight coalition carries b and none carries a; a root rounding containing that coalition is visible from

A	the singleton and the two alternating triangle edges,
B	$\{24, 134\},\ \{23, 134\},\ \{1, 23, 24\}$,
C	$\{12, 234\},\ \{12, 134\}$.

(FO-3)

The displayed choices collectively contain every possible half-weight support column.

For type D, multiply the poor loads by three. The integer $3r_b < 3$ is one or two, while $3r_a < 3r_b$. If b occurs on the weight $2/3$ triple, no pair carries b and at least two pairs carry no a; choose one of them with 234. Otherwise exactly one or two of the three pairs carry b, the triple carries no a, and one can choose a b-carrying pair that carries no a. Thus one of

$$\{234, 12\}, \qquad \{234, 13\}, \qquad \{234, 14\} \tag{FO-4}$$

is a safe root rounding.

Finally, every safe root rounding displayed above is a safe rooted tight cover in the sense of Definition 5.78, with $k = 1$. Theorem 5.79 rules it out. Hence the underloading packet cannot have type A–D and must have type E, which is exactly (FO). □

Theorem 5.103 (projective-plane trace density barrier)**.** *Under the two-poorer hypotheses of Theorem 5.101, suppose a lifted extreme packet furnished by balanced-trace transport has gaining-poor load below one and has as its positive rich traces the lines of a finite projective plane of order q, each with weight $1/(q+1)$. Let A and B be the sets of packet lines whose lifted tight coalition contains respectively the losing poor player a and the gaining poor player b, and put $C = A \cap B$, $D = A \setminus B$. The packet is impossible if either*

$$|B|(q + 2 - |B|) > |C||D| + \binom{|D|}{2} \tag{PP}$$

or

$$|D| < 2(q + 2 - |B|). \tag{PP-L}$$

In particular, every such packet with $|B| \le 6$ is impossible, regardless of q. Hence no projective-plane packet of order at most six can serve as the required underloading witness at a slack two-poorer boundary. At order seven the only count not excluded by this theorem is $(|A|, |B|) = (6, 7)$.

Proof. The transport inequality for the underloading packet, in the form (FO-1), gives

$$1 \le |B| \le q, \qquad |A| < |B|, \tag{PP-1}$$

because its poor loads are $|A|/(q+1)$ and $|B|/(q+1)$. Put $a_0 = |A|$ and $b_0 = |B|$, and let n_k count the projective-plane points incident with exactly k lines of B. Counting line–point incidences and pairs of intersecting B-lines gives

$$\sum_k k n_k = b_0(q+1), \qquad \sum_k \binom{k}{2} n_k = \binom{b_0}{2}.$$

Since $k \le 2\binom{k}{2}$ for $k \ge 2$, the number of points incident with exactly one B-line satisfies

$$n_1 \ge b_0(q+1) - 2\binom{b_0}{2} = b_0(q + 2 - b_0). \tag{PP-2}$$

Suppose no point on exactly one B-line lies on at most one A-line. At such a simple point, at most one incident A-line belongs to C, because every C-line is also a B-line. If the unique incident B-line belongs to C, map the point to one incident pair in $C \times D$; otherwise map it to a pair of incident D-lines. Different points yield different line pairs. Consequently

$$n_1 \le |C||D| + \binom{|D|}{2}, \tag{PP-2a}$$

contrary to (PP) and (PP-2).

There is also a local obstruction. Since $|A| < |B|$, choose a line $\ell \in B \setminus A$. At least $q+2-|B|$ points of ℓ are incident with no other B-line. At each one, failure of safe rounding requires at least two incident D-lines. A D-line meets ℓ at only one point, so these lines are all distinct and

$$|D| \geq 2(q + 2 - |B|),$$

contrary to (PP-L).

Under either displayed hypothesis, therefore, some point d lies on exactly one B-line and at most one A-line.

Select the $q+1$ supported lifted coalitions whose trace lines pass through d. Every rich point other than d lies on one selected line, while d lies on all of them, so their rich incidence load is

$$\mathbf{1}_H + qe_d. \tag{PP-3}$$

Their b-load is one and their a-load is at most one. Thus the selected pencil is a safe rooted tight cover with root d and multiplicity q, in contradiction to Theorem 5.79.

Finally, $a_0 < b_0$ and $q \geq b_0$, while $|C||D| + \binom{|D|}{2} \leq \binom{a_0}{2}$. For $b_0 \leq 6$, $b_0(q + 2 - b_0) \geq 2b_0 > \binom{b_0-1}{2} \geq \binom{a_0}{2}$, proving the uniform exclusion. If $q \leq 6$, then $b_0 \leq q \leq 6$. For $q = 7$, the same inequalities leave only $b_0 = 7, a_0 = 6$. □

Corollary 5.104 (all-orders projective-plane exclusion)**.** *No line-incidence packet of any finite projective plane can serve as the underloading witness at a slack two-poorer boundary.*

Proof. Let the plane have order q. Its self-dual line-incidence Lehman pair has

$$r = q + 1, \qquad k = q.$$

Write $p = |A|$ and $\ell = |B|$. Transport gives $0 \leq p < \ell < r$, hence $1 \leq \ell \leq q$. Among any ℓ lines there is a point P on exactly one of them. For example, the simple-point count (PP-2) gives $n_1 \geq \ell(q + 2 - \ell) > 0$. Let u count the A-lines through P; then $w = 1$ and the Lehman residual (LH-R) is

$$F = (r - \ell)(1 - u). \tag{PP-C}$$

If $u \leq 1$, then $0 \leq F \leq r - \ell \leq q(r - \ell) = k(r - \ell)$, so the positive band (LH-C$_+$) excludes the packet. If $u \geq 2$, then

$$0 \leq -F = (r - \ell)(u - 1) \leq (r - \ell)(p - 1) \leq q(r - p) = k(r - p).$$

Here the last inequality follows from $r - \ell \leq r - p$ and $p - 1 \leq q$. The negative band (LH-C$_-$) excludes the packet instead. □

Thus no projective-plane packet of any order can serve as the required underloading witness. The argument is incidence-theoretic and does not assume Desarguesian coordinates. This still does not assert that every higher-dimensional failure contains the four-co-singleton minor: general positive Lehman cores need not be self-dual projective-plane designs, and preserving the scaled objective gap through an arbitrary minimally-nonideal reduction remains open.

Theorem 5.105 (human two-poorer boundary threshold)**.** *Let v be a singleton-normalized exact game, let $\phi = \mathrm{Lmin}(v)$, and let H be a cumulative richer suffix at a strict payoff boundary, with poorer side $L = N \setminus H$. If $|L| = 1$, or if $|L| = 2$ and $|H| \leq 3$, then H is tight at ϕ. The conclusion is sharp: it fails when $|L| = 2$ and $|H| = 4$.*

Proof. If $|L| = 1$, slack would contradict (BT-2). Now put $|L| = 2$. If H were slack, Theorem 5.101 would furnish an underloaded minimal balanced packet containing a proper rich trace R with $3 \le |R| \le |H| - 1$. Hence $|H| \ge 4$, contradicting the hypothesis. This proves the lower bound without a payoff-stratum case split or a finite classification of three-player trace packets.

The earlier 16-query exact-rational classification remains an independent audit rather than a proof dependency:

`src/verify_n5_small_poorer_suffix.py`,
`results/n5_small_poorer_suffix.json`.

For sharpness, take the lower envelope of the following six common-total allocations:

	1	2	3	4	5	6
g^1	0	1	3	3	1	2
g^2	1	0	3	2	1	3
g^3	1	0	3	3	0	3
g^4	1	0	3	3	3	0
g^5	4	0	0	0	3	3
g^6	1	1	2	2	2	2

$$v(S) := \min_{1 \le k \le 6} g^k(S).$$

Every row has total ten, every row is a core allocation of the lower-envelope game, and every coalition attains its worth at a row. Hence v is exact directly from the definition. Put $\phi = g^6$. The generator table gives

$$\begin{aligned} v(N \setminus \{2\}) = 9, \qquad v(\{3,5,6\}) = v(\{4,5,6\}) = 6, \\ v(\{1,3,4,5\}) = v(\{1,3,4,6\}) = 7. \end{aligned} \tag{S6}$$

The point ϕ is in the core and gives every player at least one. At any core point x with $x_i \ge 1$ for all i, the first constraint in (S6) and efficiency force $x_2 = 1$. Write $y = x - \mathbf{1}_N$ and $t = y_5 + y_6$. Then $y \ge 0$, $y_2 = 0$, and $\sum_i y_i = 4$. The middle two constraints give $y_3, y_4 \ge 3 - t$. Adding the last two gives $2(y_1 + y_3 + y_4) + t \ge 6$. Since $y_1 + y_3 + y_4 + t = 4$, first $t \le 2$ and then

$$4 = y_1 + y_3 + y_4 + t \ge y_1 + 6 - t,$$

so $y_1 = 0$. Thus players 1 and 2 are pinned at the first Lmin level one. The remaining four players have total eight, so their next common level is at most two, attained by ϕ. Therefore $\mathrm{Lmin}(v) = (1, 1, 2, 2, 2, 2)$. Finally, for $H = \{3, 4, 5, 6\}$, row g^5 gives $v(H) = 6 < 8 = \phi(H)$.

The fixed generator certificate, all 62 independently recomputed exact faces, and an independent exact staged-Lmin replay are in `src/verify_n6_two_poorer_suffix_counterexample.py` and `results/n6_two_poorer_suffix_counterexample.json`. □

The six-player escape extends across an entire boundary-shape half-plane and also separates suffix tightness from equality of the two lexicographic selections.

Theorem 5.106 (internal-sacrifice half-plane)**.** *For every $m \ge 2$ and $h \ge 4$, there is a singleton-normalized integral exact game on a poorer block P of size m and a tied richest suffix H of size h such that*

$$\mathrm{Lmin}(v) = \mathrm{Lmax}(v) = D\mathbf{1}_P + 2D\mathbf{1}_H, \qquad D = (h-2)(h-3),$$

but

$$v(H) = 2D(h-1) < 2Dh = \mathrm{Lmin}(v)(H).$$

Consequently the exact-partition condition at Lmin is sufficient but not necessary for Lmin–Lmax equality, even within exact games.

Proof. Put $r = h-2$, $D = r(r-1)$, write $P = \{p_1, p_2\} \dot\cup Q$ and $H = \{a, b\} \dot\cup K$, where $|Q| = m-2$ and $|K| = r$. Define the following common-total integral allocations:

$$\begin{aligned} u &= De_{p_2} + D\mathbf{1}_Q + 3D(e_a + e_b) + (2r-1)(r-1)\mathbf{1}_K, \\ z &= 4De_{p_1} + D\mathbf{1}_Q + 2(r^2-1)\mathbf{1}_K, \\ w^k &= De_{p_1} + D\mathbf{1}_Q + 3D(e_a + e_b) + r(2r-1)\mathbf{1}_{K\setminus\{k\}} \quad (k \in K), \\ \phi &= D\mathbf{1}_P + 2D\mathbf{1}_H, \\ q^j &= \phi + D(e_{p_1} - e_j) \quad (j \in Q), \\ v(S) &:= \min\{u(S), z(S), w^k(S)\ (k \in K), \phi(S), q^j(S)\ (j \in Q)\}. \end{aligned} \tag{TF}$$

Every displayed allocation has total $D(m + 2h)$. Hence all of them belong to the core of their lower envelope, and each coalition attains its worth at one of them; the game is exact. Player p_1 is zero at u; players p_2, a, b are zero at z; each $k \in K$ is zero at w^k; and each $j \in Q$ is zero at q^j. Thus the game is singleton-normalized.

Direct substitution gives

$$\begin{gathered} v(N \setminus \{p_2\}) = D(m + 2h) - D, \\ v(H \setminus \{b\}) = v(H \setminus \{a\}) = 2D(h-1), \\ v(\{p_1, a, b\} \cup (K \setminus \{k\})) = D(2h-1) \quad (k \in K), \\ v(H) = 2D(h-1). \end{gathered} \tag{TF-T}$$

The candidate ϕ gives every player at least D. At any core point x with this property, the first constraint and efficiency force $x_{p_2} = D$. Write $y = x - D\mathbf{1}_N$, $E = y(Q)$, and $t = y(K)$. Then $y \geq 0$, $y_{p_2} = 0$, and $y(N) = hD$. The next two constraints give

$$y_a + t \geq (h-1)D, \qquad y_b + t \geq (h-1)D. \tag{TF-1}$$

The r remaining constraints give

$$y_{p_1} + y_a + y_b + t - y_k \geq (h-1)D \quad (k \in K).$$

Summing them and using $r = h - 2$ and $y_{p_1} + E + y_a + y_b + t = hD$ yields $t + rE \leq rD$. Combining this with (TF-1) gives

$$hD = y_{p_1} + E + y_a + y_b + t \geq y_{p_1} + E + 2(h-1)D - t,$$

so $y_{p_1} + E \leq t - rD \leq -rE$. Nonnegativity forces $y_{p_1} = E = 0$, hence $y_j = 0$ for every $j \in Q$. Thus the entire poorer block is pinned at the first Lmin level D. The richer block has fixed total $2Dh$, so its next common level is at most $2D$, attained by ϕ. This proves the Lmin formula, while the last line of (TF-T) gives suffix slack $2D$.

For Lmax, the coalition $H \setminus \{b\}$ has $h-1$ members and worth $2D(h-1)$, so every core point has a coordinate at least $2D$. The point ϕ attains this ceiling. On the face where every coordinate is at most $2D$, the two richer-coalition constraints force every coordinate of H to equal $2D$. The remaining m coordinates have fixed total mD, and their next common ceiling is at least D, attained by ϕ. Hence Lmax also equals ϕ.

Finally, the saturated internal-sacrifice circuit is

$$\mathbf{1}_{H\setminus\{b\}} + \mathbf{1}_{\{b\}} - \mathbf{1}_H = 0. \tag{TF-C}$$

The first coalition is tight, singleton b has slack $2D$, and the negative suffix has slack $2D$. Thus this obstruction is funded entirely inside the richer block, in contrast with the poorer-singleton funding in (UF-C). The exact parameter-grid audit and generic staged replay at $(m, h) = (2, 4)$ are `src/verify_two_poorer_rich_family.py` and `results/two_poorer_rich_family.json`.

This family also meets the necessary hypergraph obstruction in Corollary 5.77 symbolically. The coalitions in (TF-T) have every co-singleton $H \setminus \{x\}$ as an active rich trace. Weighting all h co-singletons by $1/(h-1)$ gives a fractional exact cover. Its support matrix is $J - I$, which is nonsingular, so it is a fractional vertex of the full active-trace polytope. For any three distinct $x_1, x_2, x_3 \in H$, restricting the three corresponding trace columns to the same three player rows gives

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

the smallest odd balanced-matrix obstruction. Thus the sharpness family does not merely have enough players: it realizes exactly the nonintegral trace geometry forced by every two-poorer failure. □

The co-singleton packet is the unique first obstruction, but it is not a universal minor of all larger failures. The entire next support tier admits a dimension-free classification.

The incidence geometry used in that classification is not itself new. Minimal balanced collections and their unique balancing weights are classical, and the three pair traces below are the familiar blown-up three-cycle with weights $(1/2, 1/2, 1/2)$; see Lohmann et al. (2012); Laplace Mermoud et al. (2025). The contribution of the next theorem is instead the Lmin-specific bridge: underloaded transport forces a unique labeling by the two poorer players, exact singleton faces force the two sharp atom-size bounds, and one exact-game construction realizes every admissible atom triple. Its novelty as a combined theorem remains subject to specialist review.

Theorem 5.107 (complete three-column obstruction classification)**.** *Under the hypotheses of Theorem 5.101, let an underloaded lifted extreme packet have exactly three supported tight coalitions. Up to permuting them, there is a partition*

$$H = C \dot\cup A \dot\cup B$$

into nonempty atoms such that its rich traces and weights are

$$C \cup A, \qquad C \cup B, \qquad A \cup B, \qquad \beta_1 = \beta_2 = \beta_3 = \frac{1}{2}. \tag{HT}$$

If a and b are respectively the losing and gaining poorer players, no supported coalition contains a and exactly one contains b. Labeling the last trace in (HT) by b, exactness forces $|A|, |B| \geq 2$. Consequently every such packet has $|H| \geq 5$.

Conversely, for every $t \geq 1$ and $u, w \geq 2$, there is a singleton-normalized exact game with two poorer players and

$$|C| = t, \qquad |A| = u, \qquad |B| = w,$$

such that

$$\mathrm{Lmin}(v) = \mathrm{Lmax}(v) = (1, 1, 2, \ldots, 2), \qquad v(H) = 2|H| - 1 < 2|H|,$$

and its only proper Lmin-tight coalitions lift the three traces in (HT), with the $A \cup B$ lift also containing b. Multiplying all worths and allocations by two makes the game integral. Thus the

support-three obstruction tier begins sharply at five rich and seven total players and does not contain the four-column co-singleton packet.

Proof. Let the positive packet weights be $\beta_1, \beta_2, \beta_3$. Every rich player has a nonempty membership pattern $I \subseteq \{1,2,3\}$, and its cover equation is $\sum_{j \in I} \beta_j = 1$. A singleton pattern makes one weight one. On the two remaining columns, rank two would require both singleton patterns and hence make both residual weights one; otherwise that residual system is nonextreme. A triple pattern makes every proper subset sum strictly less than one, leaving rank one. Hence a fractional rank-three vertex uses only pair patterns. Rank three requires all three of them, and their equations give $\beta_1 = \beta_2 = \beta_3 = 1/2$. Their nonempty row classes are exactly the atoms C, A, B in (HT).

Let r_a, r_b be the lifted poor loads and put $s = \phi(H) - v(H) > 0$. The packet inequality and $\Delta_a + \Delta_b = s$ give

$$s(r_b - 1) + \Delta_a(r_a - r_b) \geq 0.$$

Since $r_b < 1$ and $\Delta_a < 0$, this forces $r_a < r_b < 1$. Both are half-integers, so $r_a = 0$ and $r_b = 1/2$: no supported coalition contains a and exactly one contains b. Label that coalition over $A \cup B$. If $A = \{d\}$, selecting the coalitions over $C \cup A$ and $A \cup B$ gives rich load $\mathbf{1}_H + e_d$ and poor load e_b, a safe rooted tight cover forbidden by Theorem 5.79. Thus $|A| \geq 2$; the symmetric selection gives $|B| \geq 2$. Since C is nonempty, $|H| \geq 5$.

For the converse, put $h = t + u + w$ and let P be the polytope of nonnegative allocations with total $2 + 2h$ satisfying

$$\begin{aligned} x(C \cup A) &\geq 2(t+u), \\ x(C \cup B) &\geq 2(t+w), \\ x(\{b\} \cup A \cup B) &\geq 1 + 2(u+w). \end{aligned} \tag{HP}$$

Define $v(S) = \min_{x \in P} x(S)$. If $t = 1$, the point with $x_a = x_b = x(C) = 0$, $x(A) = 2u + 2$, and $x(B) = 2w + 2$ is feasible. If $t \geq 2$, use instead $x_a = x_b = 0$, $x(C) = 2t$, $x(A) = 2u + 1$, and $x(B) = 2w + 1$. Within every nonsingleton block, place its total away from any prescribed member. Thus every singleton has value zero. The vector $\phi = (1, 1, 2, \ldots, 2)$ is feasible and attains all three displayed lower bounds, so they are the corresponding coalition worths. The defining coalition constraints give $\mathrm{Core}(v) \subseteq P$, while the lower-envelope definition gives the reverse inclusion. Every coalition minimum is attained on the compact polytope, so v is exact.

At a vertex, at most one coordinate in each exchangeable block C, A, B is positive. After zero coordinates are removed, the remaining constraint columns are among the five player types a, b, C, A, B; every nonsingular subdeterminant of their total-plus-three-coalition incidence matrix has absolute value one or two. Thus every vertex of P is half-integral, proving the integral scaling claim.

Adding the three inequalities at any $x \in P$ and using efficiency gives

$$2x(H) + x_b \geq 4h + 1, \qquad 2x_a + x_b \leq 3. \tag{HG}$$

Thus no core point gives every player more than one. On the face where all players receive at least one, (HG) forces $x_a = x_b = 1$; the richer block then has total $2h$, whose largest common lower level is two. This proves the Lmin formula.

The feasible point

$$z_a = 0, \quad z_b = 3, \quad z(C) = 2t + 1, \qquad z(A) = 2u - 1, \quad z(B) = 2w - 1$$

attains equality in all three defining constraints and has $z(H) = 2h - 1$. Conversely, (HG) gives $x_a + x_b \le 3$ and hence $x(H) \ge 2h - 1$. Therefore the displayed suffix value and unit slack are exact.

For Lmax, the first coalition in (HT) has average two, so every core point has a coordinate at least two. On the face where all coordinates are at most two, the first two constraints force every member of H to equal two. Efficiency leaves $x_a + x_b = 2$, and the third constraint gives $x_b \ge 1$. Leximax minimization therefore equalizes the two remaining coordinates at one, proving $\mathrm{Lmax}(v) = \phi$.

It remains to identify the active traces. Since ϕ is strictly positive, the nonnegativity constraints are inactive there. If a proper coalition S is tight, LP duality writes

$$\mathbf{1}_S = \mu \mathbf{1}_N + \lambda_1 \mathbf{1}_{C \cup A} + \lambda_2 \mathbf{1}_{C \cup B} + \lambda_3 \mathbf{1}_{\{b\} \cup A \cup B}, \qquad \lambda_j \ge 0.$$

The a coordinate makes $\mu \in \{0, 1\}$. The value one gives only N; with $\mu = 0$, comparing the b, C, A, B coordinates of the zero–one vector leaves exactly the three unit choices for λ. Thus (HT) is the complete proper tight list.

Writing the three trace weights as $\beta_1, \beta_2, \beta_3$, the equations at any member of C, at any member of A, and at any member of B are

$$\beta_1 + \beta_2 = 1, \qquad \beta_1 + \beta_3 = 1, \qquad \beta_2 + \beta_3 = 1.$$

Hence their unique exact cover is $(1/2, 1/2, 1/2)$. The saturated exact-balanced identity is

$$\mathbf{1}_{C \cup A} + \mathbf{1}_{C \cup B} + \mathbf{1}_{\{b\} \cup A \cup B} + \mathbf{1}_{\{a\}} - \mathbf{1}_H = \mathbf{1}_N. \tag{HC}$$

The three nontrivial positive terms are tight, while singleton a and the negative suffix both have slack one. The exact vertex audit, complete tight list, and generic staged replay of the seven-player base case are in `src/verify_three_trace_half_integral_family.py` and `results/three_trace_half_integral_family.json`. □

The rooted residual in this family has an exact two-facet factorization. It is not an unexplained failure of the facet bridge.

Theorem 5.108 (two-facet factorization of the three-trace family)**.** *In Theorem 5.107, put*

$$T_1 = C \cup A, \quad T_2 = C \cup B, \quad T_3 = \{b\} \cup A \cup B, \quad U = H \cup \{a\}, \quad D = A \cup B.$$

The rooted obstruction identity factors as

$$\underbrace{\mathbf{1}_{T_3} + \mathbf{1}_U - \mathbf{1}_D = \mathbf{1}_N}_{\textit{complementary diamond}}, \qquad \underbrace{\mathbf{1}_{T_1} + \mathbf{1}_{T_2} + \mathbf{1}_D - 2\mathbf{1}_H = 0}_{\textit{internal triangle}}. \tag{TF}$$

Both displayed proper-coalition systems are indecomposable minimal semi-balanced systems and hence define facets of the exact-game cone. Their slack inequalities give the sharp sandwich

$$2s_\phi(H) \le s_\phi(D) \le s_\phi(U).$$

Throughout the family all three quantities equal two. Thus the rooted atom is exactly the sum of two saturated facet atoms.

Proof. Adding the two identities in (TF) and eliminating D gives

$$\mathbf{1}_{T_1} + \mathbf{1}_{T_2} + \mathbf{1}_{T_3} + \mathbf{1}_U - 2\mathbf{1}_H = \mathbf{1}_N,$$

the rooted form of (HC). Since T_1, T_2, T_3 are tight, exact balancedness applied to the two factors gives the displayed slack sandwich. The construction has $s_\phi(H) = 1$. Moreover (HG) gives $x_b \le 3$, attained by the exactifier displayed in the proof above, so $v(U) = v(N) - 3 = 2|H| - 1$ and $s_\phi(U) = 2$. The sandwich forces $s_\phi(D) = 2$.

It remains to check indecomposability without enumerating players. Within each of the nonempty blocks C, A, B all incidence rows of the internal system coincide. Any new exceptional coalition must therefore be a union of whole blocks. Players a, b lie outside every supported set, forcing the constant to be zero and excluding them from that coalition. On the three block types the supported sets are the three edges $C \cup A, C \cup B, A \cup B$ and their union H. The row equations have the unique signed dependence $(1, 1, 1, -2)$; checking the three remaining singleton block unions shows that none is exceptional. Hence the internal triangle is indecomposable.

For the complementary diamond, the three incidence row types are

$$X = \{a\} \cup C : (0, 1, 0), \qquad Y = \{b\} : (1, 0, 0), \qquad Z = A \cup B : (1, 1, 1)$$

in the columns (T_3, U, D). A new exceptional set must be a union of X, Y, Z. If it excludes Z, nonnegativity forces the empty set. If it includes Z, the four choices on X, Y give respectively D, U, T_3, N, all already supported or improper. Thus no decomposition exists. The facet conclusion is the indecomposability theorem of Studený and Kratochvíl (2022). □

Equivalently, the complementary diamond is already the $k = 2$ instance of Theorem 5.47: take root D and peripheral blocks $\{a\} \cup C$ and $\{b\}$. Only the internal triangle is an additional facet type.

A standalone derivation and exact base-case audit are in

`paper3/THREE_TRACE_FACET_FACTORIZATION.md`,
`src/verify_rooted_deficit_completion.py`,
`results/rooted_deficit_completion.json`.

There is a second external-sacrifice family. It extends the three-poorer, two-richer geometry of the five-player example simultaneously in both block sizes and closes the region between the two preceding families.

Theorem 5.109 (universal three-poorer escape)**.** *For every $m \ge 3$ and $h \ge 2$, there is a singleton-normalized integral indecomposable exact game with a strict Lmin boundary having m cumulative poorer players and a richer suffix H of size h such that*

$$\mathrm{Lmin}(v) = \mathbf{1}_{\{p_1, p_2\} \cup Q} + 2e_{p_3} + 3\mathbf{1}_H, \qquad |Q| = m - 3,$$

and

$$v(H) = 3h - 1 < 3h = \mathrm{Lmin}(v)(H).$$

Writing $\varepsilon = 1/(h+1)$, the same game has

$$\mathrm{Lmax}(v) = (2 - \varepsilon)e_{p_1} + \varepsilon e_{p_2} + (3 - \varepsilon)e_{p_3} + \mathbf{1}_Q + (3 - \varepsilon)\mathbf{1}_H. \tag{TH-M}$$

Thus the Lmin stratum sizes are $(m-1, 1, h)$ and the richer suffix has unit slack in every member of the family. Moreover Lmin and Lmax are distinct, with ℓ^1 distance $4h/(h+1)$.

Proof. Write $H = \{r_1, \dots, r_h\}$ and put $T = m+3h+1$. Take the lower envelope of the following common-total integral allocations (an indicator of an empty set is zero):

$$\begin{aligned} a &= 2e_{p_1} + 3e_{p_3} + \mathbf{1}_Q + 5e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1,r_h\}}, \\ b &= 2e_{p_2} + e_{p_3} + \mathbf{1}_Q + 4e_{r_1} + 3\mathbf{1}_{H\setminus\{r_1\}}, \\ c &= \mathbf{1}_Q + (3h+4)e_{r_h}, \\ \phi &= e_{p_1} + e_{p_2} + 2e_{p_3} + \mathbf{1}_Q + 3\mathbf{1}_H, \\ q^j &= \phi + e_{p_1} - e_j \qquad (j \in Q), \\ v(S) &:= \min\{a(S), b(S), c(S), \phi(S), q^j(S)\ (j \in Q)\}. \end{aligned} \tag{TH}$$

Every row has total T, so every row lies in the core of the lower-envelope game and every coalition attains its worth at a row. Hence v is exact. Players p_1, p_2, p_3 are zero respectively at b, a, c; every $j \in Q$ is zero at q^j; the first $h-1$ rich players are zero at c and r_h is zero at a. Thus the game is singleton-normalized.

Direct substitution gives

$$\begin{gathered} v(N \setminus \{j\}) = T - 1 \quad (j \in Q), \\ v(H \cup \{p_3\}) = 3h+2, \qquad v(H \cup \{p_1\}) = 3h+1, \qquad v(H) = 3h-1. \end{gathered} \tag{TH-T}$$

The point ϕ is in the core and gives every player at least one. At any core point x with this property, the complement constraints in (TH-T) first force $x_j = 1$ for every $j \in Q$. The constraint on $H \cup \{p_3\}$ then gives

$$x_{p_1} + x_{p_2} + x(Q) \le T - (3h+2) = m - 1,$$

so $x_{p_1} = x_{p_2} = 1$ as well. Hence the first progressive Lmin level is one and pins exactly $\{p_1, p_2\} \cup Q$.

On that face, the constraint on $H \cup \{p_1\}$ gives

$$x_{p_2} + x_{p_3} + x(Q) \le T - (3h+1) = m,$$

and therefore $x_{p_3} \le 2$. The point ϕ attains two while every member of H remains at three, so the second level pins p_3 at two. Efficiency leaves total $3h$ on H, whose common next level is at most three and is attained by ϕ. This proves the Lmin formula. The last identity in (TH-T) gives the claimed unit suffix slack.

We next prove (TH-M). Let z denote its right-hand side, put $\varepsilon = 1/(h+1)$, and first verify $z \in \mathrm{Core}(v)$. Every base row and z give each member of Q one, while the rows q^j can only lower the defining envelope. It therefore suffices to compare z with a, b, c, ϕ after deleting the common Q coordinates. Fix a coalition, write $x_i = \mathbf{1}_{\{p_i \in S\}}$, $u = \mathbf{1}_{\{r_1 \in S\}}$, $w = \mathbf{1}_{\{r_h \in S\}}$, $t = |S \cap H|$, and $y = x_1 - x_2 + x_3$. If $w = 0$, then $c(S) = 0 \le z(S)$. If $w = 1$ and $\phi(S) > z(S)$, then

$$\phi(S) - z(S) = \varepsilon t - (1-\varepsilon)y = \frac{t - hy}{h+1} > 0.$$

Since $t \le h$, this forces $y \le 0$. But then

$$a(S) - z(S) = \varepsilon(y+t) + 2u - 3 \le \frac{h}{h+1} - 1 < 0.$$

Thus at least one defining row lies weakly below z on every coalition, so z is a core point.

The tight face $H \cup \{p_3\}$ has $h+1$ members and worth $3h+2$, so every core point has a coordinate at least

$$c = \frac{3h+2}{h+1} = 3 - \varepsilon.$$

The point z attains this ceiling. On the face where every coordinate is at most c, all members of $H \cup \{p_3\}$ are forced to equal c. The tight face $H \cup \{p_1\}$ then forces $x_{p_1} \geq 2 - \varepsilon$, attained by z. Finally, for every $j \in Q$, direct substitution gives

$$v(H \cup \{p_1, j\}) = 3h + 2.$$

After the preceding coordinates are fixed, this forces $x_j \geq 1$, again attained by z. Efficiency fixes the sole remaining coordinate at $x_{p_2} = \varepsilon$. These successive ceilings prove the Lmax formula and its uniqueness. Direct subtraction gives the stated ℓ^1 distance.

It remains to verify indecomposability. In a singleton-normalized direct sum, two players with positive pair worth must lie in the same summand. If $Q \neq \emptyset$, fix $j \in Q$. Directly from (TH),

$$v(\{i, j\}) = 1 \qquad (i \neq j),$$

so the positive-pair graph is connected. If $Q = \emptyset$, positive pair worths connect p_1, p_3 and every member of H through r_h. Finally,

$$v(\{p_1, p_3, r_h\}) = 4, \qquad v(\{p_1, p_2, p_3, r_h\}) = 5.$$

If p_2 belonged to a separate summand, additivity and $v(\{p_2\}) = 0$ would make these two values equal. Thus p_2 is attached as well, and the game is indecomposable in every case.

The symbolic parameter-grid audit and independent exact staged Lmin/Lmax replays of the smallest distinct shapes are in `src/verify_three_poorer_boundary_family.py` and `results/three_poorer_boundary_family.json`. □

This whole family saturates one block-star facet. Partition $N \setminus H$ into its m singleton poorer blocks and use the spokes $H \cup \{p\}$. From (TH), their slacks in the order p_1, p_2, p_3, Q are

$$0,\ 2,\ 0,\ \underbrace{1, \ldots, 1}_{m-3},$$

whereas $s_\phi(H) = 1$. Consequently

$$\sum_{p \notin H} s_\phi(H \cup \{p\}) = (m-1) s_\phi(H),$$

which is equality in the exact-game facet inequality induced by Theorem 5.47. The five-player broken diamond and every member of this dimension-free family therefore share the same exterior facet mechanism.

The three dimension-free families and the two human impossibility arguments now give a complete cardinality classification, rather than only extremal rays.

Corollary 5.110 (complete boundary-shape phase diagram)**.** *At a strict payoff boundary of Lmin in a singleton-normalized exact game, let m be the number of cumulative poorer players and h the size of the richer suffix. A slack richer suffix is possible exactly for*

$$m \geq m_{\min}(h), \qquad m_{\min}(h) = \begin{cases} 4, & h = 1, \\ 3, & h = 2 \text{ or } 3, \\ 2, & h \geq 4. \end{cases} \tag{PD}$$

Every possible pair in (PD) *is realized by an explicit singleton-normalized integral lower-envelope game; the three universal realizing families are indecomposable. In particular, the threshold is a property of the two boundary-side cardinalities; no restriction on the number of poorer payoff strata is needed.*

Proof. Theorem 5.74 rules out $m = 1$ for every h, because its opposing-sign condition is impossible on a one-player poorer side. When $h = 1$, Theorem 5.60 rules out $m \leq 3$ and its indecomposable family realizes every $m \geq 4$. When $m = 2$, Theorem 5.105 rules out $h \leq 3$, while Theorem 5.106 realizes every $h \geq 4$. All remaining pairs have $m \geq 3$ and $h \geq 2$, and are realized by Theorem 5.109. □

Theorem 5.105 is now fully human. Its strict cover and three-player balanced-trace decomposition explain why three richer players cannot transport the slack budget, while the four-player richer block supports the internal triangle-like circuit of Theorem 5.106. Five-player slack boundaries do exist, but they require at least three cumulative poorer players. Theorem 5.109 shows that three poorer players suffice for every richer suffix of size at least two, completing the boundary-cardinality phase transition.

The earlier computer-assisted proof remains as an independent audit rather than a dependency of Lemma 5.61. One exact-rational encoding uses definition-level exact-face witnesses and stagewise KKT multipliers; a second independently formulates Lmin optimality through primal tangent cones. They agree on every payoff-stratum pattern through four players, and an exact-LP replay checks integer witnesses for every realizable pattern. The files are

`src/verify_n4_exact_lmin_pes_bridge.py`,
`src/verify_n4_exact_lmin_pes_bridge_primal.py`,
`src/verify_n4_exact_lmin_pes_bridge_witnesses.py`,
`results/n4_exact_lmin_pes_bridge.json`,
`results/n4_exact_lmin_pes_bridge_primal.json`,
`results/n4_exact_lmin_pes_bridge_witnesses.json`.

The reduced unsatisfiable cores produced by `src/verify_n4_exact_lmin_pes_bridge_cores.py` remain same-solver audit conveniences, not independent proof evidence.

Theorem 5.111 (the five-player failure is dimension-minimal)**.** *Let v, v' be exact games on the same player set N, $|N| \leq 4$, with $v' = v + d\mathbf{1}_S$ for one coalition S and some $d > 0$. Then normalized Lmin satisfies $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$. Proper-coalition monotonicity fails at five players by Theorem 5.37; hence five is the minimum possible dimension for such a failure.*

Proof. Translate an exact game by its singleton vector as in (LN); exactness and the large-core property are preserved. Exact games with at most four players have large cores (Biswas et al. 1999), and large-core games are egalitarian stable (Dietzenbacher 2021). On the latter domain the procedural egalitarian solution equals Lmax and is coalitionally monotonic (Dietzenbacher

2021). Lemma 5.61 identifies Lmin with that solution after normalization. This proves every bump that leaves the singleton vector fixed; singleton bumps are covered independently by Theorem 2.2. The five-player counterexample supplies the sharp lower bound. □

The common nonbinding case remains elementary:

Lemma 5.112 (non-binding)**.** *If $\phi(v)(S) \geq v(S) + d$ then $\phi(v') = \phi(v)$.*

Proof. $\phi(v)$ is the leximin point of $\mathrm{Core}(v) \supseteq \mathrm{Core}(v')$ and lies in $\mathrm{Core}(v')$, hence is the leximin point of the smaller set too; ϕ is single-valued. □

The separate star-failure capacity question. For a core point $x = \phi(v)$, condition $(\star_S)$ means that some $y \in \mathrm{Core}(v)$ weakly raises every coordinate in S and strictly raises $y(S)$. Its Farkas dual says that failure is equivalent to $\phi(v)$ maximizing a positive weighted sum $\sum_{k \in S} r_k x_k$ over the core. Theorem 5.37 has $(\star_S)$ throughout its binding interior, so it does not challenge the capacity results below. Instead it shows that capacity and coordinatewise monotonicity are different questions: a feasible direction that raises every target member need not be the direction selected by the next leximin stages. The remaining star-failure analysis therefore concerns when an exact bump can become binding, not whether the selector is monotone once it does.

The natural certificate is provably wrong. Where $(\star_S)$ *does* fail, one would like to prove $d_v^*(S) = 0$ and thereby certify directly that S is not exactly bumpable. This target is false. From the failure certificate, take the cover $\{\lambda_U\}$ dual to the failure, $W = \bigcup\{U : \lambda_U > 0\}$, and propose $C = W \cap S$. This route is closed by a six-player exact game (supplement, `WITNESS_KEY_FALSE_n6`): $(\star_S)$ fails at $S = \{1,3,4,5\}$ with cover $\{2,3\},\{1,2,6\},\{3,4,6\}$, yet a core point x with $x(W \cap S) = v(W \cap S)$ attains $x(S) = 24 > \phi(S) = 71/3$, so $W \cap S$ certifies nothing; the associated slack inequality $\phi(W) - v(W) \leq \phi(W \cap S) - v(W \cap S)$ fails by $1/3$. Such failures are rare ($\approx 0.1\%$ of $(\star_S)$-failures at $n \in \{6,7\}$, found only under generators biased toward thin cores with nested tight chains), which is why sampling first suggested the cover certificate was universal. In every observed failure, however, the operative bound survives: there is a coalition C with $M^{(C)} := \max\{x(S) : x \in \mathrm{Core}(v),\, x(C) = v(C)\} \leq \phi(S)$, so that S cannot be bumped past $\phi(S)$ while preserving exactness (in a $\approx 0.2\%$ minority $d_v^*(S) > 0$ — exact six-player witnesses with $d^* = 1$ and $d^* = 2$ are included in the supplement — but always $d_v^*(S) \leq \phi(S) - v(S)$). Empirically the certificate has sharp structure: C can always be taken as a subset of S contained in a *single stratum* of the leximin computation — almost always a singleton. If C is ϕ-tight, then $\phi \in F_C$, so a certificate can only be boundary-sharp, $M^{(C)} = \phi(S)$; exact boundary cases of this kind do occur, while every strict certificate is necessarily non-tight. The open problem is thus sharpened: produce, from the structure of a $(\star_S)$-failure at the leximin point of an exact game, a single-stratum $C \subseteq S$ with $M^{(C)} \leq \phi(S)$. Empirically the certificate is pinned further: writing r for the residuals of the failure's dual cover ($r_i \geq 1$ on S), a *minimum-residual* member $k \in S$ (one with $r_k = 1$) certifies in 99.6% ($n = 6$) to 100% ($n = 7$) of observed failures, the residue requiring a same-stratum pair. By strong duality the singleton form is equivalent to exhibiting nonnegative weights on ϕ-tight coalitions with coverage β on $N \setminus S$, $\beta - 1$ on $S \setminus \{k\}$, and at least $\beta - 1$ on k; weights supported inside S suffice in only about two thirds of cases, so a proof must construct this dual cover from ϕ-tight coalitions *crossing* S. The residual normal gives a possible stronger route, but not an equivalent reformulation. For a minimum-residual k, normalized by $r_k = 1$, and any core x with $x_k = v(\{k\})$, maximality of ϕ

gives

$$[x(S) - \phi(S)] + \sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \le 0.$$

Consequently the singleton face certificate would follow from the sufficient inequality

$$\sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \ge 0$$

throughout that face. The sign is essential; the opposite inequality gives no upper bound on $x(S) - \phi(S)$. Whether this stronger face inequality is forced by the leximin strata is a separate question from the carrier route. An exact audit answers that stronger question negatively. A highest-gain minimum-residual player satisfies the nonnegative-correction condition in 139/141 saved multilevel failures, but not in two seven-player cases. In one of them (`WITNESS_KEY_CX_n7_s303_1491`) the correction is identically $-3/2$ on each of the two highest-gain singleton faces, although both faces certify and both carrier inequalities hold. Thus the residual normal alone cannot prove the multilevel branch; a proof must use additional exactness geometry such as the off-side carrier comparison.

A new proof-producing exact audit isolates a still sharper target on the saved hard frontier. It covers all 141 stored three-or-more-level failures (106 at $n = 6$ and 35 at $n = 7$). In every case some minimum-residual singleton certifies, and, more strongly, some minimum-residual k satisfies the elementary carrier inequality

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \le [\phi(\{k\}) - v(\{k\})].$$

Across the 367 individual minimum-residual candidates, 282 singleton faces certify and the carrier inequality explains 280 of them; it never produces a false certificate. The remaining two seven-player singleton certificates require a richer face dual, but another minimum-residual player in the same case has the carrier certificate. This is exact finite evidence, not a theorem. Nor can it be promoted to an unconditional singleton-carrier conjecture: the exact two-level witness above has no certifying singleton; its minimum-residual pair is essential. At this stage the apparent frontier was a dichotomy. A failure dual with at most two residual levels is already handled by Proposition 5.3. On the genuinely three-or-more-level branch, these data first suggested proving the displayed singleton inequality. The whole-class audit below supersedes that target with an existential full minimum class K and removes the singleton-selection problem. Proposition 5.127 below will show that the failure cover, and hence K, still has to be selected. The saved corpus initially suggested a canonical gain selector: in all 141 cases a carrier-certifying k could be chosen to maximize $\phi_i - v(\{i\})$ inside the minimum-residual class. That refinement is false. A seeded mutation audit produced 534 further exactly certified multilevel records (505 distinct game–coalition pairs). Some minimum-residual carrier still exists in all 534 records, but a highest-gain carrier exists in only 520, and a highest-gain singleton face certifies in only 533.

Here is the sharp selector counterexample. Let the exact six-player game be the lower envelope of the four common-sum points

$$p = (-6, 4, 18, -4, 11, -10), \quad q = (-1, 4, 24, -4, 0, -10),$$
$$a = (-1, -2, 19, -4, 6, -5), \quad b = (-6, 0, 20, 0, 9, -10),$$

and take $S = \{1, 2, 3, 4, 6\}$. Its leximin point is $\phi = (-3, 0, 20, -2, 6, -8)$, the normalized failure cover has three residual levels and minimum-residual class $K = \{1, 2, 3\}$, and the singleton gains

on K are $(3,2,2)$. Thus player 1 is the unique highest-gain candidate. Nevertheless its carrier slack is $4 > 3$, and more strongly

$$\max\{x(S) : x \in \mathrm{Core}(v),\ x_1 = v(\{1\})\} = 8 > 7 = \phi(S),$$

so even its singleton face fails. Players 2 and 3 both satisfy the carrier inequality with equality and certify. Tight-hypergraph centrality and cover-weighted adjacency refinements also fail on exact mutations. Any singleton route must therefore remain existential: select *some* minimum-residual player, without imposing a gain ranking. After adding fresh three-, four-, and five-generator screens and deduplicating all corpora, this existential carrier statement holds for all 669 distinct exact multilevel game–coalition pairs (702 certified records). The whole-class carrier statement below supersedes the need to solve this singleton-selection problem.

Proposition 5.6 sharpens what remains to be proved. With $K = \{k \in S : r_k = 1\}$, a counterexample would produce simultaneously the positive-S failure normal r and carrier failure at every member of K. Proposition 5.7 shows that the resulting common separator is not a second obstruction to eliminate: it is automatically the normalized residual gap on $S \setminus K$. Thus mere coexistence of the two normals cannot yield a contradiction. A tempting coordinatewise strengthening is also false: exactness and the leximin strata need not permit a convex combination of the carrier-tight faces whose average displacement is nonnegative at every higher-residual coordinate. In the exact six-player instance `WITNESS_KEY_CX_n6_s201_10919`, the best common coordinatewise margin over all such mixtures is $-1/20$, although players 4 and 5 both satisfy the carrier inequality (each with margin $-3/4$).

The scalar condition in Corollary 5.8 is the minimal sufficient statement for a *fixed* failure cover, and it initially survives all 141 saved hard multilevel failures. Across their 367 minimum-residual faces, 277 escape; all 87 failed carriers are strictly negative, while three valid carriers do not escape but another candidate does in the same record.

Adversarial mutation shows that even this fixed-cover target is not universal. Among 505 distinct exact mutation cases, all have a minimum-residual carrier but only 123 have an escape face for the cover returned by the exact failure dual. Already trial 0 of the first seeded face-selector mutation audit has $K = \{1,2,3\}$ and normalized residual-gap weights $(2/3, 1/3)$ on players $5, 6$. Players 1 and 2 satisfy the carrier inequality, with margins -2 and 0, yet their face maxima are both $-1/3$; player 3 fails the carrier inequality and has maximum $-4/3$. Consequently neither coordinatewise neutralization nor escape for an arbitrarily chosen failure cover can be the missing theorem. A residual-based proof would have to select a favorable cover from the full dual face jointly with the carrier candidate; otherwise it must return to a cover-independent exactness exchange.

There is one dimension in which this cover-selection problem disappears.

Lemma 5.113 (three-player residual-order rigidity)**.** *Suppose $|S| = 3$ and $(\star_S)$ fails at a core point ψ. If no failure cover has at most two residual levels on S, then every failure cover has three distinct residuals, and all failure covers induce the same strict order on S. In particular, their minimum-residual class is the same singleton.*

Proof. Before the harmless positive normalization, failure covers form a polyhedral cone; the normalized residual set is convex. A residual tie on a three-player set has at most two distinct levels, so every residual vector in the stated branch has a strict order. If two covers induced different orders, their line segment would remain in the failure-cover set. Along that segment the order could change only at a residual tie, producing a cover with at most two levels, a contradiction. □

Lemma 5.114 (three-player block-star and residual-gap alternatives)**.** *Let $\mathcal{T}(\psi)$ be the proper coalitions tight at a core point ψ and put*

$$\mathcal{D}_\psi = \{d \in \mathbb{R}^N : d(N) = 0,\ d(T) \geq 0 \text{ for every } T \in \mathcal{T}(\psi)\}.$$

For a pair $\{a, b\} \subset S$ with third player c, no failure cover has $r_a = r_b$ if and only if some $d \in \mathcal{D}_\psi$ satisfies

$$d_c \geq 0, \qquad d_a + d_b \geq 0, \qquad d(S) > 0. \tag{BS}$$

Consequently, when $|S| = 3$, every failure cover has three distinct residuals if and only if all three singleton–complementary-pair versions of (BS) hold.

Now suppose failure covers exist and every cover has the fixed order $r_k < r_i < r_j$. Normalize by $r_k = 1$ and define

$$\delta = \min\{r_j - r_i : r \text{ is a failure-cover residual vector},\ r_k = 1\}. \tag{AG}$$

Then $\delta > 0$, and there is $d \in \mathcal{D}_\psi$ with

$$(d_k, d_i, d_j) = (\delta, 1, -1). \tag{GD}$$

In particular, if $P = N \setminus S$ and $A = P \cup \{k\}$, then $d(P) = -\delta$ and $d(A) = 0$.

Proof. For every failure cover and every $d \in \mathcal{D}_\psi$, the cover identity gives

$$\sum_{p \in S} r_p d_p = \mu d(N) - \sum_{T \in \mathcal{T}(\psi)} \lambda_T d(T) \leq 0. \tag{P}$$

If $r_a = r_b$, the left side is $r_a(d_a + d_b) + r_c d_c$, which is strictly positive under (BS). This proves one direction. Conversely, apply Farkas' lemma to the normalized failure-cover system with the additional equality $r_a = r_b$. The player-equation multipliers, with sign reversed, give $d \in \mathcal{D}_\psi$; the multipliers of the lower bounds on r_c and on the tied pair give respectively $d_c \geq 0$ and $d_a + d_b \geq 0$. Normalizing the Farkas right-hand side gives $d(S) > 0$. This proves the alternative.

For the second claim, the polyhedron in (AG) is nonempty. Its objective is nonnegative and attains its minimum. If the minimum were zero, an optimizer would have $r_i = r_j$, contrary to the fixed strict order; hence $\delta > 0$. Dualize this LP. The dual player multipliers again define a vector in $\mathcal{D}_\psi$. At every feasible point $r_j > r_i > r_k = 1$, so the lower bounds on r_i and r_j are slack; complementary slackness and the objective $r_j - r_i$ give $d_i = 1$ and $d_j = -1$. The dual price of the normalization $r_k = 1$, together with equality of the primal and dual values, gives $d_k = \delta$. Finally $d(S) = \delta$ and $d(N) = 0$, so $d(P) = -\delta$ and $d(A) = 0$. □

Thus the unresolved $|S| = 3$ branch has no chamber-choice ambiguity: after Proposition 5.3, it reduces to proving the carrier inequality for one uniquely determined minimum-residual player. This is strictly simpler than the four-player geometry below, where the residual cone can cross several genuinely multilevel chambers.

Lemma 5.114 replaces the universal quantifier over the cover polyhedron by three primal tangent directions. It also supplies one canonical direction parallel to the trace face $A = P \cup \{k\}$. In the six exact hard cases, independently verified rational Farkas certificates exist for all 18 pair-tie systems. Moreover every normalized adjacent gap is $\delta = 1$, and the six directions in (GD) are integral. These values are finite evidence; the block-star and residual-gap alternatives themselves are proved for every three-player target.

There is a natural stronger three-player target. If k is that unique minimum-residual player, ask whether

$$\max\{x(P) : x \in \mathrm{Core}(v),\ x(P \cup \{k\}) = v(P \cup \{k\})\} \geq \phi(P). \tag{TF}$$

Any maximizer in (TF) is an exact-face witness in Lemma 5.36, so both terms on the right side of (GC) are nonnegative and the singleton carrier inequality follows. The initial exact chamber audit contained six genuinely multilevel $|S| = 3$ cases. Each had exactly one accessible minimum-class chamber, a strict carrier margin between -8 and $-1/2$, and an exact rational primal–dual certificate for (TF). A targeted seven-player screen of 1,976 nondegenerate lower-envelope games found 649 binding star failures and seven float multilevel candidates. Four extremals were recomputed exactly: one collapsed to the proved two-level branch, while the other three supplied three of the six cases above. These counts are evidence; Lemma 5.113 is the proved dimension reduction.

Those six face duals share a sharper pattern. Each has zero grand-coalition price, and every positively weighted core constraint U satisfies $U \cap S = \{k\}$. If $b > 0$ is minus the price of the face equality and $\lambda_U \geq 0$ are the resulting core weights, stationarity reads

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}},$$

and (TF) is exactly the weighted slack inequality

$$\sum_U \lambda_U[\phi(U) - v(U)] \geq b[\phi(P \cup \{k\}) - v(P \cup \{k\})]. \tag{TS}$$

This common dual support pattern was exact finite evidence, not a theorem. The counterexample below has a positive dual coalition disjoint from S and therefore also falsifies the exact-trace-$\{k\}$ support refinement. The next reduction remains valid, but its proposed sign conclusion will likewise be falsified.

Exactness supplies a direct face-coordinate reduction. It neither assumes the observed trace-face dual nor selects a dual representative.

Lemma 5.115 (three-player residual-spread bound)**.** *Let v be exact and suppose a normalized failure cover has residuals $r_k = 1 < r_i = \rho < r_j = \sigma$ on $S = \{k, i, j\}$, and put $A = P \cup \{k\}$. For every $a \in F_A$,*

$$a(P) - \phi(P) \geq (\rho - 1)[\phi(A) - v(A)] + (\sigma - \rho)(a_j - \phi_j). \tag{FC}$$

Consequently, writing $M_j^A = \max_{a \in F_A} a_j$, the target margin satisfies

$$\begin{aligned} \max_{x \in F_A} x(P) - \phi(P) &\geq (\rho - 1)[\phi(A) - v(A)] \\ &\quad + (\sigma - \rho)(M_j^A - \phi_j) \\ &\geq (\rho - 1)[\phi(A) - v(A)] \\ &\quad - (\sigma - \rho)[\phi_j - v(\{j\})]. \end{aligned} \tag{RS}$$

In particular, (TF) holds whenever the right side of (RS) is nonnegative.

Proof. Write the tight-coalition failure cover as

$$(1, \rho, \sigma)_S = \mu \mathbf{1}_N - \sum_T \alpha_T \mathbf{1}_T, \qquad \alpha_T \geq 0,$$

where $\phi(T) = v(T)$ whenever $\alpha_T > 0$. For $a \in F_A$, put $\Delta = a - \phi$ and $s(A) = \phi(A) - v(A)$. Efficiency and tightness at A give

$$\Delta(P) + \Delta_k = -s(A), \qquad \Delta_i + \Delta_j = s(A).$$

Pairing the failure cover with Δ, and using core feasibility of a on every positively weighted ϕ-tight coalition, gives

$$\Delta_k + \rho\Delta_i + \sigma\Delta_j = -\sum_T \alpha_T[a(T) - v(T)] \le 0.$$

Substituting $\Delta_i = s(A) - \Delta_j$ and $\Delta(P) = -s(A) - \Delta_k$ proves (FC). Maximize a_j over F_A. Finally every core point has $a_j \ge v(\{j\})$, which gives (RS). □

The singleton-floor comparison in (RS) is strict in only two of the original six exact three-level cases and has lower bounds $-5/4, -5/3, -3, -2$ in the other four. The stronger face-coordinate bound (FC) proves all six. Its exact values are

$$\frac{1}{2}, \quad 1, \quad \frac{11}{4}, \quad \frac{4}{3}, \quad 0, \quad 2,$$

versus actual target margins $1/2, 1, 15/4, 4/3, 5, 2$; hence it is sharp in four cases. A natural attempted generalization was the one-coordinate inequality

$$M_j^A - \phi_j \ge -\frac{\rho - 1}{\sigma - \rho}[\phi(A) - v(A)]. \tag{HC}$$

Unlike trace extension itself, (HC) asks only for the high-residual coordinate on one exact face. It is nevertheless false.

Proposition 5.116 (genuinely multilevel face-target counterexample)**.** *Let v be the lower envelope of the three common-sum allocations*

$$\begin{pmatrix} 14 & -11 & 12 & -16 & 9 & 14 & 10 \\ 15 & -11 & 12 & -16 & 8 & 14 & 10 \\ 18 & -6 & 6 & -17 & 6 & 13 & 12 \end{pmatrix}.$$

Then v is an exact seven-player game. The game's leximin point is

$$\phi = \left(\frac{47}{3}, -9, \frac{29}{3}, -\frac{49}{3}, \frac{23}{3}, \frac{41}{3}, \frac{32}{3}\right).$$

For $S = \{2, 3, 5\}$ the core-expansion margin $\max_{x \in \mathrm{Core}(v)} x(S) - \phi(S)$ is $5/3$, $(\star_S)$ fails, and every failure cover has three distinct residuals in the order

$$r_5 < r_3 < r_2.$$

Nevertheless, for $P = N \setminus S$ and $A = P \cup \{5\}$,

$$\max_{x \in F_A} x(P) = 23 < \frac{71}{3} = \phi(P). \tag{FCX}$$

For the normalized cover $(r_5, r_3, r_2) = (1, 2, 3)$ one has $s(A) = 1/3$ and $M_2^A - \phi_2 = -1$, so both (TF) and (HC) fail by $2/3$. Trace extension itself still holds with gap zero, and the desired carrier inequality still holds with margin $s(A) - s(\{5\}) = -4/3$.

Proof. All three rows have sum 32 and are core points of their lower envelope. Every proper-coalition worth is attained by a row, proving exactness. Exact leximin LPs give successive surplus levels $2/3, 5/3, 2, 11/3$, pinning respectively $\{4,6,7\}, \{1,5\}, \{2\}, \{3\}$. A core LP gives the stated binding margin. The tight-cover weights $1,1,1,2$ on coalition masks $25, 49, 69, 104$, with grand coefficient 3, give residuals $(r_2, r_3, r_5) = (3,2,1)$ and certify star failure. Exact Farkas certificates make each of the three pair-tie cover systems infeasible, so the whole cover face is genuinely multilevel by Lemma 5.113.

On F_A, the point $(15, -10, 11, -16, 8, 14, 10)$ attains both $x(P) = 23$ and $x_2 = -10$. For the P-maximum, unit core-dual weights on masks $25, 49, 80, 104$, grand price zero, and face price -3 certify 23. A coordinate dual certifies $M_2^A = -10$: its core weights on masks $25, 49, 69, 80, 104$ are respectively $2, 2, 1, 1, 3$, with grand price -1 and face price -4. Direct substitution gives the two margins in the statement. Mask $80 = 57$ need not be ϕ-tight: complementary slackness here is with the displayed optimizer on the face F_A, not with the leximin point. The restricted A-core has the same P-maximum 23, proving zero trace gap, while the singleton and carrier slacks give $-4/3$. □

The failure is not a numerical knife edge. Replace the second generator by

$$p^2(t) = (17 - t, -11, 12, -16, 6 + t, 14, 10).$$

Exact recomputation at

$$t = 0, \frac{1}{2}, 1, \frac{4}{3}, \frac{3}{2}, 2, 3$$

keeps the leximin point, binding margin $5/3$, star opening zero, strict three-level residual order, carrier margin $-4/3$, and trace gap zero fixed. The face-target margins are respectively

$$\frac{4}{3}, \quad \frac{5}{6}, \quad \frac{1}{3}, \quad 0, \quad -\frac{1}{6}, \quad -\frac{2}{3}, \quad -\frac{2}{3}.$$

Thus an exact sum-preserving motion within A crosses the stronger target without moving the carrier comparison at all. These rational samples are a path audit, not a claim about every real parameter value.

The failed one-coordinate target has an exact dual form that separates the face price from coalition-slack compensation and diagnoses the crossing.

Proposition 5.117 (high-coordinate slack–price identity)**.** *Let F_A be nonempty, put $s(C) = \phi(C) - v(C)$, and let $M_j^A = \max_{a \in F_A} a_j$. Every optimal dual for this coordinate maximum can be written with $\lambda_C \geq 0$ and $u, b \in \mathbb{R}$ as*

$$-\mathbf{1}_{\{j\}} = \sum_{\emptyset \neq C \subsetneq N} \lambda_C \mathbf{1}_C + u\mathbf{1}_N + b\mathbf{1}_A, \tag{CD}$$

and satisfies the exact identity

$$M_j^A - \phi_j = \sum_C \lambda_C s(C) + b\, s(A). \tag{SP}$$

Consequently, for any $c \geq 0$,

$$M_j^A - \phi_j \geq -c s(A) \quad \Longleftrightarrow \quad \sum_C \lambda_C s(C) + (b + c) s(A) \geq 0. \tag{DP}$$

The equivalence holds for one, hence for every, optimal coordinate dual. In particular $b \geq -c$ is sufficient; if $b < -c$, only the positively weighted coalition slacks can supply the missing amount.

Proof. Write the core inequalities as $-a(C) \le -v(C)$ and minimize $-a_j$ subject also to grand efficiency and $a(A) = v(A)$. Reversing the signs of the nonpositive inequality-dual variables gives (CD), while strong duality gives

$$-M_j^A = \sum_C \lambda_C v(C) + u\, v(N) + b\, v(A).$$

Pair (CD) with ϕ, use $\phi(N) = v(N)$, and subtract the dual objective identity. This proves (SP); adding $cs(A)$ proves (DP). Its left side is independent of the chosen optimal dual, which proves the final assertion. □

In the original six exact hard faces, maximizing the face price over the optimal dual face gives

$$-1, \quad -\frac{2}{3}, \quad 0, \quad -2, \quad 0, \quad 0.$$

Here $c = (\rho - 1)/(\sigma - \rho) = 1$. Thus the price condition alone proves five cases. In the fourth, the maximal price is -2, but the weighted slack is $5/3$ and $s(A) = 1/3$, so the right side of (DP) is $4/3 > 0$. In Proposition 5.116, the maximal price is -4, the weighted slack is $1/3$, and $s(A) = 1/3$; hence the same right side is $-2/3$. Thus the identity distinguishes genuine compensation from a genuine sign failure, while a face-price bound by itself was already false on the positive original corpus.

The closed-form spread bound is one feasible point of a stronger finite exactness optimization. Write $s(C) = \phi(C) - v(C)$ and, for a trace-face dual (λ, b), define

$$\begin{aligned} \mathcal{B}(\lambda, A) = \sup\{(c-b)s(A) - \sum_C \eta_C s(C) : \eta_C \ge 0,\ c \ge b,\ q \in \mathbb{R}, \\ \sum_U \lambda_U \mathbf{1}_U + \sum_C \eta_C \mathbf{1}_C - c\mathbf{1}_A = q\mathbf{1}_N\}. \end{aligned} \tag{OB}$$

Proposition 5.118 (one-face exact-balance bound)**.** *Under the hypotheses of Lemma 5.115, suppose in addition that the face LP has a zero-grand dual (λ, b) satisfying*

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}}.$$

Then

$$\max_{x \in F_A} x(P) - \phi(P) \ge \mathcal{B}(\lambda, A).$$

The singleton-floor residual-spread right side in (RS) is obtained by a feasible choice in (OB).

Proof. Fix a feasible (η, c, q) and choose, by exactness, $a \in \mathrm{Core}(v)$ with $a(A) = v(A)$. Dot the incidence identity in (OB) first with a and then with ϕ. Core feasibility for every positively weighted coalition and tightness at A give

$$\sum_U \lambda_U s(U) + \sum_C \eta_C s(C) - cs(A) \ge 0.$$

Subtracting $bs(A)$ instead of $cs(A)$ proves the asserted bound and taking the supremum completes the first claim. For the second, use the tight failure-cover weights as part of η, add $(\sigma - \rho)$ on the singleton $\{j\}$, and take $c = b + \rho - 1$ and $q = \mu - \rho$. Feasibility is the incidence identity

$$\sum_U \lambda_U \mathbf{1}_U + \sum_T \alpha_T \mathbf{1}_T + (\sigma - \rho)\mathbf{1}_{\{j\}} - (b + \rho - 1)\mathbf{1}_A = (\mu - \rho)\mathbf{1}_N.$$

□

The exact LP for (OB) gives, in the six known three-level cases, respectively

$$\frac{1}{2}, \quad 1, \quad \frac{15}{4}, \quad -\frac{5}{3}, \quad -2, \quad 2.$$

It therefore proves the full face target in four cases and equals the actual target margin in all four, including the case where the closed-form spread bound is $-5/4$ but the target margin is $15/4$. Restricting the auxiliary weights η to ϕ-tight coalitions is infeasible in all six cases, so non-tight slack terms are indispensable. The last two actual target margins exceed their optimal one-face bounds by 3 and 7, but neither remains an unexplained finite case. In the latter, $s(A) = 0$, so $\phi \in F_A$ proves the face target immediately. In the former, a common-sum generator attaining $v(A)$ has outside sum $25 > 71/3 = \phi(P)$ and is the required exact-face witness. Thus all six cases now have proof-producing decompositions; only the last generator argument was specific to the finite lower-envelope model. The unconditional face-coordinate bound above now supplies exact-face proofs for both remaining cases as well, so none of the six finite proofs requires a model-specific generator argument.

The remaining dual-existence question has a clean primal form. For $A = P \cup \{k\}$ write

$$\mathrm{Core}_A(v) = \{y \in \mathbb{R}^A : y(A) = v(A),\ y(B) \geq v(B) \text{ for every } B \subsetneq A\}.$$

The projection of F_A is contained in $\mathrm{Core}_A(v)$, so always

$$\max_{x \in F_A} x(P) \leq \max_{y \in \mathrm{Core}_A(v)} y(P). \tag{TE}$$

Equality is precisely the existence of an optimal restricted allocation that extends to the full core. The corresponding dual statement is exact.

Lemma 5.119 (trace-extension dual equivalence)**.** *Equality holds in* (TE) *if and only if the full face LP has an optimal dual with zero grand-coalition price. Every such dual can be chosen with all positively weighted core constraints contained in A and has*

$$\sum_U \lambda_U \mathbf{1}_U = (b-1)\mathbf{1}_P + b\mathbf{1}_{\{k\}}$$

for some $b \geq 1$.

Proof. The restricted problem retains only the coordinates and core constraints inside A, together with efficiency on A. Any restricted dual lifts to the full face dual by assigning zero price to grand efficiency and zero weight to every other core constraint. Conversely, stationarity for a zero-grand full dual has zero coverage on $S \setminus \{k\}$. Since core weights are nonnegative, every positively weighted coalition is contained in A; deleting the unused coordinates gives a restricted dual of the same value. Strong duality proves the equivalence. Stationarity for the P-objective and the face equality gives the displayed balance, whose nonnegative coverage forces $b \geq 1$. □

Thus trace extension supplies the balance identity used in Proposition 5.118. The unconditional Lemma 5.115 no longer requires trace extension or any choice of face dual; the stronger observed fact that every positive dual coalition has trace exactly $\{k\}$ is not required there either.

For the three-player branch the primal obstruction collapses to one scalar inequality. Put $R = N \setminus A = \{i, j\}$ and $d_A = v(N) - v(A)$. For $y \in \mathrm{Core}_A(v)$ and nonempty $D \subseteq R$, define the outside demand

$$\ell_D(y) = \max_{B \subseteq A} \{v(B \cup D) - y(B)\}.$$

Proposition 5.120 (two-player-complement trace criterion)**.** *Let v be exact, let $A = N \setminus \{i, j\}$, and let $y \in \mathrm{Core}_A(v)$. Then y extends to a point of F_A if and only if*

$$\ell_{\{i\}}(y) + \ell_{\{j\}}(y) \le d_A. \tag{SD}$$

In particular, exactness rules out every obstruction involving a coalition that contains both omitted players.

Proof. An extension chooses two coordinates z_i, z_j with $z_i + z_j = d_A$. Constraints whose traces on R are $\{i\}$ or $\{j\}$ are exactly $z_i \ge \ell_{\{i\}}(y)$ and $z_j \ge \ell_{\{j\}}(y)$, so such coordinates exist exactly when (SD) holds. Constraints with empty trace already hold because $y \in \mathrm{Core}_A(v)$.

It remains to check a trace equal to R. For each $B \subseteq A$, exactness provides $a \in \mathrm{Core}(v)$ with $a(B) = v(B)$. Core feasibility and efficiency give

$$v(B \cup R) - v(B) \le a(R) = v(N) - a(A) \le v(N) - v(A) = d_A.$$

Since $y(B) \ge v(B)$, this implies $v(B \cup R) - y(B) \le d_A$. Thus every two-player-trace constraint is automatic, completing the equivalence. □

Consequently equality in (TE) is equivalent to the restricted face LP having an optimum y that satisfies (SD). This replaces a high-dimensional extension question by the exclusion of two witness coalitions, one carrying i and one carrying j.

There is an exact dual for that last selection problem. Put

$$m_A = \min_{y \in \mathrm{Core}_A(v)} y_k, \qquad \Gamma_A = \min_{\substack{y \in \mathrm{Core}_A(v) \\ y_k = m_A}} \{\ell_{\{i\}}(y) + \ell_{\{j\}}(y) - d_A\}.$$

Then equality in (TE) is equivalent to $\Gamma_A \le 0$.

Lemma 5.121 (split-obstruction dual)**.** *The number Γ_A is the maximum of*

$$\sum_C \tau_C v(C) + \sum_{p \in \{i,j\}} \sum_{B \subseteq A} \theta_{pB} v(B \cup \{p\}) - v(N) - (\alpha - 1)v(A) - \beta m_A \tag{SDD}$$

over $\tau_C, \theta_{pB} \ge 0$ and $\alpha, \beta \in \mathbb{R}$ satisfying

$$\sum_B \theta_{pB} = 1 \quad (p = i, j)$$

and the lifted incidence balance

$$\sum_C \tau_C \mathbf{1}_C + \sum_{p \in \{i,j\}} \sum_{B \subseteq A} \theta_{pB} \mathbf{1}_{B \cup \{p\}} = \mathbf{1}_N + (\alpha - 1)\mathbf{1}_A + \beta \mathbf{1}_{\{k\}}. \tag{SDB}$$

Here C ranges over the proper coalitions contained in A.

Proof. Minimize $u_i + u_j - d_A$ subject to $y \in \mathrm{Core}_A(v)$, $y_k = m_A$, and

$$y(B) + u_p \ge v(B \cup \{p\}) \qquad (B \subseteq A,\ p = i, j).$$

At optimum $u_p = \ell_{\{p\}}(y)$. In the LP dual, stationarity of u_p gives $\sum_B \theta_{pB} = 1$. Stationarity of the coordinates of y first gives the corresponding balance on the traces B; replacing each trace by $B \cup \{p\}$ adds one unit on each omitted player and yields (SDB). The dual objective, after subtracting $d_A = v(N) - v(A)$, is (SDD). Strong duality completes the proof. □

The split dual can be compared with the simpler dual for m_A. Write that dual as

$$\sum_C \kappa_C \mathbf{1}_C = c\mathbf{1}_A + \mathbf{1}_{\{k\}}, \qquad \sum_C \kappa_C v(C) = cv(A) + m_A, \tag{MD}$$

with $\kappa_C \geq 0$ on proper coalitions contained in A.

The coefficient condition also has a one-dimensional primal form. Fix a minimum-coordinate dual coefficient c and define the extended-real value

$$\begin{aligned}\widehat{\Gamma}_{A,c}(t) = \min\{u_i + u_j + t - d_A :\ & y(A) = v(A) + t,\ y_k = m_A - ct, \\ & y(B) \geq v(B) \quad (B \subsetneq A), \qquad y(B) + u_p \geq v(B \cup \{p\}) \quad (B \subseteq A,\ p = i, j)\}.\end{aligned} \tag{PT}$$

and put the value equal to $+\infty$ when the constraints are infeasible. Thus $\widehat{\Gamma}_{A,c}(0) = \Gamma_A$.

Proposition 5.122 (minimum-dual tangent criterion)**.** *The optimal split-dual face contains a member satisfying $\alpha - 1 = \beta c$ if and only if*

$$\widehat{\Gamma}_{A,c}(t) \geq \Gamma_A \qquad \text{for every } t \in \mathbb{R}. \tag{TN}$$

Equivalently, compatibility says that $t = 0$ globally minimizes the convex parametric split value in (PT)*.*

Proof. Write $\gamma = \alpha - 1$ and let J be the split-dual objective at $t = 0$. The dual feasible region is independent of t, while its objective at t is

$$J + t(\beta c - \gamma).$$

Hence $\widehat{\Gamma}_{A,c}$ is the supremum of affine functions and is convex. Its subdifferential at zero is the set of slopes $\beta c - \gamma$ attained on the optimal dual face: that face is convex and the slope map is affine, so its image is already an interval. Therefore $0 \in \partial\widehat{\Gamma}_{A,c}(0)$ exactly when some optimal dual has $\gamma = \beta c$. The convex optimality criterion identifies this with (TN). □

Proposition 5.123 (compatible minimum-cover certificate)**.** *Suppose an optimal split dual in Lemma 5.121 and an optimal minimum-coordinate dual in* (MD) *satisfy*

$$\alpha - 1 = \beta c.$$

If there is $z \in \mathrm{Core}(v)$ with $z(C) = v(C)$ whenever $\kappa_C > 0$, then $\Gamma_A \leq 0$ and trace extension holds. When v is exact and $c = 0$, the common-tightness condition is automatic.

Proof. Let ω denote the aggregate coalition weights on the left side of (SDB). The coefficient identity and (MD) give the signed incidence identity

$$\sum_C \omega_C \mathbf{1}_C - \beta \sum_C \kappa_C \mathbf{1}_C = \mathbf{1}_N.$$

Because $\omega_C \geq 0$, core feasibility can only increase the ω-sum when worths are replaced by z-values. Common tightness leaves the κ-sum unchanged, irrespective of the sign of β. Hence

$$\sum_C \omega_C v(C) - \beta \sum_C \kappa_C v(C) \leq \sum_C \omega_C z(C) - \beta \sum_C \kappa_C z(C) = v(N).$$

Using (MD) and $\alpha - 1 = \beta c$, the left side minus $v(N)$ is exactly the optimal objective (SDD), namely Γ_A. Thus $\Gamma_A \leq 0$.

If $c = 0$, the nonnegative incidence identity in (MD) forces $\kappa_{\{k\}} = 1$ and every other $\kappa_C = 0$. Exactness at $\{k\}$ then supplies the required z. □

The other hypothesis also has a scalar primal form. For a fixed optimal minimum cover κ, its positive support is jointly tight at a full-core point if and only if

$$\min_{z\in\mathrm{Core}(v)}\{cz(A)+z_k\}=cv(A)+m_A. \tag{NL}$$

Indeed, (MD) makes the left side at least the right side, and equality is equivalent to zero total κ-weighted core slack. Thus the remaining compatible-cover route is entirely primal: prove tangent non-descent (TN) and the minimum-normal lift (NL) from star failure and the leximin strata. These two statements concern the same face. Every feasible y in (PT) satisfies

$$\sum_C \kappa_C[y(C)-v(C)]=c[y(A)-v(A)]+(y_k-m_A)=ct-ct=0. \tag{MF}$$

All summands are nonnegative, so every positive κ-coalition is tight along the entire tangent LP. Hence tangent non-descent is an optimization statement on the restricted minimum-cover face, while (NL) says that this same tight face meets the full core.

This also gives an entirely primal proof of Proposition 5.123. Let z attain (NL) and put $t=z(A)-v(A)$. Equality in (NL) gives $z_k=m_A-ct$, so the restriction of z to A, together with $u_i=z_i$ and $u_j=z_j$, is feasible in (PT). Full efficiency gives $z_i+z_j=d_A-t$, hence this feasible point has objective zero. Therefore

$$\Gamma_A=\widehat{\Gamma}_{A,c}(0)\le\widehat{\Gamma}_{A,c}(t)\le 0$$

by tangent non-descent.

Exactness alone does not force the normal lift, even at unit price. Define

$$\Delta^{\mathrm{NL}}_{A,k}=\min_{z\in\mathrm{Core}(v)}\{z(A)+z_k\}-[v(A)+m_A]\qquad(c=1).$$

The most natural quantitative comparison would be

$$\Delta^{\mathrm{NL}}_{A,k}\le\eta_S(\phi). \tag{SN}$$

It is false, and its sharp-bound deformation also refutes star-conditioned trace extension itself.

Proposition 5.124 (exact star–trace counterexample)**.** *For $0\le\varepsilon\le 1$, put*

$$\begin{aligned}p&=(13,13,-12,-3,13,0,1),\\ q^{\varepsilon}&=(11,14+\varepsilon,-3,2-\varepsilon,16,-10,-5),\\ u&=(19,19,-20,-3,20,2,-12),\end{aligned}$$

and let v^{ε} be their lower-envelope game, with common grand worth 25. Let $S=\{2,4,5\}$, $k=5$, $A=\{1,3,5,6,7\}$, and $P=A\setminus\{5\}$.

At $\varepsilon=1/10$, the restricted minimum has a unit-price dual but

$$\Delta^{\mathrm{NL}}_{A,5}=1>\frac{17}{20}=\eta_S(\phi),$$

so (SN) fails. At $\varepsilon=2/3$, the star condition fails at the leximin point, S is binding, and nevertheless

$$\max_{y\in\mathrm{Core}_A(v)}y(P)=-5>-6=\max_{x\in F_A}x(P). \tag{CX}$$

Thus trace extension fails, with trace gap 1 and split obstruction $\Gamma_A=2$.

Proof. All three generators have sum 25 and belong to the core of their lower-envelope game. Every proper-coalition worth is attained by a generator, so v^ε is exact.

At $\varepsilon = 1/10$, exact primal–dual LP pairs give restricted minimum 14, the unit cover $\kappa_{\{3,5,6\}} = \kappa_{\{1,5,7\}} = 1$, normal-lift gap 1, and star opening 17/20.

At $\varepsilon = 2/3$, the leximin point is

$$\phi = \left(\frac{43}{3}, 16, -\frac{25}{2}, 0, \frac{52}{3}, -\frac{5}{2}, -\frac{23}{3}\right).$$

Its successive gain levels are $3, 10/3, 13/3, 15/2$, pinning respectively $\{2,4\}, \{1\}, \{5,7\}, \{3,6\}$; exact primal–dual pairs certify each leximin stage. The core point u has $u(S) = 36 = \phi(S)+8/3$, so S is binding. For the star LP, unit weights on coalitions $\{1,2,3,6,7\}$ and A, unit weight on the floor of player 4, and grand coefficient -2 give an exact dual upper bound $x(S) \le 100/3 = \phi(S)$. Hence the star opening is zero; the corresponding residual vector on S is $(1,2,1)$.

For trace extension, the restricted witness $(y_1, y_3, y_5, y_6, y_7) = (13, -5, 14, -8, -5)$ attains $y(P) = -5$; unit dual weights on $\{3,5,6\}$ and $\{1,5,7\}$ certify optimality. On the full face, the witness

$$(13, 47/3, -38/3, 1/3, 15, -4/3, -5)$$

attains $x(P) = -6$. Half-unit dual weights on $\{3,5,6\}, \{1,3,5,6\}, \{2,5,7\}, \{1,4,5,7\}$, together with grand and face coefficients $-1/2$ and $-3/2$, certify that this is the full-face maximum. This proves (CX); the exact split LP gives $\Gamma_A = 2$. □

The counterexample was missed by broad but non-adaptive screens. Across 10,395 arbitrary codimension-two faces of 99 nondegenerate exact seven-player lower-envelope games, 86 restricted optima did not extend, with exactly certified gaps of 3 and 6 in the two control corpora. Yet none of those gaps occurs among the 501 faces conditioned on star failure. The largest exact ratio of trace gap to star opening in the first control is 5, so the tempting quantitative bound "gap at most twice the opening" was already false. Six rational affine bridges from a genuine three-level failure to an exact trace-gap control tell the same story: every zero-opening grid point has zero gap, while the first positive gap appears only at $t = 17/20$ or $9/10$, after star opening is already positive. No gap occurs in the six exact three-player failure cases or in five additional float multilevel candidates from 1,484 nondegenerate targeted games; exactification sent the sampled extremals from the latter set to the two-level branch. A further 1,000-game local mutation screen retained 54 genuine three-level failures with a unique minimum residual. All 54 have zero trace gap and positive face-target margin; two rational extremals certify target margins 15/4 and 2.

The decisive search was instead transverse to the exact positive equality $\Delta^{\mathrm{NL}}_{A,5} = \eta_S(\phi) = 1$. Of 252 two-scale, sum-preserving generator deformations, all retained a returned unit-price cover and positive normal gap, while 88 violated (SN) in floating arithmetic. The ten closest records all recomputed to exact violations. On the rational path in Proposition 5.124, the normal gap stays 1 and Γ_A stays 2; the exact star opening is 1/40 at $\varepsilon = 13/20$ and reaches zero at $\varepsilon = 2/3$. Therefore star failure alone does not imply trace extension, and this unqualified trace route is closed. The counterexample has residuals $(1,2,1)$, however, so it belongs to the two-level branch already settled by Proposition 5.3. It does not refute the unresolved genuinely multilevel program. Proposition 5.113 then suggests the sharp repair.

Conjecture 5.125 (multilevel star-conditioned trace extension)**.** *Let v be exact, let $|S| = 3$, suppose $(\star_S)$ fails at the leximin point, and suppose no failure cover has at most two residual levels on S. Let k be the unique minimum-residual player supplied by Proposition 5.113, put $A = (N \setminus S) \cup \{k\}$ and $P = A \setminus \{k\}$. Then equality holds in (TE).*

This is now a seven-case exact frontier. An exact whole-face audit now verifies the quantifier in the conjecture, rather than merely inspecting one returned cover: for each of the three residual pairs, adding the equality $r_i = r_j$ to the rational failure-cover polyhedron is infeasible in all seven cases. Hence every cover has three distinct levels. Their strict residual orders are fixed, and after normalizing the minimum residual to one, the exact minimum gaps to the other two coordinates are one and two. The original six cases have $\Gamma_A = 0, 0, 0, -3, 0, 0$, the seventh case in Proposition 5.116 has zero trace gap, and the 54 genuine three-level local mutations all have zero trace gap. Thus the seventh case falsifies the stronger face target while continuing to satisfy this trace-extension conjecture. The earlier positive-trace-gap counterexample passes the control check: one of its pair-tie systems is feasible, so it is genuinely outside the conjectured branch.

The new counterexample shows that the multilevel hypothesis cannot simply be deleted. A targeted tie-splitting audit makes 378 exact sum-preserving deformations around the counterexample. Of these, 246 remain binding star failures; every recovered failure cover still has exactly two residual levels, and none enters the three-level chamber. A second transverse screen makes 756 deformations around two distinct whole-face multilevel fixtures. It retains 181 binding star failures and 164 returned three-level unique-minimum cases, with no positive trace gap; the extremum from each fixture recomputes exactly to gap zero and remains whole-face genuinely multilevel. Thus the positive trace gap appears locally confined to the already-settled boundary.

A separate 3,528-deformation screen targeted the high-coordinate sign rather than trace gap. It retained 1,275 binding star failures and 1,146 returned three-level failures. Two had negative float high-coordinate margins; the worst recomputed exactly to the $-2/3$ face and coordinate margins in Proposition 5.116. This closes the stronger face-target route but leaves both the minimum-residual carrier inequality and multilevel trace extension intact in the new case.

The new criterion identifies the missing step more sharply. The exact Γ_A values in the six three-level failures are $0, 0, 0, -3, 0, 0$. By contrast, the exactly certified worst-gap points in the two controls have $\Gamma_A = 3$ and 6, while the separate ratio-5 control has $\Gamma_A = 5$ even though the first returned restricted optimum has excess 6. All have zero two-player-trace excess, as Proposition 5.120 requires. The counterexample shows that star failure need not force $\Gamma_A \le 0$ without the multilevel hypothesis in Conjecture 5.125.

A tempting point-free strengthening is false even inside the hard corpus. Maximize each marginal $v(B \cup \{p\}) - v(B)$ separately over traces B that contain k and no other member of S. In one exact three-level star failure the two maxima exceed d_A by 3, although $\Gamma_A = 0$: the selected restricted optimum places exactly compensating slack on one offending trace. Thus a universal marginal-worth inequality cannot prove trace extension; selection inside the restricted optimal face, equivalently the dual balance (SDB), is essential.

The compatible-cover certificate gives proof-producing decompositions for all six exact three-level cases. Every optimal split-dual face contains a member with $\alpha - 1 = \beta c$, and every minimum-cover support has exact common-tightness gap zero. Three cases have $c = 0$ and hence follow from singleton exactness directly; the three $c = 1$ cases have exact rational common core witnesses. In the sixth case the dual first returned by the LP does not have the coefficient match, but an exact search of its optimal-dual face finds the compatible representative $\alpha - 1 = \beta = 1$.

No optimal-dual face of a positive-gap control contains a compatible member. The worst-gap controls also have common-tightness gap 2; the ratio-5 control has a common-tight minimum support but fails coefficient compatibility throughout its optimal split-dual face. Primal tangent minimization gives the same separation: the base parameter minimizes (PT) in all six hard cases, whereas the three distinct positive-gap controls have exact tangent improvements 1, 6, and 4, attained at $t = 1$, 11, and 4. The stronger demand that the split weights dominate $\beta\kappa$ coefficientwise is feasible in only one of the six hard cases, so the signed common-tightness argument is genuinely needed. The counterexample has $c = 1$, normal-lift gap 1, no compatible optimal split dual, and exact tangent improvement 2. Hence neither tangent non-descent nor the minimum-normal lift follows from star failure and the leximin strata alone. Conjecture 5.125 excludes the counterexample's two-level residual pattern; proving or breaking that repaired statement is now the precise face-lifting target.

The cover-independent audit reveals a substantially simpler target. It uses the full minimum-residual class at once.

Conjecture 5.126 (minimum-residual-class carrier)**.** *Let v be exact and let $(\star_S)$ fail at ϕ. If the failure is not already covered by Proposition 5.3, then there is a failure cover with minimum-residual class $K = \{i \in S : r_i = r_0\}$ and $P = N \setminus S$ such that*

$$[\phi(P \cup K) - v(P \cup K)] \leq [\phi(K) - v(K)], \tag{KC}$$

or equivalently

$$v(P \cup K) - v(K) \geq \phi(P).$$

This single inequality would close the multilevel capacity branch immediately: apply Proposition 5.5 with $C = K$. It requires no singleton selector and is insensitive to how payoffs are ranked inside K. A deduplicated exact audit verifies (KC) for all 669 stored multilevel game–coalition pairs (702 records): 644 inequalities are strict and 25 are equalities, with margins ranging from -28 to 0. This includes all 505 adversarial mutation pairs on which the fixed-cover escape test mostly fails. Thus (KC), rather than a singleton rule or residual-gap escape, is now the sharp computational theorem target. Deterministic replay of the historical random streams recovered and verified the discarded point generators for all 25 equality records. They represent 23 games, and every one is the lower envelope of exactly two *distinct* common-sum points; the nominal three-generator records merely repeat one point. Hence every non-strict case in this exact corpus already lies in the model class settled by Proposition 5.35. Outside that proved boundary subcase, all audited class certificates are strict.

Lemma 5.36 gives an exact-game face reduction; the finite-model audit is one way to expose its witnesses. Among the 530 deduplicated class-carrier records that retain their point generators, 461 are direct: some generator attaining $v(P \cup K)$ already has $a(P) \geq \phi(P)$. Each of the remaining 69 records has exactly three distinct generators, a unique $P \cup K$ minimizer, $|K| = 3$, and a strictly positive trace-compensated margin between $19/4$ and 8. All 69 arise from one seeded target family. These counts are exact finite evidence, not a three-generator theorem, but they isolate the only nontrivial generator mechanism in the embedded corpus.

The quantifier over failure covers in Conjecture 5.126 is essential. The next exact three-generator family has failure covers in three residual chambers. On an open parameter interval, two chambers select a class that violates (KC), while the third selects the unique carrier class. Thus no distinguished output of a failure-cover LP can be used without a separate selection argument.

Proposition 5.127 (failure-cover selection is essential)**.** *For $0 \le \kappa \le 1$, let v^κ be the lower envelope of*

$$\begin{aligned} p^\kappa &= (-4, 3-\kappa, 27, \kappa, 10, -8), \\ q^\kappa &= (-4, 1-\kappa, 29, -2+\kappa, 12, -8), \qquad v^\kappa(T) = \min\{p^\kappa(T), q^\kappa(T), u^\kappa(T)\}. \\ u^\kappa &= (-\kappa, -1, 22, \kappa, 10, -3), \end{aligned}$$

Then v^κ is exact and

$$\phi^\kappa = \left(-\frac{5}{2} - \frac{\kappa}{2}, \frac{1}{2} - \frac{\kappa}{2}, \frac{51}{2} - \frac{\kappa}{2}, -1+\kappa, 11, -\frac{11}{2} + \frac{\kappa}{2}\right).$$

For $S = \{2,3,4,6\}$ and $P = \{1,5\}$, the coalition S is binding by $5/2 - \kappa/2$, but $(\star_S)$ fails throughout the interval. Moreover every normalized failure cover has, for some

$$a \ge 1, \qquad t - a \ge 1,$$

the residual vector

$$(r_2, r_3, r_4, r_6) = (a, t, t-a, a+t), \tag{RC}$$

and every vector in this cone is attained by a failure cover. Its minimum-residual class is

$$K = \begin{cases} \{2\}, & t > 2a, \\ \{2,4\}, & t = 2a, \\ \{4\}, & a+1 \le t < 2a. \end{cases}$$

Writing

$$\Delta_\kappa(K) = [\phi^\kappa(P \cup K) - v^\kappa(P \cup K)] - [\phi^\kappa(K) - v^\kappa(K)],$$

the three possible margins are

$$\Delta_\kappa(\{2\}) = -\frac{3}{2} + \frac{\kappa}{2}, \qquad \Delta_\kappa(\{4\}) = \Delta_\kappa(\{2,4\}) = \frac{1}{2} - \frac{\kappa}{2}.$$

Thus, for $0 \le \kappa < 1$, the normalized cover with residuals $(1,2,1,3)$ selects the noncarrier class $\{2,4\}$, whereas the cover with residuals $(1,3,2,4)$ selects the carrier class $\{2\}$. At $\kappa = 1$ the former class reaches carrier equality while the latter remains strict.

Proof. All three generators have total 28. Each is therefore efficient and dominates v^κ on every coalition, hence belongs to the core; a generator attains $v^\kappa(T)$ for every T, proving exactness. The singleton floors are

$$m^\kappa = (-4, -1, 22, -2+\kappa, 10, -8).$$

We verify the leximin point directly. If every gain $x_i - m_i^\kappa$ is at least s, the core inequality $x(\{1,2,3,6\}) \ge 18 - \kappa$ and efficiency imply $x_4 + x_5 \le 10 + \kappa$, while the floors imply $x_4 + x_5 \ge 8 + \kappa + 2s$. Thus $s \le 1$, and equality pins $(x_4, x_5) = (-1+\kappa, 11)$. On that face, $x(\{3,4,6\}) \ge 19 + \kappa$ gives $x_1 + x_2 \le -2 - \kappa$; hence the next gain is at most $(3-\kappa)/2$, with equality pinning $(x_1, x_2) = (-5/2 - \kappa/2, 1/2 - \kappa/2)$. On the resulting face, $x(\{1,3,4\}) \ge 22$ gives $x_6 \le -11/2 + \kappa/2$, so the next gain is at most $5/2 + \kappa/2$. Equality pins x_6, and efficiency fixes $x_3 = 51/2 - \kappa/2$. Direct affine endpoint substitution shows that the displayed point is in the core throughout $0 \le \kappa \le 1$, so these successive bounds prove the formula for ϕ^κ.

The proper coalitions tight at ϕ^κ are exactly

$$134, \quad 125, \quad 45, \quad 1236, \quad 346,$$

where strings denote player sets. For each other proper coalition, one fixed generator lies strictly below ϕ^κ at both parameter endpoints, so the same affine strict inequality holds throughout the interval. The exact verifier records all 62 selectors. The multipliers

$$\lambda_{134} = 1, \qquad \lambda_{125} = 2, \qquad \lambda_{45} = 1, \qquad \mu = 3$$

cover the two players in P exactly μ times and leave residuals $(1, 2, 1, 3)$ on S. If d were a direction in $(\star_S)$, then

$$r \cdot d_S = \mu d(N) - \sum_T \lambda_T d(T) \le 0,$$

but $d_i \ge 0$ on S and $d(S) > 0$ make the left side positive. Hence $(\star_S)$ fails. Also $v^\kappa(P) = 6$, so every core point has $x(S) = 28 - x(P) \le 22$; the generator p^κ attains equality, while $\phi^\kappa(S) = 39/2 + \kappa/2$. This proves the binding margin $5/2 - \kappa/2$.

It remains to describe the whole cover cone. Put multipliers A, B, C, D, E on the five tight coalitions in the displayed order. Exact coverage of players 1 and 5 gives

$$\mu = A + B + D = B + C, \qquad \text{so} \qquad C = A + D.$$

Setting $a = A$ and $t = B - E$ now gives

$$(r_2, r_3, r_4, r_6) = (a, t, t - a, a + t).$$

The normalization $r_i \ge 1$ reduces to $a \ge 1$ and $t - a \ge 1$. Conversely, every such pair is realized by taking $D = E = 0$, $A = C = a$, and $B = t$. Comparing the four coordinates yields the three stated residual chambers. Direct substitution in the worth function gives the displayed affine values of $\Delta_\kappa(K)$. Finally, $(a, t) = (1, 2)$ and $(1, 3)$ give the two explicit covers claimed. □

A dedicated screen of 500 six-player, three-generator nearby instances retained 496 games with three distinct generators. It found 30 multilevel binding star failures and one positive margin for the cover returned by the search LP; exactification produced the $\kappa = 0$ member of Proposition 5.127. An exact prescribed-class audit then tested every proper class and found the favorable $K = \{2\}$ cover above. Two targeted mutation sweeps around that member retained 1,998 nondegenerate games and found 28 multilevel binding star failures. The returned cover failed in 16, but full chamber enumeration found a carrier cover in every case; the closest exact mutation is the $\kappa = 1$ endpoint. These screens are finite discovery evidence; the proposition itself is analytic and exhaustive over the full cover cone for every parameter value.

A separate targeted screen of 3,000 four-generator lower-envelope games found 1,123 star failures and 25 genuinely multilevel failures, with no positive class-carrier margin; the closest boundary case was recomputed exactly and has margin zero. The broader screen is discovery evidence only, whereas the 669-pair audit is fully exact. The boundary admits an exact parametric analysis.

Proposition 5.128 (class-carrier boundary coupling)**.** *For $0 \le \varepsilon \le 1/3$, let v^ε be the lower envelope of the four common-sum points*

$$p = (1, 1, 1, 2, 1, 2), \quad q^\varepsilon = (0, 1, 2 - \varepsilon, 0, 4 + \varepsilon, 1),$$
$$r = (2, 1, 2, 1, 2, 0), \quad s = (1, 2, 0, 2, 1, 2).$$

Put $S = \{2,4,5,6\}$, $P = \{1,3\}$, *and* $K = \{4,6\}$. *Then* v^ε *is exact,*

$$\phi(v^\varepsilon) = (1,2,1,1,2,1),$$

the class-carrier failure margin is exactly

$$[\phi(P \cup K) - v^\varepsilon(P \cup K)] - [\phi(K) - v^\varepsilon(K)] = \varepsilon,$$

and the star opening is exactly

$$\eta_S(\phi) = 3\varepsilon.$$

Thus crossing the class-carrier boundary in this exact family restores $(\star_S)$ *at three times the crossing rate.*

Proof. The generators have common total 8, so their lower envelope is exact. The singleton floors are $(0,1,0,0,1,0)$. Direct endpoint substitution shows that $\phi = (1,2,1,1,2,1)$ belongs to the core throughout the interval. Its six gains all equal one, while the total gain is six; hence it is the leximin point. Moreover,

$$v^\varepsilon(K) = 1, \qquad v^\varepsilon(P \cup K) = 3 - \varepsilon, \qquad \phi(P) = 2,$$

which proves the carrier-margin formula.

The allocation

$$y^\varepsilon = \phi + \varepsilon(-1,0,-2,1,0,2)$$

belongs to the core for $0 \le \varepsilon \le 1/3$, dominates ϕ on S, and satisfies $y^\varepsilon(S) = 6 + 3\varepsilon$. Core membership is a finite affine endpoint check: for each proper coalition the same one of the four generators lies weakly below y^ε at both endpoints, hence throughout the interval. The standalone exact verifier records all 62 selectors.

For the reverse inequality, let x be any core point with $x_i \ge \phi_i$ for $i \in S$. The three core constraints and two coordinate floors

$$x_{34} \ge 2 - \varepsilon, \quad x_{145} \ge 4, \quad x_{136} \ge 3 - \varepsilon, \quad x_2 \ge 2, \quad x_5 \ge 2$$

give, after taking multiplicities $(1,1,2,2,1)$,

$$3x(P) + 2x(S) = 24 - x(S) \ge 18 - 3\varepsilon.$$

Thus $x(S) \le 6 + 3\varepsilon$, and y^ε attains equality. □

The recovered two-generator boundary has a second exact local test. Six of the 25 equality records had stored a duplicated generator. A 1,620-direction single-copy splitting screen found positive minimum-class margin in exactly those six records. In the worst screened direction for each record, exact constraint-generation certificates give class margin $1/100$ and star opening $1/50$, hence the same ratio 2 in all six cases. These are class-specific comparisons: another carrier has margin zero in every deformation, so the global γ_S defined above remains zero. The calculation therefore supplies further boundary coupling for (KC), not an improved coefficient in (Q). The exact leximin verifier begins with singleton and co-singleton core constraints and adds every violated omitted coalition until its rational optimizer satisfies the full core.

Proposition 5.3 explains an important part of this structure exactly. If the residuals have only the levels 1 and $R > 1$, the entire minimum-residual class $C = \{i \in S : r_i = 1\}$ is itself

the required face certificate; no crossing-cover construction is needed. The rare same-stratum pair exceptions to singleton certification are examples with $|C| = 2$. Accordingly, all unresolved certificate geometry lies in duals with at least three distinct residual levels. Proposition 5.5 removes a second large class: it is enough that an off-side carrier have no more slack than its trace inside S; tightness of the carrier is a particularly simple sufficient condition. In particular, the three-level witness `WITNESS_KEY_FALSE_n6` is certified by $T = \{1, 2, 6\}$ and $C = \{1\}$ (player labels one-indexed), even though the union-of-cover certificate described above fails.

Proposition 5.2 removes the frequency language from the logical frontier. Every pair (v, S) falls into the exact dichotomy

$$d_v^*(S) \le \sigma_S \quad \text{or} \quad d_v^*(S) > \sigma_S.$$

The first side admits no binding exact bump and is finished by Lemma 5.112. On the second side a binding exact bump exists, and the missing structural statement is precisely

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Proposition 5.3 proves this implication whenever the failure dual has at most two residual levels, and Corollary 5.4 covers every $|S| = 2$. The remaining capacity frontier therefore requires both $|S| \ge 3$ and a genuinely three-or-more-level dual. After Proposition 5.5, a counterexample to the capacity implication must additionally have no off-side slack-dominating carrier. The singleton/pair certificate statistics motivate how to attack those cases, but neither their observed percentages nor sampling rarity enters the conditional theorem.

The sharpened computational target is correspondingly narrow. The broad slack-domination condition covered all 141 saved hard star-failure cases at $n = 6, 7$ and all 744 fresh binding star failures in a screen of exact lower-envelope games at $n = 4, 5, 6$, with no exception. The new class-carrier audit finds a successful full minimum-residual class in all 669 distinct exact pairs. Proposition 5.127 shows that this class need not come from an arbitrarily returned failure cover. This remains evidence, not a theorem, but proving Conjecture 5.126 would close that branch with a single application of Proposition 5.5. Proposition 5.35 proves the broader existence of some slack-dominating carrier for every two-generator lower-envelope game. Thus within the finite lower-envelope model class, any counterexample must use at least three genuinely active generators; this is a structural restriction, not a sampling observation.

The obvious three-generator escape from the sharp two-generator boundary is itself blocked, but for an instructive reason. In the exact equality fixture, let $S = \{1, 2, 3, 4, 6\}$ and transfer $a = 1/10$ from the off-side player 5 of the second generator equally among the five members of S, retaining both original generators. All three generators are genuinely active. The leximin point remains $(1, 4, 3, 5, 7, 4)$ and the minimum off-side slack comparison becomes $1/50 > 0$, so the carrier condition has indeed been crossed. However, $(\star_S)$ is simultaneously restored: the core point

$$(26/25, 4, 153/50, 253/50, 171/25, 4)$$

dominates ϕ on S and has S-sum $429/25 > 17$. Thus the star opening is $4/25$, eight times the carrier escape. This is an exact boundary calculation, not a general theorem; a finite simplex-grid deformation screen found the same coupling throughout the tested local family. It suggests proving the carrier conjecture contrapositively: show that escaping every off-side carrier inequality constructs a core direction nonnegative on S, rather than trying to track one distinguished minimizing generator.

The carrier lift suggests a sharp quantitative form of that contrapositive:

$$\eta_S(\phi) \geq \gamma_S(\phi). \tag{Q}$$

Proposition 5.25 disproves (Q) in unrestricted dimension, while Proposition 5.24 proves it through six players. In view of Proposition 5.13, this sharp finite-dimensional theorem applies to every $\psi \in \mathrm{Core}(v)$, not only to $\psi = \phi$. Proposition 5.17 proves it for $|S| = 2$, and the coefficient one is already sharp there. Proposition 5.16 proves the stronger uniform lift when $|N \setminus S| = 1$; hence Corollary 5.18 settles (Q) for every exact game with at most four players, and Proposition 5.19 closes the remaining three-versus-two case by exact dual enumeration, even under the weaker assumption of total balancedness. Finally, Proposition 5.24 closes both new six-player splits by complete minimal-balanced enumeration. The three-generator lower envelope in Proposition 5.25 then violates (Q) at seven players at its leximin point with $(\gamma, \eta) = (3/4, 0)$ (and at the core reference zero with $(1, 0)$), making the six-player theorem sharp. Earlier, six exact lower-envelope screens covering 3,650 games and 35,629 binding pairs (v, S), all 746 cases with $\gamma_S(\phi) > 0$ satisfied (Q); the minimum in every screen was recomputed exactly and equalled one. These counts are now historical cross-checks, not part of the proof. The exactness hypothesis is load-bearing even for the coefficient: a balanced non-exact five-player game has leximin point $(-3/2, 10, 6, 17/2, -10)$ and, at $S = \{2, 3, 4\}$, $\gamma_S = 7/2$ but $\eta_S = 1/2$. Thus the dimension-free target is a convex-combination statement for the carrier-tight points supplied by Lemma 5.12, not an inequality valid on arbitrary balanced cores.

Earlier independent screens anticipated the theorem and tested whether it depended on the leximin reference. At the leximin point, 10,000 four-player games yielded 4,473 positive-γ three-player targets, including exact equality, and 3,000 five-player games yielded 3,362 positive-γ targets of size at least three; none had $\eta/\gamma < 1$, and the exact five-player minimum was 91/83. More strikingly, the leximin hypothesis was then discarded: 10,000 exact four-player core references yielded 19,521 positive-γ cases with exact minimum one, while 6,400 exact five-player references yielded 3,732 positive-γ targets of size at least three, again with no violation. Finally, 800 exact references drawn from four saved five-player games with genuinely non-large cores produced 729 positive-γ targets of size at least three, no violation, and exact minimum ratio one. Thus the pattern is not explained merely by the implication from exactness to largeness that holds in smaller player sets. These finite campaigns do not add logical scope to Propositions 5.19 and 5.24, but they helped identify the selection-free exact-core closure inequality (CG)–(CE).

The common-coordinate strengthening $\kappa_S(\psi) \geq \gamma_S(\psi)/|S|$ also survived dense, sparse, generator, and non-large exact screens in the first open regime $|S| = 3$, $|N \setminus S| = 2$. The largest generator campaign contained 60,418 positive-γ targets and had exact minimum normalized ratio 47/45; the non-large audit attained equality. By contrast, a targeted 26,471-case screen drove the corresponding outside-slack ratio down to 3/112 while retaining carrier-normalized ratio 3/2. Thus a proof cannot replace the carrier slack system by a single outside-coalition cap; the exact dual in Proposition 5.14 is the sharper remaining object.

At six players the totally balanced analogue already fails, with $(\gamma, \eta) = (1, 0)$, but exact closure removes the gap. The resulting three-block identity, Lemma 5.21, first extends to the complete unit-triple layer of Proposition 5.22. More decisively, Lemma 5.20 converts every relevant eta-dual vertex into a minimal balanced collection containing S. Exhausting all 1,489 symmetry orbits and certifying every exact-cone optimum proves the quantitative carrier inequality for every exact game through six players (Proposition 5.24). The earlier direct sample of 505 orbits is recovered in full by the census and now serves as an independent cross-check

rather than merely finite evidence. The three-generator exact game in Proposition 5.25 has $(\gamma, \eta) = (3/4, 0)$ at its leximin point, so this finite theorem is dimension-sharp.

Remark 5.129. *The monotonicity picture is now sharp in minimum dimension. Singleton bumps are monotone by Theorem 2.2; grand-coalition resource monotonicity fails by Theorem 3.2; and proper nonsingleton coalitional monotonicity fails on exact five-player games by Theorem 5.37, at the minimum possible size by Theorem 5.111. The exact counterexample also shows why the earlier cone heuristic failed: an all-target-improving feasible direction exists throughout the binding path, but leximin re-equalization selects a different direction that lowers two previously advantaged target players.*

References

Björner A, Las Vergnas M, Sturmfels B, White N, Ziegler GM (1999) *Oriented Matroids*, 2nd ed. Cambridge University Press.

Dutta B, Ray D (1989) A concept of egalitarianism under participation constraints. Econometrica 57:615–635. https://doi.org/10.2307/1911055

Dietzenbacher B (2021) Monotonicity and egalitarianism. Games Econ Behav 127:194–205. https://doi.org/10.1016/j.geb.2021.03.006

Arin J, Iñarra E (2001) Egalitarian solutions in the core. Int J Game Theory 30:187–193. https://doi.org/10.1007/s001820100073

Arin J, Kuipers J, Vermeulen AJ (2008) An axiomatic approach to egalitarianism in TU-games. Int J Game Theory 37:565–580. https://doi.org/10.1007/s00182-008-0133-6

Mauri Masdeu L (2018) Essays on egalitarianism-based solution concepts for cooperative TU-games. Doctoral thesis, Universitat Rovira i Virgili. https://repositori.urv.cat/estatic/TDX0011/en_TDX2707.html

Biswas AK, Parthasarathy T, Potters JAM, Voorneveld M (1999) Large cores and exactness. Games Econ Behav 28:1–12. https://doi.org/10.1006/game.1998.0683

Llerena F, Mauri L (2017) On the existence of the Dutta–Ray's egalitarian solution. Math Soc Sci 89:92–99. https://doi.org/10.1016/j.mathsocsci.2017.07.002

Dietzenbacher B, Yanovskaya E (2020) Antiduality in exact partition games. Math Soc Sci 108:116–121. https://doi.org/10.1016/j.mathsocsci.2020.04.006

Young HP (1985) Monotonic solutions of cooperative games. Int J Game Theory 14:65–72. https://doi.org/10.1007/BF01769885

Housman D, Clark L (1998) Note: Core and monotonic allocation methods. Int J Game Theory 27:611–616.

Schmeidler D (1972) Cores of exact games, I. J Math Anal Appl 40:214–225. https://doi.org/10.1016/0022-247X(72)90045-5

Csóka P, Herings PJJ, Kóczy LÁ (2011) Balancedness conditions for exact games. Math Methods Oper Res 74:41–52. https://doi.org/10.1007/s00186-011-0348-3

Lohmann E, Borm P, Herings PJJ (2012) Minimal exact balancedness. Math Soc Sci 64:127–135. https://doi.org/10.1016/j.mathsocsci.2012.01.002

Laplace Mermoud D, Grabisch M, Sudhölter P (2025) Minimal balanced collections and their applications to core stability and other topics of game theory. arXiv:2507.05898. https://arxiv.org/abs/2507.05898

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Math Methods Oper Res 95:35–80. https://doi.org/10.1007/s00186-022-00770-4

Studený M, Kratochvíl V (2017) Linear core-based criterion for testing extreme exact games. Proceedings of Machine Learning Research 62:313–324. https://proceedings.mlr.press/v62/studen%C3%BD17a.html

Fujishige S (1980) Lexicographically optimal base of a polymatroid with respect to a weight vector. Math Oper Res 5:186–196. https://doi.org/10.1287/moor.5.2.186

Frank A, Murota K (2022) Decreasing minimization on base-polyhedra: relation between discrete and continuous cases. arXiv:2202.06561. https://arxiv.org/abs/2202.06561

Miranda E, Montes I (2023) Centroids of the core of exact capacities: a comparative study. Ann Oper Res 321:409–449. https://doi.org/10.1007/s10479-022-05097-1

Berge C (1972) Balanced matrices. Mathematical Programming 2:19–31. https://doi.org/10.1007/BF01584535

Cornuéjols G, Guenin B, Tunçel L (2009) Lehman matrices. J Combin Theory Ser B 99:531–556. https://doi.org/10.1016/j.jctb.2008.06.009

Cornuéjols G (2001) Combinatorial Optimization: Packing and Covering. SIAM, Philadelphia. https://doi.org/10.1137/1.9780898717105

Király T, Pap G (2016) An extension of Lehman's theorem and ideal set functions. Discrete Appl Math 209:251–263. https://doi.org/10.1016/j.dam.2015.07.035